WOBBLEBEANS

Master Game Design Document

Original online party game for children aged 6–12 · PC (Windows) first · free MVP



Version 1.0 · 23 October 2026 · Status: merged draft for owner review

Merged from ten design documents, built on the core requirements v3 (Wobblebeans_Requirements_v3.docx).

All imagery in this document is a placeholder. Final art will come from the owner's own GitHub asset repository (UD-3, Q-D2).

Document control

Version history

Version	Date	Status	Change
1.0	23 Oct 2026	Merged draft for owner review	First master document. Merges design documents 01–10 into one file, removes duplicate content, reconciles conflicts (Appendix B), consolidates the design issues (Appendix A) and the open questions (Appendix C).

Source documents

The ten source documents (all dated 23 October 2026) and the requirement baseline they trace to:

Section	Source document	Owner (as stated)	Version (as stated)
1	`01-core-gdd.md` — Core Game Design Document	Principal Game Designer	1.0 — 23 Oct 2026; design-ready for MVP pre-production
2	`02-rounds-levels.md` — Round & Level Design (MVP round set)	Senior Level Design	Draft v1.0 (23 Oct 2026)
3	`03-character-controls.md` — Character Movement, Physics Feel & Controls	Gameplay/Feel Design	Draft v1 (2026-10-23)
4	`04-ux-ui.md` — UX Flows & UI Design	UX/UI design (kids 6–12)	Draft v1 (2026-10-23)
5	`05-art-direction-assets.md` — Art Direction Framework & Asset Pipeline	Art Direction	Proposal v0, framework only (23 Oct 2026)
6	`06-audio.md` — Audio Design	Audio Director	Draft v1 (2026-10-23)
7	`07-progression-rewards.md` — Progression, Rewards & Customisation Systems	Not stated	Systems/economy design (23 Oct 2026)
8	`08-technical-architecture.md` — Technical Architecture Design	Technical/systems architecture	23 Oct 2026 (no version stated)
9	`09-multiplayer-safety.md` — Multiplayer, Social & Child Safety Design	Multiplayer / Trust & Safety design	Design v1 (23 Oct 2026)
10	`10-creative-liveops.md` — Builder (Creative Level Editor) & Live Ops Design	Senior Designer, UGC & Live Ops	23 Oct 2026 (no version stated)
—	`Wobblebeans_Requirements_v3.docx` — core requirements v3 (FR/NFR, AD, UD, DEP, gapfill-A to D)	Requirements analyst	v3

Order of authority used in the merge

Where two source documents conflicted, the merge followed this order. Every reconciliation is logged in Appendix B.

Topic	Authoritative source
Game rules	Section 1 — Core Game Design (`01-core-gdd.md`)
Physics and controls	Section 3 — Character Movement, Physics and Controls (`03-character-controls.md`), with gapfill-A of the requirements
Systems	Section 8 — Technical Architecture (`08-technical-architecture.md`)
Child safety	Section 9 — Multiplayer, Social and Child Safety (`09-multiplayer-safety.md`)
Everything else	The section that owns the topic; the core requirements v3 remain the baseline for all sections

Sign-off

Role	Name	Decision	Date
Owner			
Requirements analyst			
Privacy counsel			
Technical lead			

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Executive summary

Wobblebeans is an original online party game for children aged 6–12. Up to 32 wobbly bean characters tumble through obstacle rounds in a short "show"; players are eliminated round by round until a Final crowns a winner. The vision (Section 1): *a bright, giggly, 15-minute show where every child walks away with a laugh, a reward and a reason to try again — never a reason to feel bad.*

This document merges the ten design documents into one master design. It keeps every requirement ID from the core requirements v3, so each design rule can be traced back to the requirement it meets.

Scope at a glance

Topic	MVP position	Where
Audience and rating	Children 6–12, every player treated as a child; target PEGI 3 / ESRB E (UD-1, AD-01, AD-24)	§1.2, Section 9
Platform	PC (Windows 10/11, 64-bit) first; consoles and mobile Later (AD-51)	§1, §8.0
Money	Free; no real-money purchases in the MVP; Coins are earned only and buy cosmetics only (AD-12)	Section 7
Main Show	Lobby target 32 (config 8–40); up to 5 rounds, typically 3 qualifiers + a Final; median show 12–16 min	§1.4
MVP levels	Eight original levels (FR-02-11): Knock-Knock Run, Bumpy Parade, Slime Rise, Beam Hop, Wall Dodge, Tail Chase, Tile Drop, Summit Grab, built from an 11-piece obstacle kit	Section 2
Movement	Physics-driven bean on one shared 50 Hz simulation step; run, jump, dive, grab; kid-friendly assists	Section 3
Progression	XP, a free 70-tier Season Path over an 8-week season, Coins, cosmetic slots; no chance-based rewards	Section 7
Systems	Dedicated, server-authoritative game servers with client-predicted own movement; Unity 6 LTS recommended for the M0 engine decision (pending Q-A1); MVP regions EU-West, EU-Central, NA-East, NA-West; planning load 10,000 peak players, load-tested to 10×	Section 8
Child safety	No free text or voice chat; preset quick-chat and emotes; friends only through codes with parent approval; block, mute and report in ≤ 2 interactions; verifiable parental consent for profile features	Section 9
Builder and live ops	Builder (level editor) is Later; MVP live ops cover the season container, News Board and level rotation	Section 10

What the merge did

- **Ten sections in source order**, with section and issue references rewritten so they point inside this document (see *How to read this document*).
- **Duplicate content removed.** Rules that two documents stated twice are now stated once, in the section with authority, and referenced from the other (Appendix B.2). One duplicate diagram was removed.
- **26 reconciliations.** Where documents conflicted, the order of authority on the Document control page was applied; each case is logged in Appendix B.1.
- **One Design Issues register.** The 126 issues listed across the ten documents contained 10 duplicates, leaving **116 unique issues** (Appendix A.1). The merge raised 6 more issues, listed separately (Appendix A.2).
- **23 diagrams** rendered from the Mermaid source and embedded; four diagrams needed syntax fixes (Appendix B.3).
- **One Open Questions & Sign-offs list** of 15 items for the owner, counsel and analyst (Appendix C).

Decisions needed first

The sources mark these as needed before or at **M0** (prototype and movement lock): the meaning of "MiniMax" and the engine (Q-A1), the PC storefront (Q-C1), the peak-player forecast and server budget (Q-C3) and the

parental-consent vendor. Counsel sign-off (DEP-10) is needed before closed beta (M3), and the trademark check (Q-D1, DEP-12) before launch (M4). See Appendix C.

Art. Every image in this document and every art slot in the design is a placeholder. The owner will supply the art from their own GitHub repository later (UD-3, Q-D2, Section 5).

How to read this document

- **Section references.** "§4.2.5" means section 4.2.5 of this document. Each source document became one section, and its own numbering was kept under the section number, so source document 04 §2.5 is §4.2.5 here. Numbers are missing where a source's issue, traceability or open-question section moved to the appendices.
- **References to the requirements.** "B §9", "gapfill-A §5.3", "Ch. 15" and similar refer to the core requirements v3 and its gapfill annexes A–D, not to this document. "src 04 §1.2" refers to section 1.2 of source document 04 (here §4.1.2).
- **Requirement IDs** (FR-xx-yy, NFR-xx-yy, with ACn for an acceptance criterion), analyst decisions (AD-n), user decisions (UD-n), dependencies (DEP-n), user questions (Q-xn) and gapfill defaults (D-n) are unchanged from the requirements v3. Appendix D.2 lists every requirement cited.
- **Issue IDs** from the source documents are prefixed with the source document number, so "I-5" in document 02 is **02-I-5** here. Each source issue maps to one register entry **DI-nnn** (Appendix A.3).
- **REC-nn** is a reconciliation between documents (Appendix B.1). **MRG-nn** is an issue raised by the merge (Appendix A.2). A ⚑ marks a rule that depends on an issue or reconciliation.
- **Design estimates.** Numbers marked as design estimates, defaults or tunables are starting values for playtest, not confirmed facts.
- **Placeholders.** "(img)" slots, the cover image and all named colours, icons and art are placeholders until the owner's asset repository is available.

1 Core Game Design

Source: `01-core-gdd.md` (1.0 — 23 Oct 2026; design-ready for MVP pre-production). **Authority:** Game rules (highest authority for rules of play).

Scope defaults (from the GDD header): engine-neutral (NFR-10-02, Q-A1; see REC-26) · PC (Windows) first (NFR-10-01, AD-51) · free, no real money in MVP (AD-12, Q-B1 default) · child-directed 6–12, PEGI 3 / ESRB E (UD-1, AD-01, AD-24) · no open chat (FR-11-03, AD-02) · placeholder art only (UD-3, FR-12-06, FR-12-10).

All numbers are **design estimates** inherited from the requirements unless marked *new (GDD)*; every new value is a tunable default for playtest. Every design rule ends with a bracketed trace, e.g. `[FR-01-05 AC3; AD-10]`. Where the GDD resolves a conflict in the requirements, the rule is marked ⚑ **Resolution** and listed in Appendix A. Reference-game (Fall Guys) facts are research only and never appear in product names, art or layouts (UD-2, AD-25, AD-38).

1.1 Vision and design pillars

Vision statement. *Wobblebeans is a bright, giggly, 15-minute show where 32 wobbly beans tumble through obstacle rounds, and every child walks away with a laugh, a reward and a reason to try again — never a reason to feel bad.* [UD-1, UD-4, US-01-1]

1.1.1 Pillars

#	Pillar	What it means	What it rules out	Trace
P1	**Falling is funny**	Failure is slapstick, readable and fast to recover from. Beans always get up; there is no injury, damage state or scary content.	Long stun-locks, punishing respawns, violent framing.	FR-03-03 AC4, FR-04-05, AD-07, AD-24, FR-11-18
P2	**Everyone goes home with something**	Every round played pays XP and Coins; the Final always crowns at least one winner; copy is always kind.	Win-only rewards, zero-winner shows, "you lost" copy, quit punishments that feel like penalties.	FR-01-09 AC1, FR-01-05, AD-10, FR-08-03, AD-37, FR-06-13 AC2
P3	**Read it in one look**	Rules, hazards and rewards are understood from pictures, icons and motion, not reading. A 6-year-old can play without an adult.	Text-heavy rules, colour-only signals, hidden mechanics that punish first contact.	FR-08-19, FR-03-17, FR-08-11, FR-01-01 AC4, FR-02-05 AC2
P4	**Fair chaos**	Randomness and crowding create comedy, but outcomes are decided by the server, timers are visible and rules are deterministic.	Hidden rubber-banding, pay-to-win, off-by-one disputes, client-side authority.	FR-02-02 AC4, FR-01-04, AD-28, FR-05-01, FR-07-12, NFR-10-13
P5	**Safe by design, calm by design**	Play together without strangers talking; no pressure loops (no countdown offers, streaks, auto re-queue, loss framing).	Open chat, streak rewards, FOMO, "one more game" traps.	FR-11-03, FR-07-14, FR-08-18, AD-15, FR-11-15, FR-11-19

Pillar priority when they collide: P5 > P2 > P3 > P1 > P4. Example: if a fairer tie-break would yield a zero-winner Final, P2 wins (→ `final_draw_policy = all_win`, AD-10).

1.1.2 Design tests (apply to every feature)

1. *Would a 6-year-old understand it muted, without reading?* (P3)
2. *After the worst possible outcome, what does the child get?* — must be non-empty (P2)
3. *Does it create pressure to keep playing or to spend?* — must be "no" (P5, FR-08-18)
4. *Is the outcome decided by a rule a child could explain afterwards?* (P4)

1.2 Player fantasy and audience

1.2.1 Player fantasy

"I'm a contestant on the silliest TV game show ever." The player is a squishy, wobbly bean (silhouette and slot design only — final look comes from the owner's asset repository, FR-12-04, FR-12-10) who runs, jumps, dives

and grabs through a giant colourful set. The show is the star: bright stages, big obstacles, a cheering crowd mood carried by music and visuals (no voiced presenter, AD-53).

Fantasy beat	Delivered by	Trace
"I'm on stage"	Round intro fly-through + picture objective card	FR-08-04, gapfill-B §1.1, SC-04
"I'm wobbly and it's hilarious"	Ragdoll stumbles, bumpers that bounce, auto get-up	FR-04-05, FR-03-03, AD-07
"I made it!"	Big "Qualified!" slate, checkpoint chimes	SC-06, FR-03-12 AC2
"I'm the champion"	Victory screen with equipped celebration, Trophy	FR-01-05 AC6, FR-06-07, AD-35
"This is *my* bean"	Split upper/lower costumes, colours, patterns, faceplates (slots, not specific characters)	FR-05-02, FR-05-03, FR-05-17

1.2.2 Audience segments (all players get child protections — AD-01)

Segment	Age	Needs	Key design responses
Little Beans	6–8	Pictures not words, short sessions, high success, adult nearby but not required	Picture-led tutorial ≥ 80% solo completion (FR-08-17 AC2), Year 2 reading level (FR-08-19), new-player lobbies (FR-01-02 AC5), R1 survive ≥ 55% for new cohort (gapfill-B §9 AT)
Big Beans	9–12	Mastery, showing off, playing with friends	Dive/grab skill ceiling (FR-04-03, FR-04-04), Trophy Rank (FR-06-07), parties of up to 4 approved friends (FR-07-03)
Parents	adult	Trust, control, no spending surprises	Parent Account, controls, playtime limits applied at show end (FR-11-02, FR-11-08, FR-11-15)

1.3 Core loop

1.3.1 Loop layers

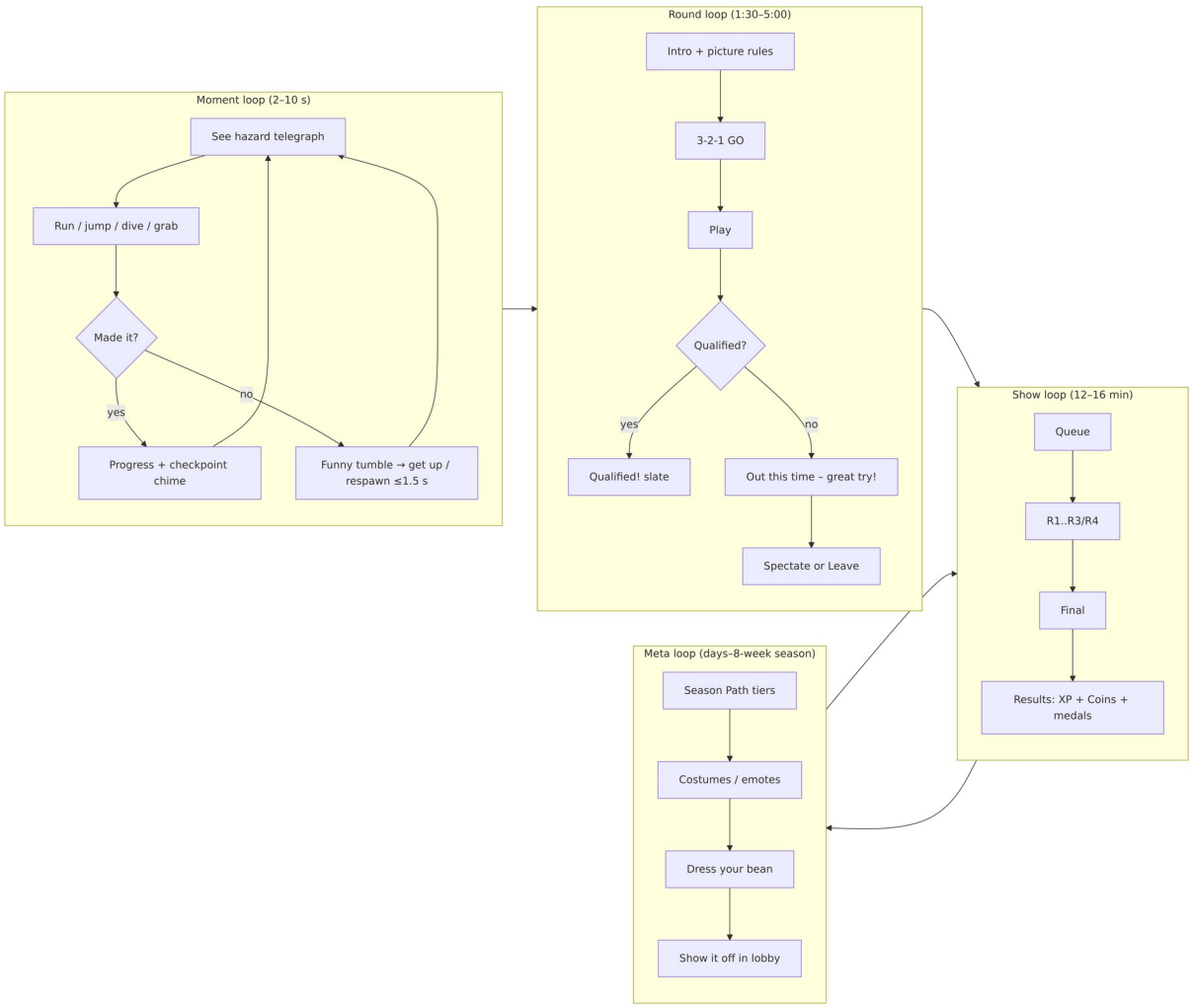


Figure 1.1 — Core loop layers: moment, round, show and meta loops

Layer	Duration	Goal	Reward	Trace
Moment	2–10 s	Pass one obstacle section	Progress, checkpoint cue; fall costs ≤ 12 s	FR-03-12, FR-03-13, gapfill-B §6.2
Round	1:30–5:00 + 22 s overhead	Qualify (finish quota / survive / score)	Qualified slate; +150 XP per qualified round	FR-01-04, FR-02-02..04, gapfill-B §1, gapfill-C §1
Show	12–16 min median	Reach the Final, win a Trophy	Participation XP, Final bonus, win bonus, Coins, medals	FR-01-03, FR-01-05, FR-01-09, gapfill-B §9
Session	~1–4 shows	Have fun, progress a bit	Season Path tier progress, challenges	FR-06-01, FR-06-13, FR-11-15
Season	8 weeks	Unlock free-track cosmetics	70-tier free Season Path	FR-06-02, AD-11, AD-14

1.3.2 Moment-to-moment verbs (MVP)

Move, Jump, Dive, Grab, Emote — and nothing else. Every MVP level must be completable with Move + Jump; Dive and Grab are *advantage* verbs, never *gate* verbs (*new (GDD)*, supports P3; physics per FR-04-01..04, AD-31). Grab can be toggle or hold (FR-08-12). Emotes cancel on any input and are not usable airborne/grabbing/ragdolled (AD-34).

1.4 Show structure

1.4.1 Main Show (MVP) at a glance

- Lobby: 32 target (config 8–40) [AD-05, FR-01-02, NFR-10-03]
- Rounds: up to `MAX_ROUNDS = 5`; typical 3 qualifiers + Final [gapfill-B §3.4, US-01-1]
- Final when `N ≤ FINAL_MAX (10)` or `round_idx = MAX_ROUNDS` [AD-29]
- No repeated level in a show; no same round type twice in a row unless forced [FR-01-03 AC2–AC3]
- Show length median 12–16 min; Final 2–4 min [gapfill-B §9]

1.4.2 Show state machine

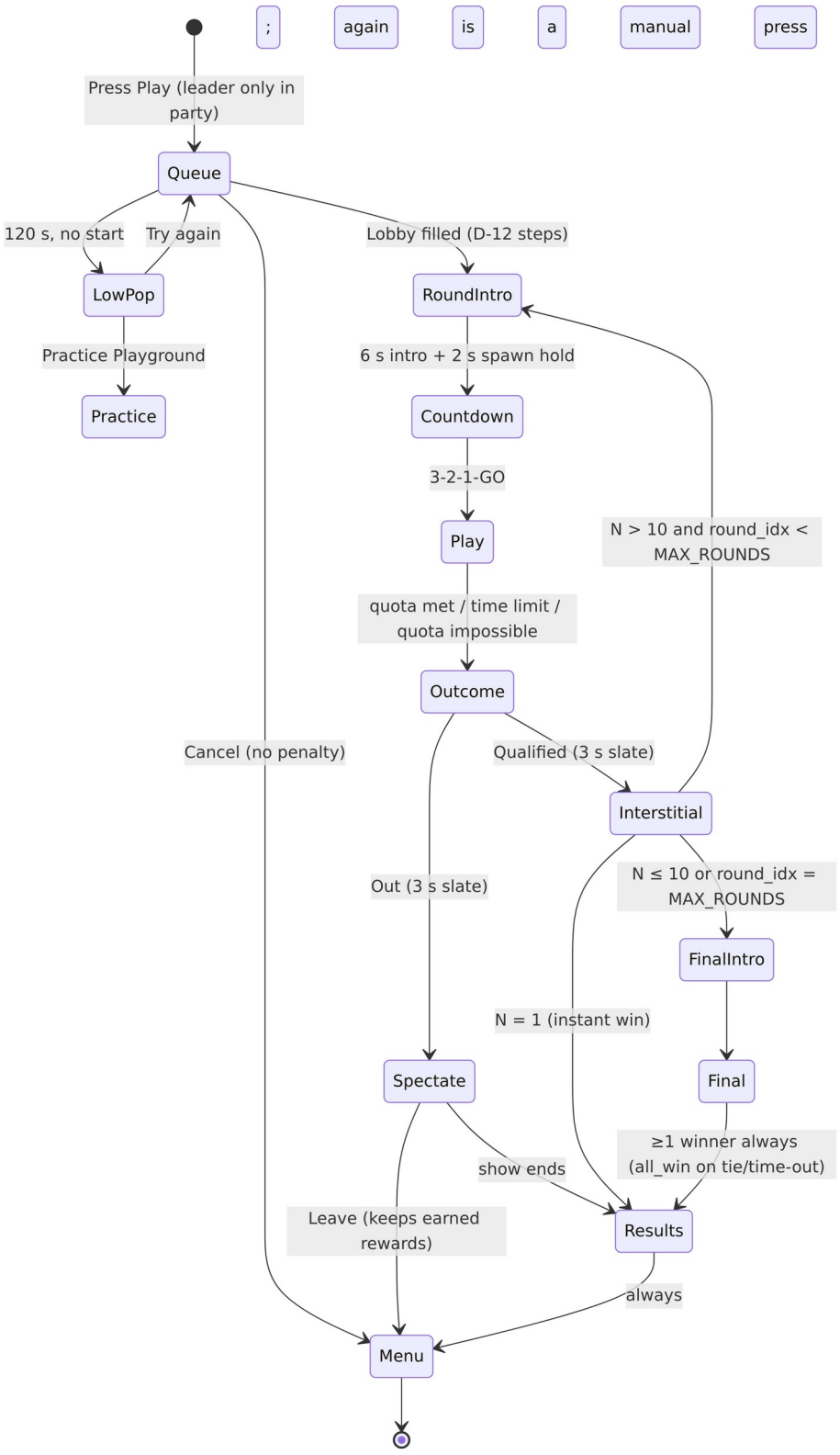


Figure 1.2 — Show state machine

Trace: FR-01-02 AC1/AC8, FR-01-03, FR-01-04 AC1, FR-01-05 AC2–AC5, FR-07-11, FR-08-04, FR-08-18 (no auto re-queue), FR-07-03 (leader queues), AD-29.

1.4.3 Round phase timeline (every round)

Phase	Default	Purpose	Trace
Intro fly-through + picture "How to play" card	6.0 s	Readability, loading cover	gapfill-B §1.1, SC-04
Spawn hold (input locked)	2.0 s	Net settle	gapfill-B §1.1
3-2-1-GO	3.0 s	Fair start, 0 early-movement events	gapfill-B §1.1 AT
Play	per type (§1.4.5)	—	gapfill-B §1.2
Final-seconds banner	last 15 s (solo)	Visible, not flashing > 3 Hz	gapfill-B §1.1, FR-08-02, NFR-10-19
Qualified / Out slate	3.0 s	Emotion beat	SC-06
Between-round results	8.0 s	Read next quota; loading	gapfill-B §1.1

Fixed overhead ≈ 22 s per round (target median ≤ 25 s).

1.4.4 Slot planning and round selection

Slot	When	MVP type weights (renormalised, ⚑ 01-ISS-05)	R2+ type weights (requirement)
R1	round 1, N > 10	Course 80 · Survival 10 · Points 10	same
R2	round 2, N > 10	Course 39 · Survival 33 · Points 28	Course 35 · Survival 30 · Points 25 · Logic 10
R3/R4	round 3–4, N > 10	Survival 44 · Points 34 · Course 22	Survival 40 · Points 30 · Course 20 · Logic 10
F	N ≤ 10 or round 5	Final pool, filtered by maxP	same
—	N = 1	instant win	same

Selection: valid candidates by slot, min/max players and team divisor, minus already-played levels, minus last round's type unless that empties the set; weighted by `type_weight × level.weight × freshness` (× 0.5 if > 50% of the lobby played it in their last 3 shows); look-ahead guarantees a valid Final; fallback level if empty. [FR-01-03 AC1–AC7, FR-02-01, gapfill-B §3.3, AD-29]

Worked example (32 players, MVP defaults): R1 Course 70% → 22 · R2 Survival 60% → 13 · R3 Points 60% → 8 (≤ 10) → **R4 = Final**. Four rounds, ~14 min. [gapfill-B §2.1, §1.2.2; AD-28 round-half-down applies only to exact .5 ties]

1.4.5 Round types and how qualification works

Type	Release	Qualify rule	Default limit	Fall rule	Trace
Course (race)	MVP	First `quota` finishers	3:00 (2-lap 3:30, rising slime 2:20)	Respawn at checkpoint (slime variants eliminate)	FR-02-02, FR-03-13, FR-03-14
Survival	MVP	Survive timer, or round ends when alive ≤ quota	1:45–2:00	Eliminate	FR-02-03
Points (hunt)	MVP	Top `quota` by score at time	2:00	Respawn	FR-02-04
Final — survival / trophy-grab	MVP	Last standing / first grab; ties & time-out → all remaining win	5:00 (tunable 3–6)	Eliminate	FR-01-05, AD-10
Team	R2	Bottom `max(1, floor(teams/3))` teams out	2:00	per level	FR-02-06, FR-01-06, AD-28
Logic	R2 (Could)	Survive picture prompts; answers ≤ 20	1:50	Eliminate	FR-02-05
Peekaboo Teams	Won't	—	—	—	FR-02-08, AD-48

Quota: `quota = clamp(round_half_down(N × pct), 1, N − 1)`; disconnects count as eliminated and do not shrink the quota; if remaining-able ≤ remaining spots, all qualify immediately. [FR-01-04, gapfill-B §2.1, AD-28]

1.4.6 Finals — "there is always a champion"

- Trophy-grab: first server-timestamped grab wins (FR-01-05 AC1).
- Survival: last standing wins; exact sub-tick tie → all tied win (AC2, AD-10).
- Time-out: every player still in wins (AC3, AD-10).

- Bots never enter a Final; bots rank below humans before the Final (FR-01-02 AC7, AD-17). If only bots remain, the show closes with no winner recorded (FR-01 edge case 9) — players who left already have their rewards, so no child sees a zero-winner screen.
- Winners see SC-08 victory with equipped celebration for 8 s (FR-01-05 AC6, AD-35).

1.5 Modes — MVP vs R2 vs Later

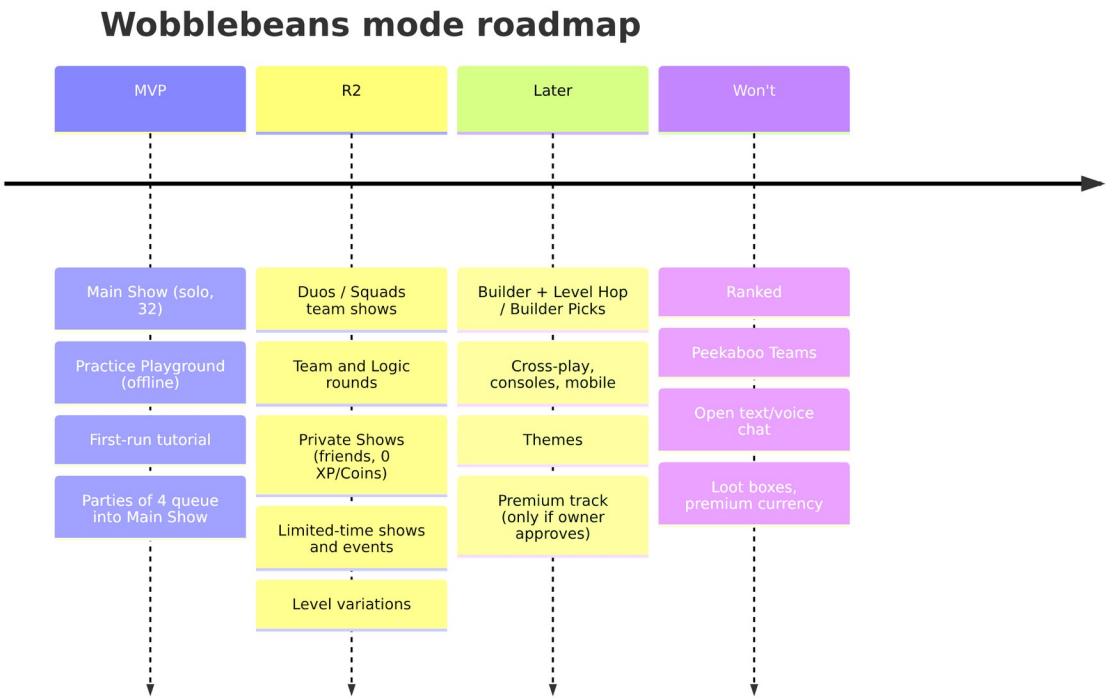


Figure 1.3 — Mode roadmap (MVP, R2, Later, Won't)

Mode	Release	Players	Rewards	Design purpose	Trace
Main Show (solo)	MVP · Must	8–32 (target 32)	Full XP/Coins/Trophy	The game	FR-01-01, FR-01-02..05, FR-01-09, AD-05
Practice Playground	MVP · Should	1 + 0–7 practice bots, offline	None (anti-farm)	Learn safely, low-pop fallback, accessibility	FR-01-12, FR-08-12, AD-18
First-run tutorial	MVP · Must	1, offline	Small cosmetic on first completion *new (GDD), tunable; never XP*	Teach verbs	FR-08-17, AD-44
Party queue into Main Show	MVP · Should	up to 4 approved friends	Normal	Play with friends	FR-07-03, AD-52
Duos / Squads	R2 · Should	32 in teams of 2/4	Team win counts for all; best team medal	Cooperative belonging	FR-01-06, FR-02-06, US-01-2
Private Shows	R2 · Should	2–32 approved friends	0 XP / 0 Coins	Birthday-party play	FR-07-08, AD-19
Limited-time shows	R2 · Should	per playlist	Event rewards earnable in ≤ 4 h; no countdown UI	Freshness without FOMO	FR-09-23, FR-01-01 AC3, AD-15
Level Hop / Builder Picks	Later · Could	≥ 20	Capped XP	UGC	FR-01-08, FR-09-01..18
Ranked	Won't	—	—	Competitive pressure unsuitable	FR-01-07, AD-16

MVP design rule: the Show Picker exists but shows one card (Main Show) — the hub's big Play button goes straight to it (FR-01-01, SC-01/SC-02). Grouping appears only when > 4 shows exist.

1.6 MVP level set and round mix

⚑ **Merge reconciliation (REC-01).** The level list below is the FR-02-11 set specified in Section 2; the original GDD list is recorded in Appendix B. The GDD rules on slots, fall rules and R1 content still apply.

8 original levels built from the 11 MVP obstacles (Swing Ball, Spin Bar, Spin Mallet, Tilt Plank, Conveyor, Bumper, Pusher Block, Launch Arm, Trick Door, Drop Tile, Slime). Names are placeholders; layouts are original and pass legal review. [FR-02-11, FR-03-01, AD-38, AD-40, AD-25] Level specifications: Section 2; registry: §2.10.

#	Placeholder	Type	Slots	Players (registry)	Core obstacles	Fall rule
L1	Knock-Knock Run	Course	R1, R2	8–40	Trick Door, Bumper, Pusher Block	Respawn at checkpoint
L2	Bumpy Parade	Course	R1 (also Level Hop fallback)	8–40	Swing Ball, Tilt Plank, Bumper, Conveyor	Respawn at checkpoint
L3	Slime Rise	Course (rising slime)	R2, R3	6–32	Conveyor, Swing Ball, Spin Mallet, Launch Arm, Slime	Respawn only if the checkpoint is above the slime; slime contact = out
L4	Beam Hop	Survival	R2, R3	6–32	Spin Bar, Slime	Eliminate
L5	Wall Dodge	Survival	R2, R3	6–32	Pusher Block (wall variant)	Eliminate
L6	Tail Chase	Points (hold a tail)	R2, R3	6–32	Bumper (grab is a player verb)	Railed arena, no falls
L7	Tile Drop (fallback)	Final — survival	F (+ fallback)	2–20	Drop Tile, Slime	Eliminate
L8	Summit Grab	Final — trophy grab	F	2–10	Swing Ball, Spin Bar, Launch Arm	Respawn at checkpoint (no slime)

Coverage check (FR-02-11 AC3: ≥ 2 candidates per non-final slot, 2 finals, N = 8–32): R1 → L1, L2; R2 → L1, L3–L6; R3 (and R4, which shares the R3 slot, 01-ISS-04) → L3–L6; F → L7, L8 (Summit Grab maxP is 10, 02-I-2). ✔️ R1 has Course candidates only, so the R1 type weights renormalise to 100% Course (01-ISS-05 / 02-I-1). The GDD content rule of 01-ISS-11 (no grab-steal Points level in R1) holds because Tail Chase is tagged R2/R3 only.

1.7 Session flow

1.7.1 Screen flow

The screen-level navigation is specified once, in the navigation map of §4.1.2 (Figure 4.1). The GDD session-flow diagram duplicated it and was removed in the merge (Appendix B.2); its session rules are kept below.

Session-flow trace: FR-08-04, FR-08-17, FR-01-12, FR-07-11, FR-01-09, FR-11-15 AC1–AC2, FR-08-16 (offline states, SC-14).

1.7.2 Session pacing targets

Moment	Target	Trace
Launch → first input in tutorial	≤ 60 s *(new GDD target)*	NFR-10-18
Press Play → round start (P50 / P95)	≤ 20 s / ≤ 60 s queue + ~11 s intro	D-12 AT
Show length	12–16 min median	gapfill-B §9
Typical session	2–4 shows (~30–60 min); break reminder at 60 min	FR-11-15, AD-23
Limits never interrupt a round	100%	FR-11-15 AC2

1.7.3 Ending a session well

- After SC-08 the child returns to the hub; **no auto re-queue**, no "one more" prompt, no purchase prompt (FR-08-04 AC2, FR-08-18, FR-01-09 AC3).
- Leaving mid-show is always allowed and never punished; earned per-round rewards are kept (AD-37, FR-01-04 AC2; ⚑ 01-ISS-01).

- Unfinished daily challenges bank for up to 7 days; rested XP welcomes the child back — absence never costs anything (FR-06-13, gapfill-C §1, AD-15).

1.8 Onboarding and tutorial for young kids

1.8.1 Principles

1. **Show, don't tell** — no voice-over, single words only; every instruction is an animated pictogram of the input + the bean doing it (FR-08-17, FR-08-19, AD-53).
2. **Do it, then name it** — the child performs the verb before any icon label appears.
3. **Can't fail the tutorial** — no timers, no elimination; falling respawns in place within 1 s.
4. **≤ 5 minutes median, offline** (FR-08-17 AC1, AC3; AD-44).
5. **Skippable without shame** — "Play now" is always available on first run; the tutorial is offered again next launch (FR-08-17). ⚑ 01-ISS-07.

1.8.2 Tutorial path (runs inside the Practice Playground space — ⚑ 01-ISS-02)

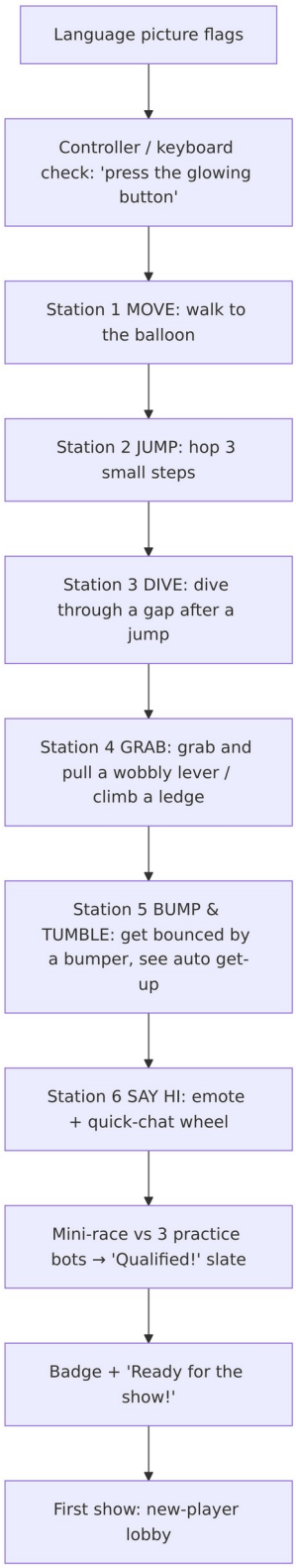


Figure 1.4 — Tutorial path through the Practice Playground

Station	Teaches	Success condition	Assist after 3 tries / 20 s	Trace
Move	stick/WASD	reach marker	glowing footprints path	FR-04-01
Jump	jump	3 hops	hops get lower	FR-04-02, AD-31
Dive	jump + dive	clear 4 m gap	gap shrinks to 3 m	FR-04-03
Grab	hold/toggle grab	pull lever 1 s	grab auto-toggles on; hint to Settings	FR-04-04, FR-08-12

Station	Teaches	Success condition	Assist after 3 tries / 20 s	Trace
Bump & tumble	ragdoll is OK	get bounced once	— (automatic)	FR-03-03 AC2, FR-04-05, P1
Say hi	emote, quick-chat	send one	—	FR-05-06, FR-07-14
Mini-race	full loop	cross finish	bots slow down (practice only)	FR-01-12 AC3

1.8.3 First five shows (new-player protection)

- Matched with other new players + disclosed bots (FR-01-02 AC5, AD-17); 20 s fallback to normal queue.
- R1 for this cohort targets ≥ 55% survival (gapfill-B §9 AT). R1 type weight Course 80 favours the most-learnable type.
- Round intro card always shows the type icon + quota pictogram ("first 22 beans through!").
- After show 1 the hub highlights the Dressing Room once (non-blocking, not a pop-up during play) so the child learns rewards are cosmetic (FR-05-01, FR-08-18).

1.9 Fail-forward and "never feel like a loser" design

1.9.1 The emotional curve we design for

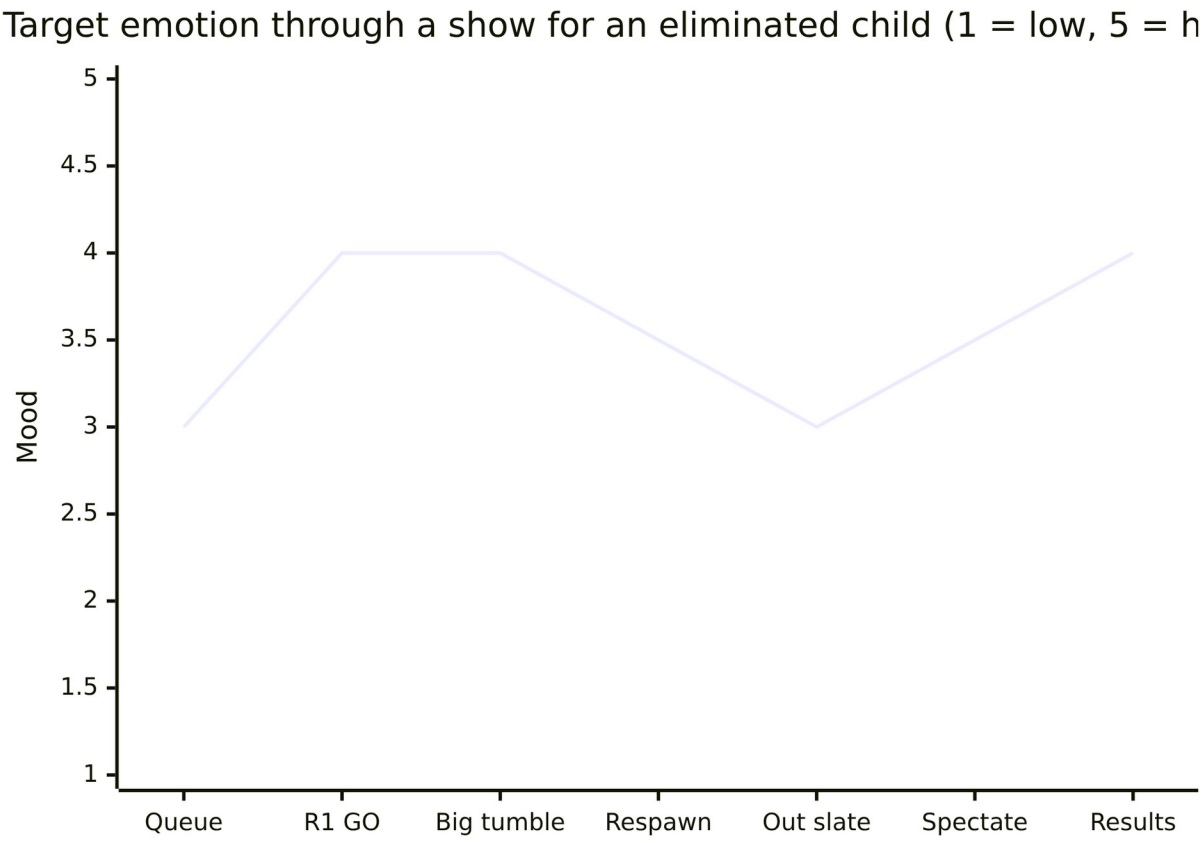


Figure 1.5 — Target emotion through a show for an eliminated child (1 = low, 5 = high)

Rule: no step in a show may drop below "3 — okay". Elimination is a pause, not a punishment.

1.9.2 Fail-forward toolkit

Lever	Rule	Trace
Fast recovery	Knockdown 1.2–3.0 s, 0.5 s re-knockdown immunity; ≤ 2 knockdowns in 4 s	FR-04-05, gapfill-A §5.3
Cheap falls	Respawn 1.5 s + 1 s protection; ≤ 12 s progress lost; checkpoints never move you back	FR-03-12, FR-03-13

Lever	Rule	Trace
Gentle bumps	Bean-to-bean bumps never knock down (Δv ≤ 5); bumpers never knock down	FR-04-05 AC1, AD-07
Rewards for trying	+100 XP participation, +150 per qualified round, Coins per round; itemised ("Qualified round 2: +N Coins", rates per §7.3; REC-25)	FR-01-09 AC1, FR-06-06, gapfill-C §1
Medals for progress	Results show a medal per round reached, not a rank number out of 32 *(GDD: placement number hidden for eliminated players; shown for finalists only)*	FR-01-09, FR-08-03
Kind copy	No "lose/lost/loser/fail/dead/die/kill"; "Out this time – great try!"	FR-08-03
Something to do	Spectate with party members first, quick-chat cheers ("Nice!", "Wow!")	FR-07-11, FR-07-14
Champions always exist	all_win ties and time-outs	FR-01-05, AD-10
Non-win goals	0 challenges require a win; 0 reward grabbing/knocking others	FR-06-13 AC2
No leaver penalty	Leave any time, keep what you earned, return to party after the show	AD-37, FR-07-11 AC3
Personal bests *(new GDD, R2 Should)*	"Furthest round" sticker on SC-08 when beaten — local only, no streaks	FR-08-18 compliant

1.9.3 Anti-patterns banned in core design

Leaderboards of strangers · placement shaming ("32nd") · streaks · "you missed out" · countdown offers · pop-ups during play · win-gated cosmetics on the free path · elimination by another player's grab in R1 (*GDD content rule: R1 levels must not have grab-to-eliminate mechanics*). [FR-08-18, FR-06-02 AC2, AD-15]

1.10 Difficulty and fairness

1.10.1 Difficulty curve across a show

Slot	Target eliminated	Hazard density	speed_mult	Fail risk / obstacle	Variants
R1	~30%	low–medium, wide lanes	1.0	5–10%	none
R2	~40%	medium, mixed	1.0–1.15	10–15%	"Faster" 20% (R2 release)
R3/R4	~40%	medium–high	1.1–1.25	15–20%	"Faster" 20% (R2 release)
Final	→ winner(s)	high, escalates in-round (smoothstep)	escalating	—	—

Trace: gapfill-B §9, B §5; FR-02-03 AC2; FR-03-04; FR-03-15 (variants R2).

1.10.2 In-level difficulty rules (level design contract)

1. **Telegraph everything:** motion visible before contact, ≥ 3:1 outline contrast, ≥ 80% of 6–10s can explain a hazard after one encounter (FR-03-17).
2. **Never by colour alone:** tiles, teams, hazards use pattern/icon too (FR-08-11).
3. **Multiple routes per section:** an easy-but-slow lane and a fast-but-risky lane in every Course section *(GDD rule, supports P3/P4)*.
4. **Reach limits:** gaps ≤ 3 m (jump) / ≤ 5 m (jump+dive) (AD-31).
5. **Hidden-info is fair-on-contact:** trick doors/fake tiles revealed only on server contact, randomised per match (FR-02-02 AC4).
6. **No checkpoint loops:** never place a checkpoint where a launcher failure returns you > 3 times in a row (gapfill-B §6.2).
7. **Escalation, not ambush:** survival hazards ramp so < 25% of eliminations occur in the first third (gapfill-B §5 AT).

1.10.3 Fairness model

Fairness promise	Mechanism	Trace
Same rules for everyone	Cosmetics have no gameplay effect	FR-05-01

Fairness promise	Mechanism	Trace
Server decides	Authoritative 50 Hz sim, sub-tick tie-breaks, anti-cheat speed check	FR-07-12, AD-06, AD-09, gapfill-B §2.3
Deterministic counts	Round-half-down quota, visible quota on intro card and HUD	AD-28, FR-08-01
Fair start	Input locked until GO	gapfill-B §1.1
Leavers don't distort	Quota not recomputed on disconnects; 60 s reconnect, 2 per show	FR-01-04 AC3, FR-07-16
Honest bots	Disclosed, ≤ 40%, never in Final, rank below humans	FR-01-02 AC7, AD-17
Matching without stress	Loose 3-band hidden MMR only with big pools; no Ranked	D-13, AD-16
Variety is fair	No repeats, freshness weighting	FR-01-03
Assist without secret rubber-banding	Assists are *visible and opt-in* (Practice, grab toggle, hold-to-mash) — no hidden catch-up in competitive rounds	FR-08-12, P4

1.10.4 Accessibility as difficulty control

Grab toggle, hold-to-mash, text 100/125/150%, visual audio cues, motion comfort (blur off, shake toggle, FOV 60–85°), photosensitivity limit, full rebinding. Option defaults: §3.5; presentation: §4.6. [FR-08-12, FR-08-13, FR-08-20, NFR-10-19, FR-04-12]

1.11 Social play inside the core loop (child-safe)

Social rules are specified once in Section 9 (child-safety authority): parties of up to 4 **parent-approved** friends, leader-only queue, never split (§9.5); preset positive phrases and emotes only, ≤ 2 inputs to send, mute in ≤ 2 interactions, parent can disable (§9.6); generated display names (§9.2); report and block in ≤ 2 interactions, picture-led (§9.7). [FR-07-03, FR-07-07, FR-07-14, FR-07-15, FR-11-03, FR-11-04, FR-11-06, AD-03, AD-04, AD-43, AD-52]

- Team shows (R2): one team win counts for all; non-qualifying member gets the team's best medal — the purest expression of P2 (FR-01-06 AC3).

1.12 Tuning, telemetry and playtest gates

Gate	Metric	Pass	Trace
G1 Tutorial	6–8s complete unaided; median time	≥ 80%; ≤ 5 min	FR-08-17
G2 Readability	hazard explained after 1 encounter	≥ 80%	FR-03-17 AC2
G3 Falls feel funny	survey "funny" / "unfair"	≥ 70% / ≤ 20%	gapfill-A §5.2 AT
G4 Fall cost	median progress lost per fall	≤ 12 s	FR-03-12 AC3
G5 Show length	median show	12–16 min	gapfill-B §9
G6 New-player R1	survival of < 5 h cohort	≥ 55%	gapfill-B §9
G7 Downtime	median non-play overhead/round	≤ 25 s	gapfill-B §1 AT
G8 Sequencing	Monte-Carlo 10k shows	no repeats; fallback < 2%	FR-01-03 AC7
G9 Kind copy	banned-word & urgency lint	0 hits	FR-08-03, FR-08-18
G10 Emotion	eliminated children rating "I want to play again" (no prompt shown) *new GDD*	≥ 75%	P2

All values live in data config (NFR-10-20). Telemetry is pseudonymous and minimal (NFR-10-12, AD-21).

2 Rounds and Level Design

Source: `02-rounds-levels.md` (Draft v1.0 (23 Oct 2026)). **Authority:** Level content and level-building rules; yields to Section 1 on game rules and to Section 3 on physics.

Scope: the 8 MVP levels (FR-02-11, AD-40), the MVP obstacle kit (FR-03-01), and level-building guidelines.

All numbers are **v3 tuning-baseline design estimates** (Chapter 13, gapfill-A/B). None of them is a confirmed Fall Guys fact. They stay data, not code (NFR-10-20), and they lock only after movement gate M0 (DEP-02). Only Wobblebeans placeholder names are used (AD-25). No layout copies a reference map (AD-38, FR-02-11 AC2). Every blockout below is an original design.

2.0 Shared rules for every round

2.0.1 Show flow and slots (FR-01-03, AD-29, B §3, §2.9)

Lobby of 32 (AD-05). The quota formula and rounding rule are defined once in §1.4.5 (FR-01-04, AD-28).

Round	Slot	N in	Pct	Quota out	MVP candidates (type)	Target elim (B §9)	speed_mult
1	R1	32	70%	22	Knock-Knock Run (Course), Bumpy Parade (Course)	30%	1.0
2	R2	22	60%	13	Knock-Knock Run, Slime Rise, Beam Hop, Wall Dodge, Tail Chase	40%	1.0–1.15
3	R3	13	60%	8	Slime Rise, Beam Hop, Wall Dodge, Tail Chase	40%	1.1–1.25
4	F (N ≤ 10)	8	—	≥ 1 winner	Tile Drop, Summit Grab	1 winner	escalates (B §5)

At 32 players this gives 4 rounds and about 12–16 min (US-01-1, B §9). The per-round phase timeline (22 s of fixed overhead) is defined once in §1.4.3.

2.0.2 Common fall and respawn rules (FR-03-12, FR-03-13, B §6)

- **Course:** a fall respawns you at your furthest checkpoint after 1.5 s, with 1.0 s of protection. Respawn points are spread at least 1.2 m apart.
- **Survival, Final and slime contact:** you are out. Slate copy: "Out this time – great try!" (FR-08-03).
- **Kill plane** at level_min − 10 m. **Soft out-of-bounds:** a warning at 3 s and the fall rule at 5 s.
- **Chain-hit guard:** 0.5 s of knockdown protection after getting up, and a 3.0 s ragdoll cap (A §5.3).

2.0.3 Kid-readability rules used on every page (FR-03-17, FR-08-19, FR-08-20, NFR-10-19)

- Rules fit on one picture card: 1 icon goal plus at most 8 words.
- Every hazard telegraphs before it can hit: a wind-up, a glow, or a visible arc.
- Hazard outlines and walkable edges have a contrast of at least 3:1.
- Colour is never the only cue (FR-08-11). The player always gets shape plus motion plus sound.
- Nothing flashes more than 3 times per second. Camera shake respects the comfort setting.
- No weapon imagery: mallets are foam and the ball is soft (AD-24).

2.1 Knock-Knock Run — COURSE (R1, R2)

Trace: FR-02-02, FR-03-06, FR-03-12, FR-02-11 · B §1.3 (3:00) · Obstacles: Trick Door, Bumper, Pusher Block

Goal (card): 🚪 "Bump doors. Find the open ones!"

Blockout (about 230 m, original 5-row layout; widths in metres):

```
START (32 spawns, 24 m wide pad)
   │ ramp 20 m
[ROW A] 8 doors / 5 break  ── CP1
   │ Bumper meadow 30 m (6 moving bumpers, 1.5 m/s)
[ROW B] 7 doors / 3 break  ── CP2
   │ Pusher lane 35 m (3 lanes × 4 Pusher Blocks, staggered phase)
[ROW C] 6 doors / 3 break  ── CP3
   │ Bridge 25 m (side rails, 2 static bumpers)
[ROW D] 5 doors / 2 break  ── CP4
   │ Uphill 30 m
[ROW E] 4 doors / 1 break  ── CP5
   │ sprint 20 m
FINISH (arch, 18 m wide)
```

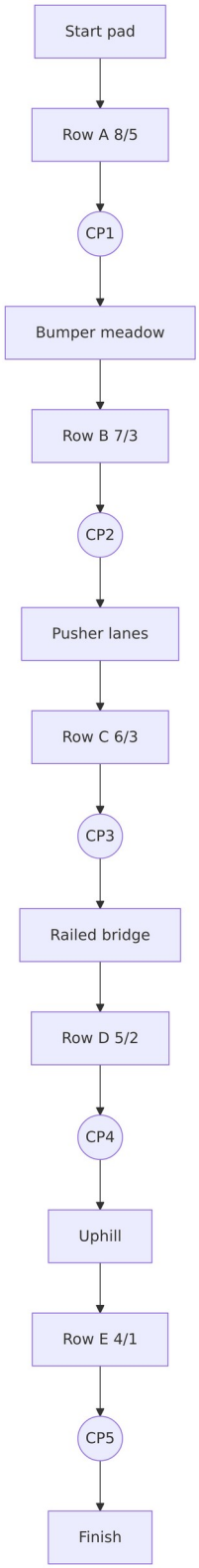


Figure 2.1 — Knock-Knock Run: door rows and checkpoints

Obstacle placement and timings

Element	Count	Params (source)
Trick Door	30 in 5 rows	Real/fake randomised per match on the server; identical meshes; breaks on a hit at ≥ 3 m/s; despawns after 0.3 s (FR-03-06, B §4.2)
Bumper (moving)	6	Reflect + 10 m/s, ragdoll:none (AD-07), travel 1.5 m/s
Pusher Block	12	Stroke 3.0 s (out 1.0, hold 0.5, in 1.0, hold 0.5), Δv 10 m/s, extend 4 m/s; phases staggered by 0.75 s per lane

Each door row is the checkpoint area: the checkpoint sits behind it, so passing a row is banked progress.

Qualification: first to finish. R1 70% (22 of 32), R2 60%. Limit 3:00 (B §1.3). The round ends at the quota, the timer, or when the quota becomes impossible (FR-01-04). **Difficulty variants** (FR-03-04 in MVP; named sets FR-03-15 in R2): *Easy R1* uses 2 extra real doors per row and slower pushers (×0.8). *R2* uses the base layout. *Faster* (R2 only, weight 20%) runs pushers at ×1.25. **Failure states and mitigations**

- Fake-door jam: the crowd piles at a solid door. Mitigation: at least 1 real door per row and a 2 m door spacing so the crowd spreads.
- Spawn fling: avoided with the 24 m pad and 1.2 m spawn spacing.
- Pusher shoves a player off the bridge: avoided by side rails on the bridge.
- A door left unbroken with no players nearby: the server despawns it at the quota or timer.

2.2 Bumpy Parade — COURSE (R1; also the Level Hop fallback)

Trace: FR-02-02, FR-03-05, FR-03-12, FR-01-03 AC4 · Obstacles: Swing Ball, Tilt Plank, Bumper, Conveyor

Goal: 🏁 "Run, hop, reach the finish!"

Blockout (about 250 m; A §1.1 skilled run 45–70 s):

```
START ─ 25m warm-up (2 static bumpers) ─ CP1
 ─ SWING ALLEY 40m: 4 Swing Balls across a 6 m path (period 4.0 s) ─ CP2
 ─ TILT GARDEN 40m: 3 rows × 3 Tilt Planks over a soft-respawn pit ─ CP3
 ─ CONVEYOR HILL 45m: uphill belt at 3.5 m/s with 2 side rails ─ CP4
 ─ BUMPER BALL-PIT 40m: 8 moving bumpers ─ CP5
 ─ FINAL DASH 30m (wide; 2 Swing Balls, offset phase) ─ FINISH
```

Timings: Swing Ball period 4.0 s, Δv 14 + 5 up, safe window ≥ 1.2 s, phase randomised. Tilt Plank max 25°; 3 beans tip it in about 1.0 s. Conveyor 3.5 m/s (the 3.6 m/s cap, AD-08), which still nets at least 5 m/s uphill (A §5.5). Bumpers as in §2.1. Five checkpoints (12 s rule, B §6.2). **Qualification:** first to finish, R1 70%. Limit 3:00 (B §1.2). In Level Hop (Later) everyone qualifies with a 30-minute limit (FR-01-08). **Kid-readability:** swing arcs are shown with a dotted floor shadow, and tilt planks have a pivot stripe. **Variants:** *Easy* runs Swing Balls at period 5.0 s. *Base*. *Swap* (R2+, FR-03-15) replaces Swing Alley with a Spin Mallet row. **Failure states:** a 32-bean pile-up on Tilt Garden holds the planks flat or tipped. Mitigation: 3 parallel planks per row, and the pit respawns at CP2 with a fall cost of 12 s or less.

2.3 Slime Rise — COURSE, rising slime (R2, R3)

Trace: FR-02-02 AC2, FR-03-14, FR-03-13, B §4.5 · Obstacles: Conveyor, Swing Ball, Spin Mallet, Launch Arm, Slime

Goal: ⬆️🟢 "Climb up! Stay above the slime!"

Blockout (original switchback climb, 60 m of height, about 260 m of path):

```
y=60  FINISH ═══  (end flood here at T−2 s)
y=50  ← SLIME-SIDE STRETCH: 3 Swing Balls + 2 Spin Mallets ←   CP4
y=38  → LAUNCH ARM DECK (1 arm, edge rails; launch fwd or back) → CP3
y=24  ← CONVEYOR SWITCHBACK (2 belts against travel, 3.5 m/s) ←  CP2
y=10  → MALLET RAMP (4 Spin Mallets, 90 °/s) →                   CP1
y=0   START ▓▓▓ slime starts after 3 s grace ▓▓▓
```

Timings: limit 2:20, 15 s countdown banner, end flood at T−2 s, 3 s grace. Rise rate = 60 / (140 − 3 − 2) ≈ **0.44 m/s** (B §4.5). Spin Mallet 90 °/s, Δv 12 + 4 up; the overclocked variant runs at ×1.4. Launch Arm 120 °/s, tip 20 m/s, with a 1,000-launch bounds test (FR-03-10 AC2). **Fall rule:** a fall that lands on a lower tier respawns at the checkpoint, but only if that checkpoint is still above the slime. If the slime has already covered your checkpoint, you are out. Slime contact always puts you out. **Qualification:** finishers qualify. The quota is 60% for R2–R3, so the round ends at the quota (13 of 22, or 8 of 13). Acceptance: when no quota is set, 50–70% of players finish, and a skilled run beats the slime by at least 25 s. **Readability:** the slime line has a bubbling edge, and a ground

arrow shows the climb direction. The countdown banner is an icon plus a number. **Variants:** *R2 base*, then *R3 Faster* (mallets ×1.4, weight 20%). *Easy* adds a 6 s grace (config). **Failure states:**

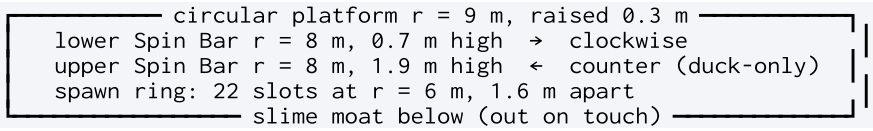
- Standing on a side wall at the end flood is allowed (`allow_wall_stand`, B §6.3).
- A Launch Arm that throws a player back 3 or more times in a row breaks the B §6.2 placement rule. Mitigation: CP3 sits after the arm, not before it.

2.4 Beam Hop — SURVIVAL (R2, R3)

Trace: FR-02-03, B §1.3 (1:30), §2.5 · Obstacles: Spin Bar, Slime

Goal: 🔄 "Jump the spinning bar!"

Blockout:



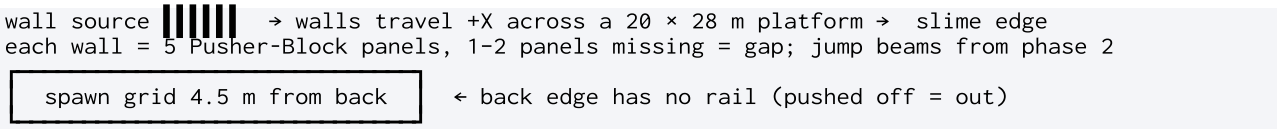
Timings: the lower bar starts at 40 °/s (rim ≈ 5.6 m/s) and escalates by smoothstep from 10 s to 85 s up to ×2.6 (104 °/s). The upper bar runs at 25 °/s, counter-rotating with a ±10% sine. Spin Bar Δv is 9 tangential: a hit while grounded is a stumble, a hit in mid-air is a knockdown (A §5.4). **Qualification:** quota 60% (13 of 22, or 8 of 13). The round ends when players still in ≤ quota, or at 1:30, when everyone still in qualifies (FR-02-03 AC1). **Acceptance:** fewer than 25% of eliminations happen in the first third, and every third has at least one; the quota is met at 70–100% of the limit in at least 60% of runs (B §5). **Readability:** the bar casts a bright sweep shadow, and a "whoosh" pitch rises with speed. The upper bar is clearly higher and coloured differently, and the "duck" icon only appears in R3. **Variants:** *R2 single bar* (upper bar off) and *R3 double bar*. **Failure states:** a player stands on the bar's pivot hub, which is a safe-spot exploit. Mitigation: the hub is a slippery dome that counts as soft out-of-bounds after 3 s.

2.5 Wall Dodge — SURVIVAL (R2, R3)

Trace: FR-02-03, B §1.3, §2.5 · Obstacle: Pusher Block (wall variant, see Issue 02-I-4)

Goal: 🧱 "Find the gap in the wall!"

Blockout:



Timings (B §5, total 1:45): Phase 1 runs 0–45 s with walls 6.0 → 5.0 s apart at 3.0 m/s. Phase 2 runs 45–87 s at 4.5 s spacing and adds 0.6 m jump beams. Phase 3 runs 87–105 s: side blocks close 30% of the platform, spacing tightens 4.5 → 3.0 s, and speed rises to 5.0 m/s. Push Δv is 10 m/s. **Qualification:** quota 60%. The round ends at the quota, or at 1:45 when everyone still in qualifies. **Readability:** each gap is lit 2 s before its wall moves, with a floor arrow. The walls are tall and solid, with a striped leading edge. **Variants:** *R2* runs phases 1–2 only, holding phase 2 to time. *R3* runs the full three phases. **Failure states:** a crowd blocks a gap and pushes players into the wall. This is intended chaos, but player bumps are capped at 5 m/s. Two gaps per wall from phase 2 make sure a gap always exists.

2.6 Tail Chase — POINTS (R2, R3)

Trace: FR-02-04 AC3, FR-04 (grab), FR-08-02 · Obstacles: Bumper (grab is a player verb, not an obstacle; see Issue 02-I-6)

Goal: 🎀 "Hold a tail when time's up!"

Blockout (original "picnic hill" arena, 40 × 40 m):

```
┌──────────────────────────────────────┐
│  ○ bumper   ▲ ramp    ○ bumper       │
│        ┌─ central mound ─┐           │
│  ○     │ 2 tunnels underneath │    ○  │
│        └──────────────────────┘       │
│  ▲ ramp     ○ bumper   ▲ ramp        │
└──────────────────────────────────────┘
```

edges railed (no fall-out)

Timings: limit 2:00. The final-seconds banner shows at 15 s. Six bumpers (reflect + 10 m/s, ragdoll:none) act as escape tools. There are no falls: the arena is fully railed, and the respawn is 1.0 s if someone cheeses out (B §6.1). **Qualification:** at the start, tails are given to `quota` random players (13 of 22, or 8 of 13). A grab steals a tail, confirmed by the server. At 0:00, the holder on the server's last tick qualifies (FR-02-04 AC3). The quota is exact, so no tie rule is needed. **Readability:** a tail-holder has a glowing tail and a ring icon over their head. A "!" warning plus a pulse of 3 Hz or less shows when someone is within 2 m (FR-08-02). **Variants:** *R2* starts tails at 50% of players, then they are redistributed at GO. *R3* adds 2 extra bumpers. **Failure states:** grab spam, handled by the grab cooldown (A, FR-04). A player who never moves (AFK) gets the standard idle kick. A stolen tail always gives 1.0 s of grab immunity, so a tail cannot be snatched straight back.

2.7 Tile Drop — FINAL (survival) and Main Show fallback

Trace: FR-01-05 AC2/AC3, FR-03-07, B §4.3, FR-01-03 AC4 · Obstacles: Drop Tile, Slime

Non-final fallback (REC-12): in a non-final slot, Tile Drop runs with a 60% qualifier quota and a 1:45 limit on the same layer table (01-ISS-10); owner confirmation is open (Appendix C).

Goal: ⬡ "Tiles drop! Be last on top!"

Blockout (original circular stack, layers 8 m apart; hex tiles 1.6 m across):

```
L1 (top)   ⬡⬡⬡⬡⬡⬡⬡  radius 14 m, spawn ring
L2         ⬡⬡⬡⬡⬡⬡⬡⬡  radius 16 m
L3         ⬡⬡⬡⬡⬡⬡⬡⬡⬡ radius 18 m
L4 (13–20 players only)
▓▓▓▓▓▓▓▓ slime (out) ▓▓▓▓▓▓▓▓
```

Timings: a tile vanishes 0.8 s after the first touch: dip 0–0.12 s, white 0.12–0.56 s, bounce-back 0.56–0.72 s, gone at 0.8 s. The first contact is authoritative. The client hides the tile within 100 ms of the server's removal event (FR-03-07 AC2). Layers: 3 for 8–12 players and 2 for 7 or fewer (B §4.3). Limit 5:00. **Winner:** the last bean standing. Simultaneous final eliminations resolve by sub-tick timestamp, and an exact tie means all win (AD-10). At the 5:00 time-out, everyone still in wins. **Acceptance:** with 12 players the median round lasts 2:00–3:30, and fewer than 5% of rounds time out. **Readability:** the bounce-back "last call" is also a sound tick. Lower layers glow so they stay readable from above. **Variants:** the layer count follows N. *Tile Return* (tiles come back after 6 s) is the team-show fallback in R2. **Failure states:** a perfectly synced slow-jump stalemate is answered by the 5:00 all-win time-out. A desync where a tile has already gone on the server is handled by the server's authority (src 03 §4).

2.8 Summit Grab — FINAL (trophy grab)

Trace: FR-01-05 AC1/AC3, FR-03-10, FR-03-12 · Obstacles: Swing Ball, Spin Bar, Launch Arm

Goal: 🏆 "Climb up! Grab the trophy!"

Blockout (original spiral hill, about 140 m of path, 30 m rise):

```
        🏆 trophy on 3 m plinth (grab volume)          CP3
      / Spin Bar ring (2 bars, 45 °/s) on summit plateau
    / LAUNCH ARM BEND (1 arm, rails; lands fwd or back)  CP2
  / SWING BALL STAIRS (3 balls, period 3.5 s offset 1.2 s)  CP1
START: 10 spawns, staggered 1.2 m
```

Timings: Swing Ball period 3.5 s (range 3–6), Δv 14 + 5. Spin Bar 45 °/s. Launch Arm 120 °/s with a 20 m/s tip. Limit 5:00. **Fall rule:** there is no slime. A fall respawns you at your checkpoint after 1.5 s (fall rule per level,

FR-03-13 AC1; see Issue 02-I-3). **Winner:** the first server-confirmed grab of the trophy wins. At the 5:00 time-out, all remaining finalists win (AD-10). **Readability:** a light beam rises from the trophy so it can be seen from the start, and a path arrow is painted on the ground. **Variants:** *Base*, and *Windy* (R2, FR-03-15 Swap: Spin Mallets instead of Swing Balls). **Failure states:**

- A Launch Arm puts someone on top of the summit and skips the climb. The bounds test makes sure no launch reaches the plateau unless that path is marked as intended.
- Plinth camping: the plinth's grab volume is 360°, and bumping a player off it is allowed.

2.9 MVP Modular Obstacle Kit (FR-03-01 – FR-03-17, A §5.4–5.5, B §4)

Every piece is a data-driven prefab with `period`, `phase`, `speed`, `dv`, `trigger` and a **required** `ragdoll` tag (CI fails if the tag is missing, FR-03-01 AC2). Motion runs on the server time base, with drift of 50 ms or less at p95 (FR-03-02). Every piece has a `speed_mult` (FR-03-04).

#	Piece	Ragdoll tag	Key defaults (range)	Footprint / snap	Used in
K1	Swing Ball	knockdown	period 4.0 s (3–6); Δv 14 + 5 up (10–18)	6 m path module; pivot height 7 m	Bumpy Parade, Slime Rise, Summit Grab
K2	Spin Bar	stumble (grounded) / knockdown (air)	40 °/s (30–60), Δv 9 (7–12); escalation per B §5	r 4–9 m; heights 0.7 m low, 1.9 m high	Beam Hop, Summit Grab
K3	Spin Mallet (foam)	knockdown	90 °/s (60–150), Δv 12 + 4; overclock ×1.4	4 m disc	Slime Rise
K4	Tilt Plank	none (physics)	max 25° (15–35); 3 beans tip it in ≈ 1.0 s	3 × 8 m	Bumpy Parade
K5	Conveyor	none (floor)	3.5 m/s, **≤ 3.6 cap** (AD-08)	2 m tiles	Bumpy Parade, Slime Rise
K6	Bumper	**none** (AD-07)	reflect + 10 (7–14); travel 1.5 m/s	1.5 m round	Knock-Knock Run, Bumpy Parade, Tail Chase
K7	Pusher Block (+ wall variant)	stumble	stroke 3.0 s (2–4.5), Δv 10 (7–14), extend 4 m/s; wall 3→5 m/s	2 × 2 m panel	Knock-Knock Run, Wall Dodge
K8	Launch Arm	knockdown	120 °/s (90–180), tip 20 m/s (15–26)	6 m arm, must pass bounds test	Slime Rise, Summit Grab
K9	Trick Door	none	break at ≥ 3 m/s; despawn 0.3 s; identical meshes	2 m door; rows of 2–10	Knock-Knock Run
K10	Drop Tile	none	vanish 0.8 s (0.5–1.5), white 0.5 s; layers 8 m apart	1.6 m hex	Tile Drop
K11	Slime (out / slippery)	none	out on contact; rising rate per B §4.5	plane	Slime Rise, Beam Hop, Wall Dodge, Tile Drop
support	Checkpoint, rail, ramp, spawn pad, kill plane, soft out-of-bounds volume, finish arch	—	B §6	1 m grid	all

Kit conventions

- 1 m grid. Path modules are 6, 12 or 24 m wide; R1 uses lanes at least 6 m wide ("wide lanes", B §9).
- Every hazard has a telegraph state and a cue: an anticipation pose of at least 0.25 s, plus a sound.
- Collision never changes with skin (FR-03-16).

2.10 Level-building guidelines

1. **Movement budget (A §1–3):** run 9.0 m/s, jump apex 1.5 m, air time ≈ 0.73 s. Gaps are at most **3 m for jump only** and at most **5 m for jump plus dive** (AD-31). Ledges taller than 1.5 m need a ramp or bounce.
2. **Course length:** about 250 m, which is 45–70 s for a skilled player and fits a 3:00 limit (B §1.2).

3. **Checkpoints:** the **12 s rule** governs (B §6.2). Place one after each high-variance section, typically 4–6 per course. Never place one right before a launcher that can loop a player back.
4. **Difficulty by slot (B §9):** follow the difficulty curve in §1.10.1 (target eliminations, fail risk per obstacle, speed_mult and "Faster" variants by slot).
5. **Crowd design:** the spawn pad holds max players at 1.2 m spacing. Make a choke point no narrower than 1/3 of the start width, and avoid choke points in the first 20 m.
6. **Safe windows:** each periodic hazard leaves a window of at least 1.2 s. Neighbouring hazards on the same path use different phases.
7. **Readability:** apply the rules of §2.0.3 and §1.10.2; in a child first-encounter test, at least 80% must describe the hazard correctly (FR-03-17).
8. **Tone:** slapstick only, with no blades or injury (AD-24). Out-copy is positive (FR-08-03).
9. **Originality:** a legal and design sign-off confirms the layout is not a 1:1 copy of any reference map (AD-38).
10. **Data:** each level record follows the FR-02-01 schema (id, name key, type, min/max players, rule, pct, slots, divisor, weight, limit, variants, enabled flags). It must pass the name lint and the schema check in CI.
11. **Exit tests per level:** the B §1.2 or B §5 timing test, the 1,000-respawn overlap test, a wall-hug soft-lock pass (0 soft-locks), the launch bounds test, flash analysis, and the client perf capture on min-spec (NFR-10-04).
12. **Practice Playground (FR-01-12):** this offline space includes one station per kit piece, K1–K11.

Registry sketch (FR-02-01)

id	type	slots	minP–maxP	pct / rule	limit	weight
knock_knock_run	COURSE	R1,R2	8–40	70 / 60	3:00	1.0
bumpy_parade	COURSE	R1	8–40	70	3:00	1.0
slime_rise	COURSE_SLIME	R2,R3	6–32	60	2:20	1.0
beam_hop	SURVIVAL	R2,R3	6–32	60 / timer	1:30	1.0
wall_dodge	SURVIVAL	R2,R3	6–32	60 / timer	1:45	1.0
tail_chase	POINTS	R2,R3	6–32	60 (tails = quota)	2:00	1.0
tile_drop	FINAL	F (+ fallback)	2–20	last standing	5:00	1.0
summit_grab	FINAL	F	2–**10**	trophy grab	5:00	1.0

3 Character Movement, Physics and Controls

Source: `03-character-controls.md` (Draft v1 (2026-10-23)). **Authority:** Physics and controls, together with gapfill-A.

Audience: kids 6–12, PC (Windows) MVP (AD-51). All numbers are **design estimates (not confirmed from Fall Guys)** from gapfill-A §8, locked only after movement-lock gate M0 (Ch. 17, DEP-02). No Fall Guys names, art or audio are used (UD-2, AD-25).

Feel pillars: (1) *Floppy but controllable* — wobble is visual, control is crisp (≤0.25 s to full speed). (2) *Falls are funny, never unfair* — slapstick tumbles, short and capped. (3) *Forgiving for small hands* — buffers, coyote time, fixed jump, assist options. (4) *Readable in a crowd of 32* — clear silhouettes, cues and camera.

3.1 Character state machine

Simulation runs on one 50 Hz fixed step on client and server (AD-06, FR-07-12). The local client predicts; the server validates and owns grabs, knockdowns and respawn.

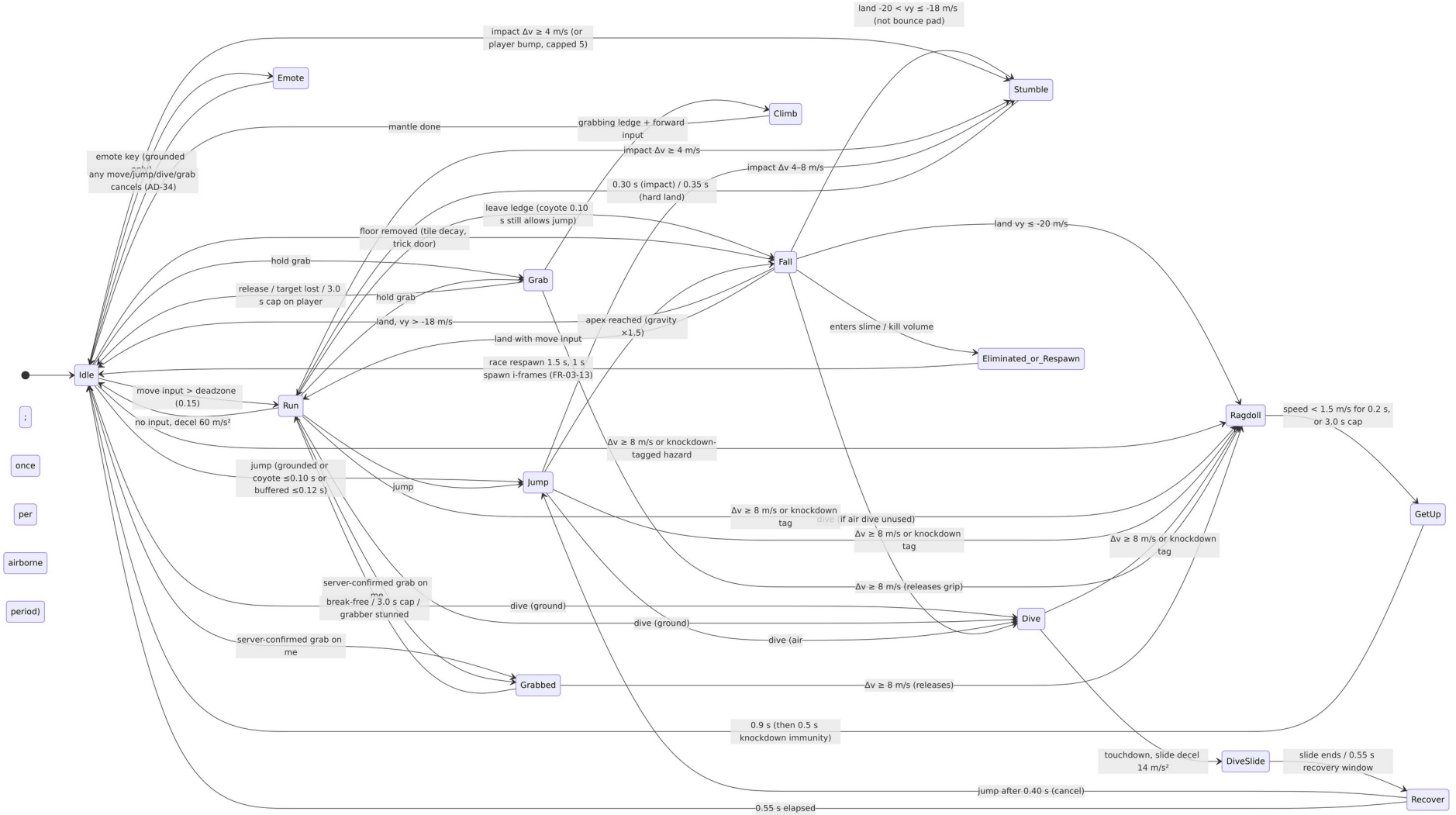


Figure 3.1 — Character state machine

`Fall` and `Jump` share the *Airborne* super-state (air accel 18 m/s², one air dive). `Climb`, `DiveSlide`, `GetUp`, `Grabbed` and `Emote` are sub-states needed for animation and gating, not new requirements.

3.1.1 State rules and gating

State	Player control	Can enter Jump / Dive / Grab	Exit	Trace
Idle	full	Y / Y / Y	input	FR-04-01
Run	full, 9.0 m/s, analog ≥2.0 m/s	Y / Y / Y	input	FR-04-01
Jump (rise)	air accel 18 m/s²	– / Y (once) / N (grabs objects/ledges only on contact, see 03-I-07)	apex	FR-04-02
Fall	air accel 18 m/s²	coyote 0.10 s / Y (once) / ledge only	land	FR-04-02
Dive	none (committed)	N / N / N	touchdown	FR-04-03
DiveSlide + Recover	none until 0.40 s, then jump-cancel	jump after 0.40 s	0.55 s	FR-04-03 AC3
Grab (holding)	speed ×0.6, no jump/dive	N / N / –	release, 3.0 s cap (players only), Δv ≥ 8	FR-04-04, FR-04-01 AC3
Climb	none	N	mantle complete	FR-04-04 AC1
Grabbed	speed ×0.45, pulled by spring 200 N/m; break-free input	N / N / N	free, cap, knockdown	FR-04-04
Stumble	40 % control	N	0.30 s / 0.35 s	FR-04-05, AD-30
Ragdoll	none	N (inputs ignored, not buffered)	settle or 3.0 s cap	FR-04-05
GetUp	none	N	0.9 s; 0.5 s knockdown immunity after	FR-04-05
Emote	none	any action cancels	input	FR-05-06, AD-34

Priority rule when several triggers land on the same tick: **Knockdown > Eliminate > Stumble > Grabbed > Dive > Jump > Grab > Move**. Server outcome wins over local prediction; corrections are smoothed over ≤100 ms (net.interpBuffer).

3.2 Parameter set (tuning baseline, gapfill-A §8)

Exposed as data (config keys), hot-reloadable in dev builds; live values versioned with the build (FR-13 config). Tuning order: speed → jump → air control → dive → grab → ragdoll → hazard Δv → camera. Changing `move.maxSpeed` or gravity re-validates gap rules (jump-only ≤ 3 m, jump+dive ≤ 5 m).

3.2.1 Locomotion (FR-04-01)

Key	Default	Range	Feel note
move.maxSpeed	9.0 m/s	7–11	100 m in 11.1 s ± 0.3
move.minWalkSpeed	2.0 m/s	1.5–3	analog floor
input.stickDeadzone	0.15	0.10–0.25	no drift on worn sticks
move.accel / decel / reverseAccel	45 / 60 / 90 m/s²	30–70 / 40–90 / 60–120	90 % speed in 0.15–0.25 s
move.turnRate	720 °/s	480–1080	visible turn, dive aim ≤10° error
move.slopeLimit / stepOffset	45° / 0.35 m	35–55 / 0.25–0.45	

3.2.2 Jump & air (FR-04-02)

Key	Default	Range
jump.height / timeToApex	1.5 m / 0.40 s	1.2–1.9 / 0.32–0.50
physics.gravity (derived) / jump.velocity (derived)	18.75 m/s² / 7.5 m/s	–
jump.fallGravityMult / terminalVelocity	1.5× / 25 m/s	1.2–2 / 18–35
jump.releaseCutMult	1.0 (off, AD-31)	0.4–1.0

Key	Default	Range
air.accel	18 m/s² (40 %)	11–27
jump.coyoteTime / input.jumpBuffer	0.10 / 0.12 s	0.06–0.15 / 0.08–0.20
jump.landCooldown / land.hardStumbleTime	0.05 / 0.35 s	0–0.15 / 0.2–0.6

3.2.3 Dive (FR-04-03)

Key	Default	Range
dive.horizSpeed / vertSpeed	max(current, 11) / +3.0 m/s	9–14 / 1.5–4.5
dive.slideDecel	14 m/s²	8–20
dive.recoveryTime / cancelToJumpAfter	0.55 / 0.40 s	0.35–0.9 / 0.25–0.7
input.diveBuffer	0.10 s	0.08–0.20

3.2.4 Grab (FR-04-04)

Key	Default	Range
grab.range / coneHalfAngle	1.0 m / 45°	0.7–1.4 / 30–60
grab.maxHoldPlayer / regrabCooldown	3.0 / 1.5 s	2–5 / 1–3
grab.breakFreePerInput	0.25 s	0.1–0.4
grab.victimSpeedMult / grabberSpeedMult	0.45 / 0.6	0.3–0.7 / 0.4–0.8
grab.pullSpring	200 N/m	100–400
grab.throwForward (+2 m/s up)	4 m/s	0–6
grab.ledgeMantleTime *(new, 03-I-08)*	0.45 s	0.3–0.7
grab.ledgeMaxHeight *(new, 03-I-08)*	ledge top ≤ 0.4 m above feet ("lip catch"; max jump-reachable ledge ≈ 1.9 m, so 2.0 m ledges stay gated)	0.2–0.45 m

3.2.5 Collision, stumble, ragdoll (FR-04-05, FR-04-07)

Key	Default	Range
body.mass	60 kg	40–80
ragdoll.stumbleDv / knockdownDv	4 / 8 m/s	3–6 / 6–11
ragdoll.playerBumpCapDv	5 m/s	3–7
ragdoll.stumbleTime *(new key, 03-I-02)*	0.30 s, 40 % control	0.2–0.5
land.knockdownVy *(new key, 03-I-02)*	−20 m/s	−18 to −25
ragdoll.settleSpeed (+0.2 s hold)	1.5 m/s	0.8–2.5
ragdoll.getUpTime / maxDuration / iFrames	0.9 / 3.0 / 0.5 s	0.6–1.4 / 2–4 / 0.3–1.0
respawn.delay (+1 s spawn i-frames)	1.5 s	1–3

Floor and zone modifiers (conveyor 3.5 m/s, ice accel ×0.25 / decel ×0.1, water ×0.7, low-g ×0.4, fan 12 m/s², bounce pad vy 14 ragdoll:none) are owned by the obstacles doc (FR-03-05, AD-07, AD-08); the controller must expose hooks: `surfaceAccelMult`, `surfaceDecelMult`, `speedMult`, `gravityMult`, `externalAccel`, `platformVelocity` (inherited on landing).

3.2.6 Camera and network

Camera (FR-04-08): see §3.6. Net (FR-07-12): tick 50 Hz, interp 100 ms, ragdoll replicated as root + 3 key bones (pelvis, spine_02, head; §3.3.1). The systems budget in §8.3 allows up to 5 key bones only if D-07 bandwidth allows (REC-11).

3.3 Animation needs per state (asset slots)

Art arrives later from the owner's repository (UD-3); every slot ships with a placeholder grey "PH" bean (FR-12-06). Naming follows FR-12-02: `wb_emote_*` for emotes; locomotion uses category **`anim`** (see 03-I-09) e.g. `wb_anim_base_full_run_v01.fbx`.

3.3.1 Rig requirements (FR-12-04, FR-05-02)

- One shared base rig: `root, pelvis, spine_01, spine_02, head`, 2 arm chains, 2 leg chains, face bones/blend shapes (eyes, mouth). Waist seam edge loop at fixed height; **upper** costume skinned pelvis-and-above, **lower** pelvis-and-below.
- **Layered animation:** locomotion on a *lower-body layer* (pelvis down) and actions on an *upper-body layer* (spine_01 up, masked) so the bean can run while reaching for a grab, waving or carrying an egg. Pelvis is the shared pivot; additive *wobble* layer on spine_01/02 + head driven procedurally (spring, velocity-based) to sell "floppy" without new clips.
- Active ragdoll: physics bodies on pelvis, spine_02, head, upper/lower arms, upper/lower legs (11 bodies). Network replicates **root, pelvis, spine_02, head** (the "root + 3 key bones" of gapfill-A §7); limbs local-simulated.
- Costumes never change the collision capsule (1.6 m × r 0.5 m) (FR-12-04 AC3); accessory leaf bones (≤8/part) are spring-simulated, cosmetic only, and excluded from ragdoll bodies.
- Every clip must pass the seam test in 5 poses (FR-12-04 AC1); clip list below names the 5 test poses ★.

3.3.2 Clip slots

Slot ID	Layer	State	Loop	Notes / events
idle_01 ★, idle_fidget_01–03	full	Idle	Y	fidgets after 6 s idle
walk_fwd, run_fwd ★	lower (+upper arm swing)	Run	Y	blend by speed 2→9 m/s; footstep events L/R
run_lean_L/R (additive)	full	Run turning	Y	lean from turn rate
jump_takeoff, jump_rise, jump_apex ★, fall_loop	full	Jump/Fall	rise/fall loop	takeoff event (SFX)
land_soft, land_hard	full	land / hard stumble	N	hard = 0.35 s
dive_start, dive_air ★, dive_belly_slide, dive_getup	full	Dive → Recover	slide loop	getup cancel mark at 0.40 s
grab_reach, grab_hold_loop, grab_release, grab_throw	upper	Grab	hold loop	arms IK to target point
grab_carry_loop (object)	upper	Grab object	Y	egg/ball/tail/trophy socket `hand_r_socket`
grabbed_struggle_loop	upper + face	Grabbed	Y	intensity param from break-free rate
ledge_hang, ledge_mantle	full	Climb	N	root motion 0.45 s
stumble_fwd/back/L/R	full (additive over locomotion)	Stumble	N	0.30 s, pick by impact dir
ragdoll (physics) + ragdoll_pose_blend	full	Ragdoll	–	procedural; face "dizzy"
getup_front, getup_back ★	full	GetUp	N	0.9 s, chosen by pelvis facing
respawn_pop	full	Respawn	N	0.3 s appear
emote slots ×4 (Wave, Thumbs Up, Happy Dance, Cheer)	full	Emote	varies	FR-05-06; friendly only (AD-24)
celebrate_qualify, celebrate_win, sad_eliminated (gentle)	full	round end	N	non-shaming tone (UD-4)
face_states: happy, effort, surprised, dizzy, determined	face	all	–	blend shapes

3.4 Input mappings

All actions rebindable (FR-04-12), reset to defaults, per-profile persistence; prompts follow last-used device within 1 s, hot-plug within 2 s, no forced pause (FR-04-15).

3.4.1 Keyboard/mouse and gamepad (MVP)

Action	Keyboard / Mouse	Xbox-layout	PlayStation-layout	Switch Pro	Type
Move	W A S D (+ arrow keys alt)	Left stick	Left stick	Left stick	analog / digital = full speed
Camera	Mouse	Right stick	Right stick	Right stick	
Jump	Space	A	Cross	B	press, buffered 0.12 s
Dive	Left Ctrl (alt: Right mouse)	X	Square	Y	press, buffered 0.10 s
Grab	Left Shift (alt: Left mouse)	RT	R2	ZR	hold (toggle option)
Break free	any of Jump/Dive/Grab or stick wiggle	same	same	same	mash (hold option)
Show names	Tab	LT	L2	ZL	hold (FR-04-09)
Emotes 1–4	1 2 3 4	D-pad	D-pad	D-pad	press
Quick-chat wheel	Q (hold)	LB (hold)	L1	L	Ch. 7/11 owns content
Re-centre camera	Middle mouse / R	R3	R3	R-stick click	press
Settings (no pause online)	Esc	Menu	Options	+	FR-08-12 AC2
Walk modifier	— unbound	—	—	—	Won't (AD-48)

3.4.2 Touch (Later — mobile)

Left virtual joystick (floating, dead-zone 0.15); right-half drag camera (0.25 °/px); double-tap re-centre; on-screen Jump / Dive / Grab buttons (≥ 12 mm, thumb arc); emote button; repositionable layout with reset. Grab defaults to **toggle** on touch. Jump buffer 0.12 s covers ~50–80 ms touch latency. Grab-lock is **not** adopted as specified (see 03-I-06).

3.5 Kid-friendly assist options

Defaults chosen for ages 6–12; all persist per profile, reachable mid-show (FR-08-12), none gives a competitive edge online beyond accessibility parity.

Option	Default	Effect	Trace
Fixed jump height	ON (releaseCutMult 1.0)	tap = full jump	AD-31
Grab: hold / toggle	Hold (PC), Toggle (touch)	toggle respects 3.0 s cap and cooldown	FR-04-04 AC4, FR-08-12
Break-free: mash / hold	Mash	hold = 1 input / 0.25 s	FR-08-12
Auto-recentre camera	ON, 2.0 s	off for advanced players	FR-04-08
Camera shake	Reduced (50 %)	off / reduced / full	FR-08-20
Screen flashes & speed lines	ON (≤3 flashes/s)	off / reduced	FR-08-20, NFR-10-19
Motion blur	OFF		FR-08-20
FOV	70°	60–85°	FR-08-20
Camera sensitivity / invert X / invert Y	1.0 / off / off	0.1–3.0×	FR-04-08
Stick deadzone	0.15	0.10–0.25	gapfill-A §1.2
Controller vibration	ON, 70 %	off – 100 %	03-I-05 (new)
Audio-cue visual indicators	ON		FR-08-12 AC4
Practice Playground	menu	offline, all mechanics with signage	FR-01-12, FR-08-16

Option	Default	Effect	Trace
Tutorial prompts	first 3 shows	contextual "Hold Shift to grab!"	onboarding (Ch. 8)

Built-in (always-on) forgiveness: coyote 0.10 s, jump buffer 0.12 s, dive buffer 0.10 s, player bumps never knock down, ragdoll cap 3.0 s + 0.5 s immunity, grab cap 3.0 s + 1.5 s cooldown, level gaps ≤ 3 m / ≤ 5 m (margin under real reach).

3.6 Camera design (FR-04-08, FR-08-20)

- **Rig:** third-person orbit; pivot 1.2 m above feet; boom 6.0 m; default pitch 15° down; vertical FOV 70° (≈103° h at 16:9; mobile 75°); pitch clamp −30° to +60°.
- **Input:** pad yaw 180 °/s, pitch 120 °/s, curve exp 2.0, deadzone 0.12; mouse 0.12 °/count; touch 0.25 °/px; sensitivity 0.1–3.0×; separate invert X/Y; persisted.
- **Collision:** sphere-cast r 0.3 m; snap in instantly, ease out over 0.3 s; never inside geometry (30-min soak). Other beans do **not** pull the camera in (they fade to 30 % alpha within 1.0 m of camera) — avoids pops in crowds.
- **Auto-recentre:** after 2.0 s without camera input while moving, yaw eases behind travel direction at ≤ 90 °/s; never during manual input; toggle.
- **Framing aids:** slight look-ahead (pivot offset up to 0.8 m in velocity direction) so a hazard 15 m ahead stays on screen (AC1); airborne: vertical follow damped (0.15 s) to avoid bob; on ragdoll: camera keeps tracking pelvis with 0.25 s damping, no spin with the body.
- **Comfort:** no roll, no FOV kick by default (speed-FOV +3° only if speed lines enabled), shake capped at 0.15 m / 0.25 s and scaled by setting; no forced camera cuts mid-round except respawn (fade 0.2 s).
- **Spectate/eliminated:** see §4.2.6; uses same rig settings.

3.7 Feedback & juice (VFX / SFX / haptics)

Merge note (REC-13, REC-14): the SFX column below is a feel brief; the authoritative audio event IDs and briefs are the §6.3 event list (FR-08-08 AC1). The haptic column is the authoritative haptic list.

Rules: all cues friendly and non-violent (UD-4: "boing", "bonk", stars, no impact gore/"hit" sparks); every gameplay audio cue has a visual (FR-08-12 AC4); flashes ≤3/s (NFR-10-19); shake respects FR-08-20; VFX budgets scale for 32 beans (local player full, others LOD). Asset slots use `wb_vfx_*` / `wb_sfx_*` (FR-12-02).

Event	VFX	SFX	Haptic (gamepad)	Camera
Footsteps (run)	small dust puffs every 2nd step	soft squishy steps, pitch by surface	—	—
Jump takeoff	squash 0.9/stretch 1.1 over 60 ms, dust ring	"boing" (rising)	light 40 ms	—
Land soft	squash 1.15 for 80 ms, dust ring	soft "pomf"	light 30 ms	—
Land hard (stumble)	bigger dust, wobble	"thud-oof"	medium 80 ms	shake 0.05 m
Dive start / slide	stretch + speed streaks, slide trail (surface tinted)	"whoosh" + slide hiss	medium 60 ms	—
Dive get-up cancel ready (0.40 s)	small sparkle on bean	tiny "tick"	—	—
Grab reach / connect	hand glow; connect = ring pulse on target	"grab-pop"	medium 50 ms on connect	—
Grab whiff	puff at hands	soft "fwip"	—	—
Being grabbed	struggle lines, break-free meter ring around bean	wobbly struggle loop	pulsing light while held	—
Break free / auto-release	burst of confetti dots	"pop!"	medium 60 ms	—
Grab cap 3.0 s nearing (2.5 s)	grabber hand ring blinks (≤2 Hz)	—	—	—

Event	VFX	SFX	Haptic (gamepad)	Camera
Pick up object (egg/tail/trophy)	sparkle, object outline on	cheerful "plink"	light 40 ms	—
Player bump	soft ring at contact	rubbery "bonk"	light 30 ms	—
Stumble	tilt wobble, sweat drops	"whoa!" vocal (non-verbal)	medium 80 ms	shake 0.05 m
Knockdown	cartoon stars + swirl, comic "POW" puff shape (no text)	"bonk-boing" + giggle-y vocal	strong 150 ms	shake 0.15 m / 0.25 s (scaled)
Ragdoll tumbling	dizzy stars loop	tumble squeaks on each bounce (rate-limited 4/s)	—	track pelvis
Get-up / immunity 0.5 s	soft shimmer outline 0.5 s	"hup!"	light 30 ms	—
Bounce pad / bumper (ragdoll:none)	spring ring	big "BOING"	medium 70 ms	—
Conveyor/ice/water entry	surface-specific particles	surface loop	—	—
Fall into slime / out of bounds	splash (bright, cartoon)	"sploosh"	medium 100 ms	fade 0.2 s
Respawn	pop-in sparkle	"pop"	light	—
Qualify	confetti burst	jingle	double light pulse	—

Haptics are short, never continuous > 3 s, disabled when vibration = 0, and only for the local player's own events.

4 UX Flows and UI Design

Source: `04-ux-ui.md` (Draft v1 (2026-10-23)). **Authority:** Screens and flows; yields to Section 1 on rules and Section 9 on child safety.

Scope: screen map, navigation, user flows, low-fi wireframes, HUD, kid readability, accessibility, localisation, component library.

Imagery: every picture, icon and bean shown here is a placeholder (`[img:...]`). Final art comes from the user's asset repo through the Chapter 12 pipeline (AD-26, UD-3).

Key constraints this design follows:

- Everyone is a child. There is no age gate and no teen or adult path; adults use the separate Parent Account (AD-01, FR-11-01).
- No free text or voice anywhere in a child profile, and no keyboard or text field (FR-11-03 AC3). Names are generated (FR-11-04).
- No purchases, ads or dark patterns (AD-12, FR-11-09, FR-11-14, FR-08-18).
- MVP platform is Windows PC: mouse, keyboard and controller, played at desk or TV distance (AD-51). Touch is designed for but ships Later.
- Original look. The reference game's pink, yellow and blue UI is not reused (AD-26).

4.1 Screen map

4.1.1 Screen inventory

SC-01…SC-15 come from src 08 §1. **SC-16…SC-22 are new UX IDs** proposed here for screens the requirements need but do not number (see Issue 04-UX-I3).

ID	Screen	Requirements	Offline?
SC-01	Main menu (hub)	src 08 §1, FR-08-04 AC2	Partial (online tiles greyed out with a cloud icon)
SC-02	Show Picker	src 08 §1	No
SC-03	Matchmaking / lobby	AD-17, FR-07-14	No
SC-04	Round intro	FR-08-04	No
SC-05	In-round HUD	FR-08-01, FR-08-02	No
SC-06	Qualified / Out	FR-08-03	No
SC-07	Spectate	FR-07-11, AD-36	No
SC-08	Results / victory	AD-35, FR-08-18	No
SC-09	Dressing Room + Coin Shop	FR-05-17, FR-05-10, FR-11-09	Dressing Room yes (cached items), Shop no
SC-10	Settings	FR-08-09/11/12/20, FR-04-12	Yes
SC-11	Parent gate + Parent Account (consent, controls, dashboard, data)	FR-11-02/07/08/10/12	Gate yes, Account no
SC-12	First-run onboarding + tutorial	FR-08-17, AD-44	Yes
SC-13	Report a problem	FR-11-06, FR-07-15	No (queued)
SC-14	Error / offline state	FR-08-16	Yes
SC-15	Practice Playground	FR-01-12, AD-18	Yes
SC-16	Player card (overlay: Mute, Block, Report; no Add friend on a stranger's card — REC-18)	FR-07-15, FR-11-06, AD-04	No
SC-17	Friends and Party	FR-07-06, AD-04	No
SC-18	Season Path	06	No

ID	Screen	Requirements	Offline?
SC-19	Break reminder / playtime limit	FR-11-15, AD-23	Yes
SC-20	Reconnecting overlay	FR-07-16	–
SC-21	Name picker (re-roll)	FR-11-04	Yes
SC-22	Kids' privacy explainer (picture-led)	FR-11-13	Yes

4.1.2 Navigation map

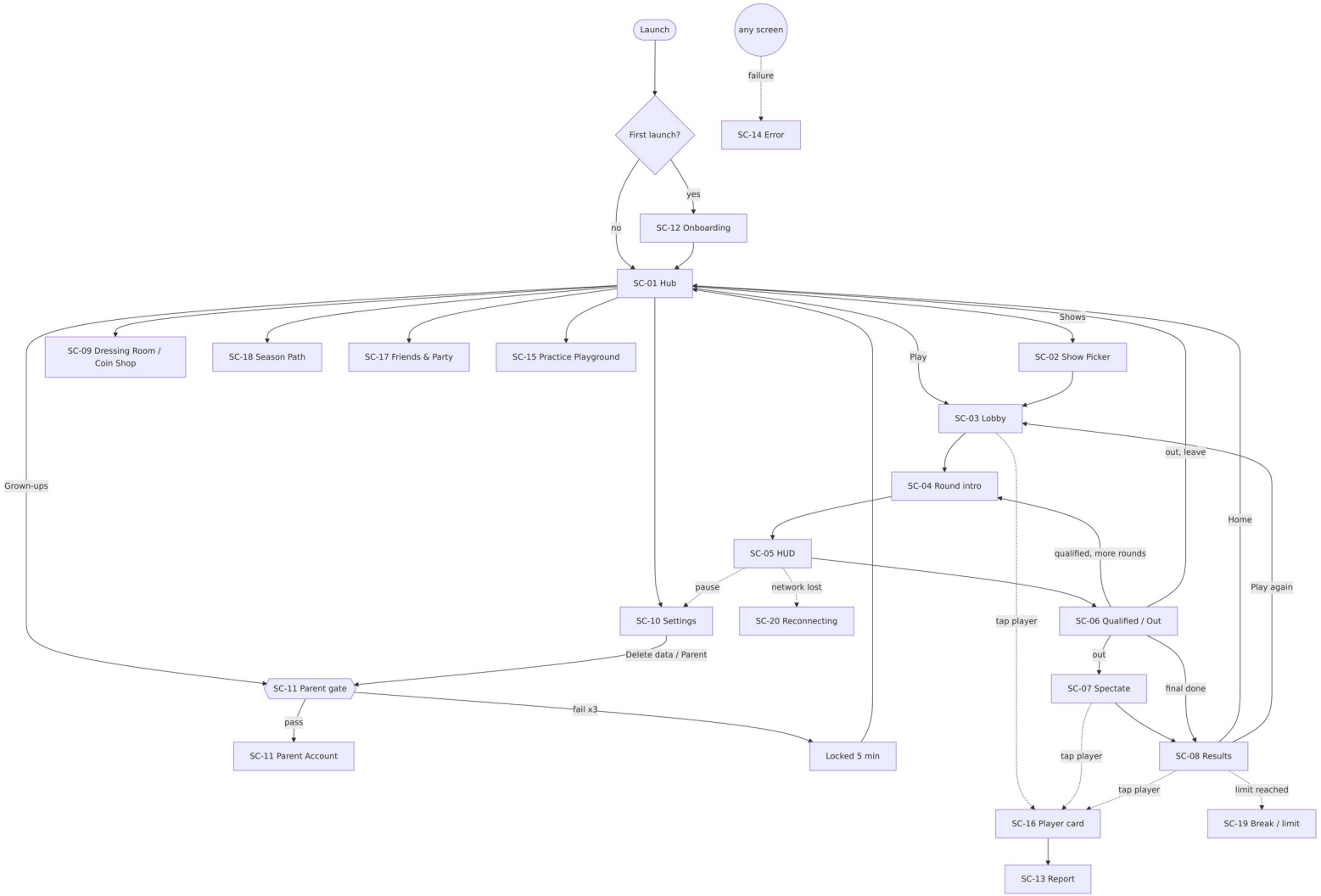


Figure 4.1 — Navigation map (SC-01 … SC-22)

Navigation rules:

- **N1:** Every screen has one Back control in the same place (top-left), mapped to Esc / B. Back never skips the parent gate (FR-11-07 AC1).
- **N2:** The hub is at most 2 steps from any non-game screen.
- **N3:** The show flow SC-03 → SC-08 never returns to the hub between rounds (FR-08-04 AC1). After SC-08 the player goes to the hub unless they press "Play again". There is no auto re-queue (FR-08-18).
- **N4:** Settings opens as an overlay during a show and does not pause the game (FR-08-12 AC2).
- **N5:** Focus is always visible, and every screen is fully usable with controller only, keyboard only, or mouse only.

4.2 User flows

4.2.1 First launch (FR-08-17, AD-44, FR-11-01)

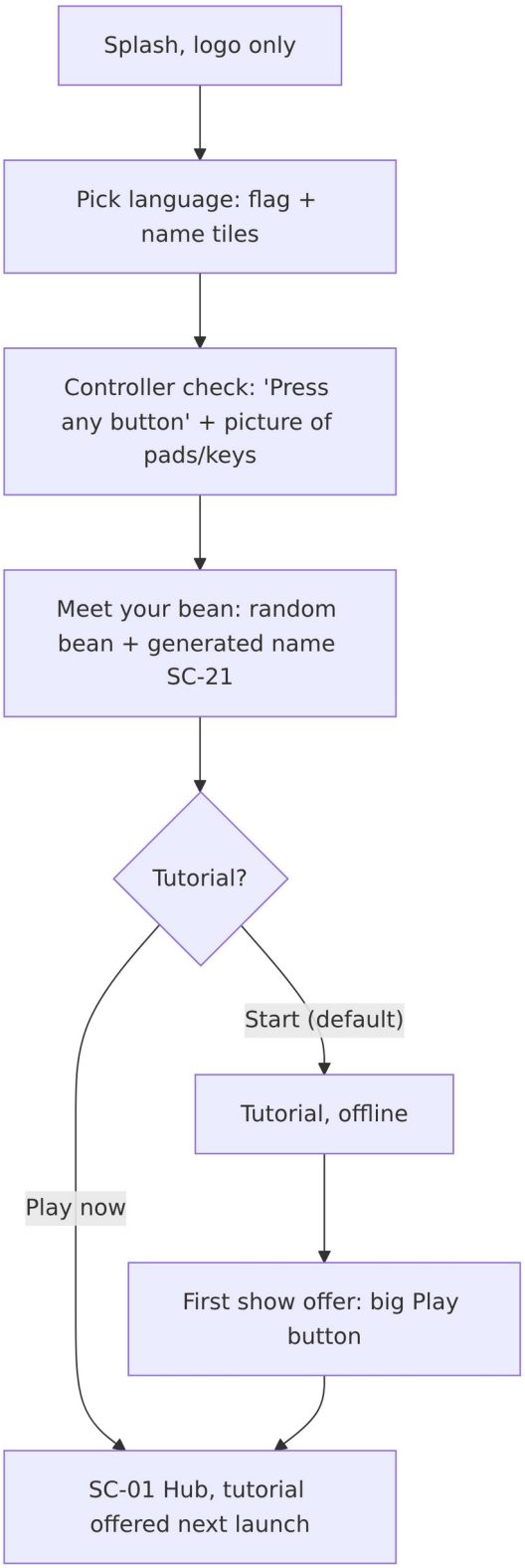


Figure 4.2 — First-launch flow

- Play starts as a **guest** at once. Nothing is asked: no name, email or age (US-11-4). Only a pseudonymous ID is sent (FR-11-01 AC2).
- Language defaults to the OS locale, and the child confirms it with one tap.

4.2.2 "Age gate" → grown-up fork (AD-01, FR-11-07)

The brief asked for an age gate. **Per AD-01 there is no age gate**: a mixed-audience gate is weaker under COPPA and the Children's Code. Instead, a "Grown-ups" button (padlock icon) on the hub and in Settings opens the parent gate. See Issue 04-UX-I1.

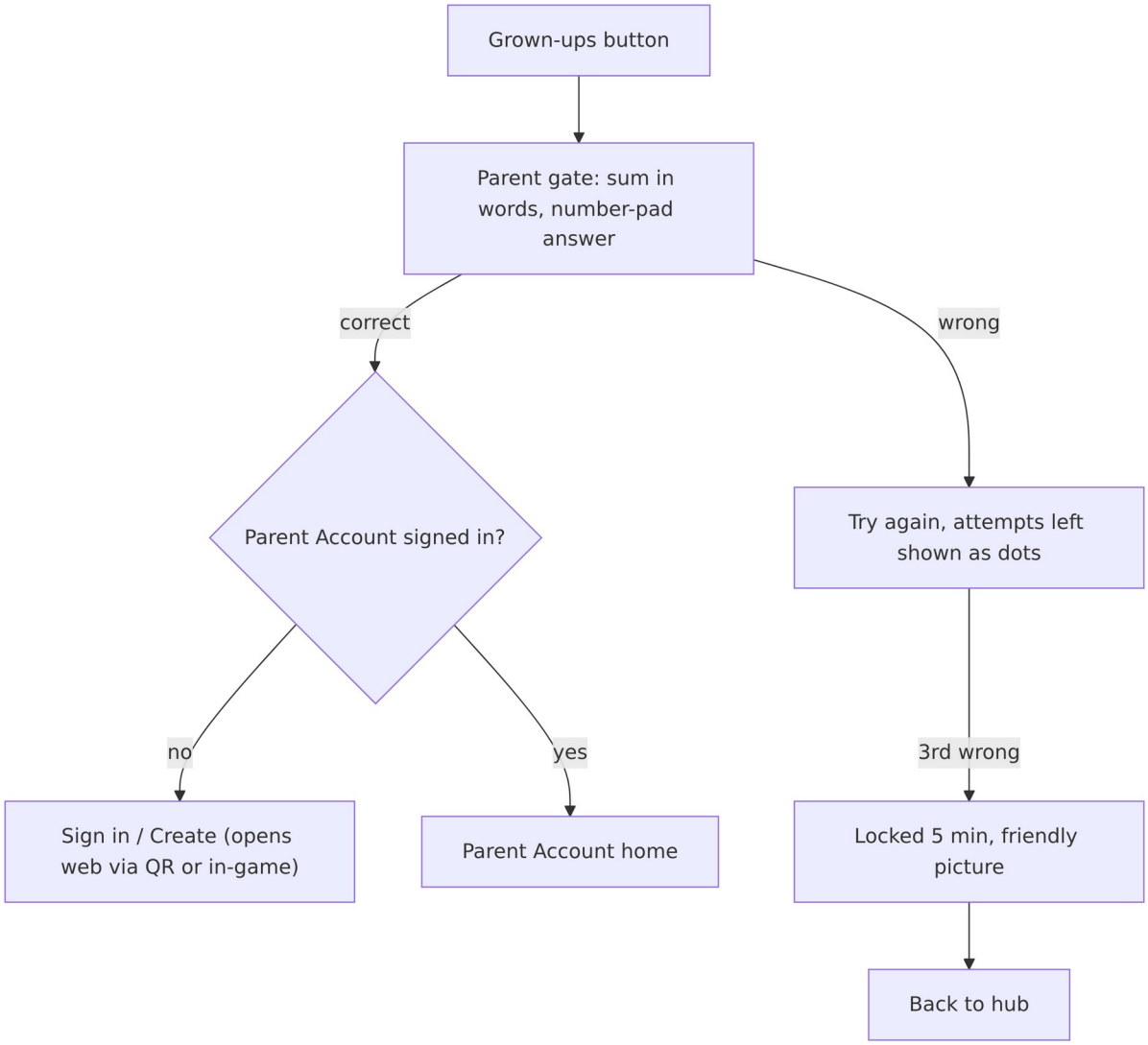


Figure 4.3 — "Grown-ups" fork to the parent gate

4.2.3 Parent consent (FR-11-02, AD-22, FR-07-06 AC2/AC3)

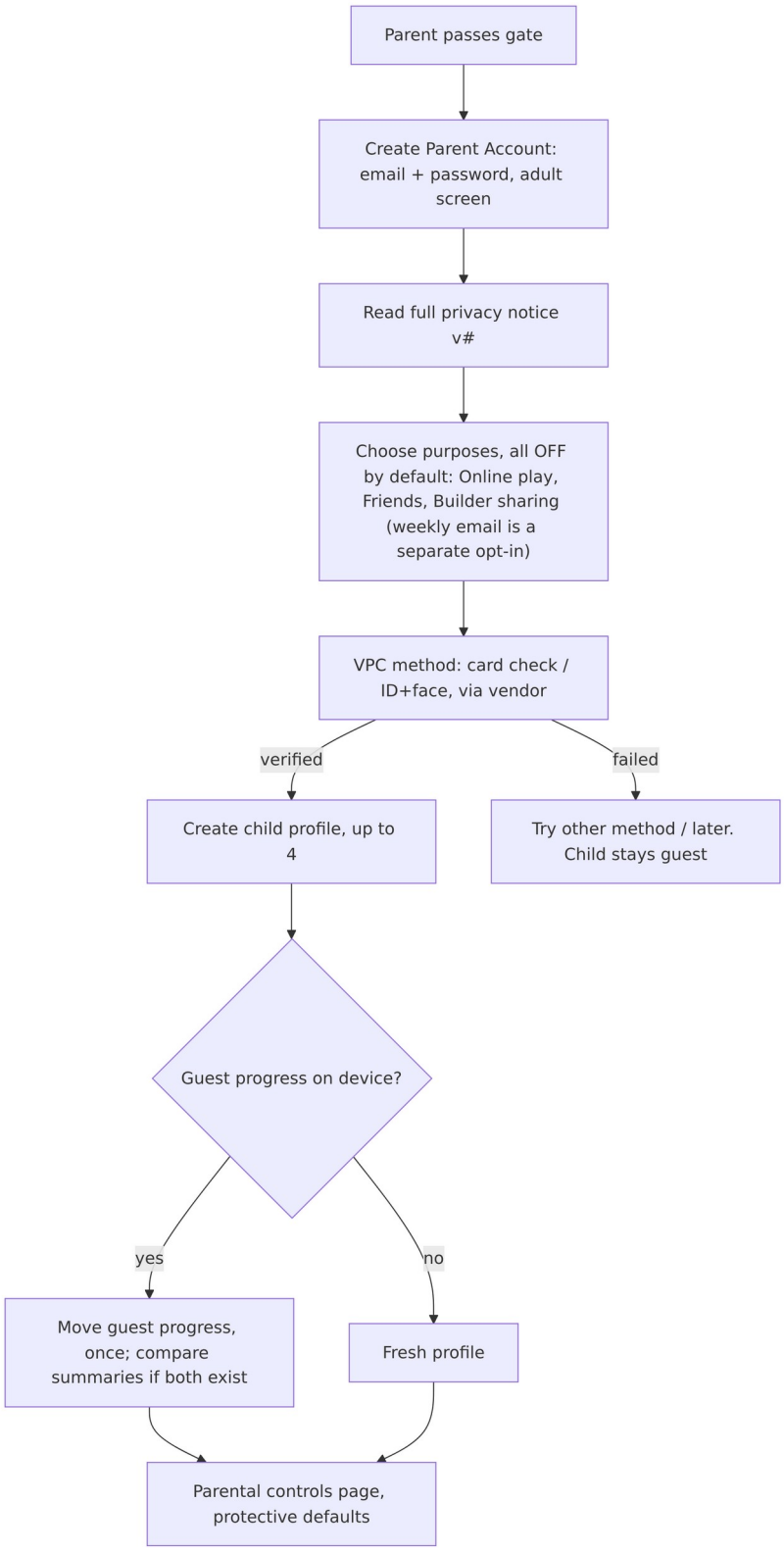


Figure 4.4 — Parent consent flow (screens)

- Each consent record stores method, timestamp, purposes and notice version (FR-11-02 AC3). Withdrawing consent shows "Online play will lock after this show" (src 11 §5).
- Adult screens can use normal text input and smaller text (≥ 24 px). Child readability rules do not apply to them, but contrast and localisation rules do.

4.2.4 Tutorial (FR-08-17, FR-01-12)

The tutorial path, stations, assists and skip rule are defined once in §1.8 (game-rule authority; REC-07, REC-06). UI presentation rules: each station has an animated picture of the input, a single-word label ("Jump!"), a ghost bean demonstrating it, and a gate that opens on success. There is no fail state. After 20 s with no progress the bean demo replays. Target: 5 minutes or less (AC1).

4.2.5 Play a show (FR-08-04)

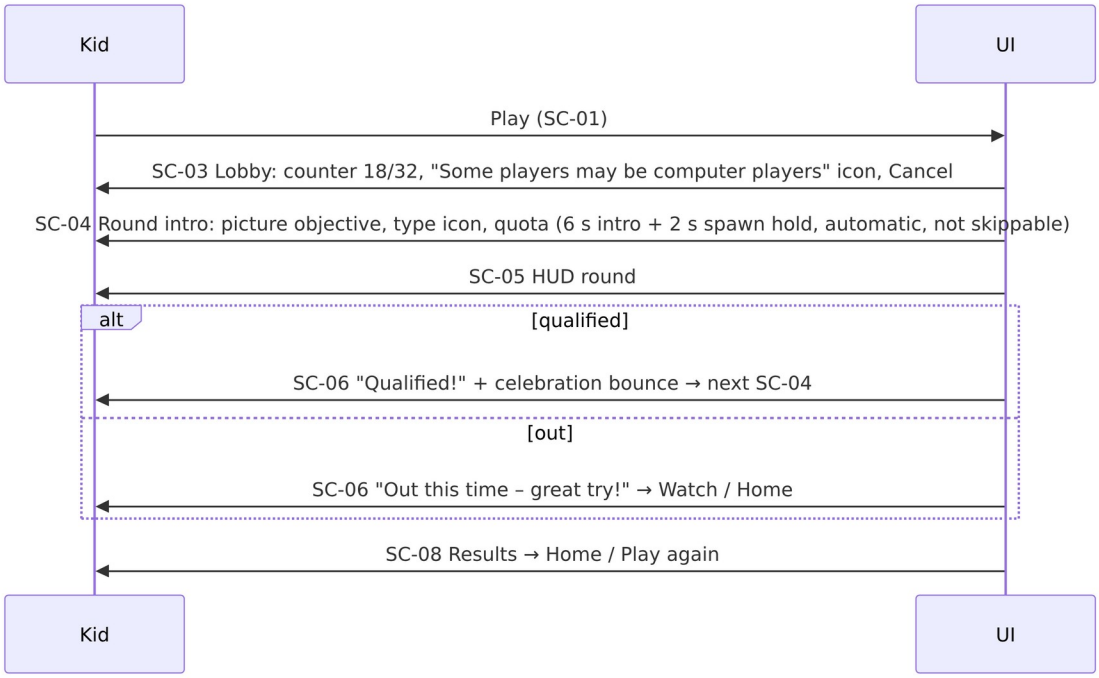


Figure 4.5 — Play-a-show sequence

4.2.6 Eliminated → spectate (FR-07-11, AD-36, src 08 §4)

SC-06 Out shows two big buttons: **[eye] Watch** (focused by default) and **[house] Home**. Neither has a countdown (FR-08-18). In SC-07: ◀️ ▶️ cycles players, with party members first (AC2). When the watched player finishes, the camera moves on after 2 s. When everyone has left, the player goes to SC-08 at round end.

4.2.7 Results (AD-35, FR-08-18)

Order: winner celebration (equipped celebration plays) → my result (medal for the furthest round reached; a placement number is shown to finalists only — REC-05) → itemised rewards (XP bar, Coins and Trophy tally, counted up in under 3 s and skippable) → **Home** (default focus) and **Play again**. No streaks, no "you'll lose", no timed buttons. The break reminder or limit (SC-19) shows here, after the show only (FR-11-15 AC2).

4.2.8 Dressing Room (FR-05-17, FR-05-10, FR-11-09)

Tabs (icons + label): Colour, Pattern, Faceplate, Upper, Lower, Emote, Celebration, Banner, Title. Tapping equips at once and the bean preview reacts. Unowned items show a lock and a **"How to get it"** chip (for example "Season Path step 12"). The Coin Shop tab shows Coin prices only: no real money and no countdown or "rotating soon" timer (FR-08-18, see 04-UX-I6).

4.2.9 Settings

Tabs: Sound, Controls, Screen, Easy-play (accessibility), Language, Help & Safety. Destructive and adult options ("Delete my data on this device", "Parent controls") sit behind the parent gate (FR-11-12).

4.2.10 Parental dashboard (FR-11-08, FR-11-10, FR-11-12)

Per-child tabs: **Activity** (30-day playtime bar chart; Trophies and Season Path), **Friends** (approve or remove; pending first), **Safety** (reports made, blocks, and moderation notices and appeals — 09-I-10), **Controls**, **Purchases** (empty state: "No purchases. Spending is off." in MVP), **Data** (view, export JSON, delete profile). The content, defaults and timing of every control are specified once in §9.11 (child-safety authority; REC-21). A "Data as of hh:mm" stamp shows the time (no more than 15 minutes old, FR-11-10 AC1). Controls show "Applies from the next show" (FR-11-08 AC1).

4.3 Low-fidelity wireframes (ASCII, 16:9, 1920×1080 reference)

Legend: `[ ]` button · `(img)` placeholder art · ◉ default focus · ▣ icon.

SC-12a Language

```
+-------------------------------------------------------------+
|                    (img: logo placeholder)                  |
|          [▣ flag English UK ◉]   [▣ flag English US]        |
|          [▣ flag ...R2]          [▣ flag ...R2]              |
|                        [ ✓ OK ]                             |
+-------------------------------------------------------------+
```

SC-12b Controller check

```
+-------------------------------------------------------------+
|        (img: gamepad)      (img: keyboard+mouse)            |
|                 "Press any button!"                         |
|           ✓ detected: Gamepad   [ ▶  Go ◉ ]                 |
+-------------------------------------------------------------+
```

SC-21 Name picker

```
+-------------------------------------------------------------+
| [◀️]            Your name                                    |
|              (img: bean, waving)                            |
|                                                             |
|              ┌──────────────────┐   [🎲 New name] ●●●○○      |
|              │  BouncyMuffin42  │    (5 re-rolls/day)        |
|              └──────────────────┘                           |
|                    [ ✓ I like it ◉ ]                        |
+-------------------------------------------------------------+
```

SC-12c Tutorial station

```
+-------------------------------------------------------------+
| [⏸️]                                  Station ●●○○○○         |
|                                                             |
|   ┌──────────┐   JUMP!                                      |
|   │(img: A   │                                              |
|   │ button)  │                                              |
|   └──────────┘                                              |
|                        (3D: ghost bean jumps over gap)      |
|                        (player bean)       ⇒ (gate)         |
+-------------------------------------------------------------+
```

SC-01 Hub

```
+-------------------------------------------------------------+
| [▣ Grown-ups]   BouncyMuffin42  ⭐Lv 7   🪙 1,250   [⚙ Settings] |
|                                                             |
|  [👕 Dressing]                                [👥 Friends]   |
|  [🗺️ Season  ]        (img: my bean, idle)    [🎯 Practice]  |
|  [🛍️ Shop    ]                                              |
|                                                             |
|          [ ▶  Shows ]        [  ◉   PLAY    ]  (large)     |
|  ☁️ offline? → online tiles greyed with cloud icon          |
+-------------------------------------------------------------+
```

SC-02 Show Picker

```
+-------------------------------------------------------------+
| [◀️]                  Pick a show                           |
|                                                             |
|  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐       |
|  │(img: show)   │  │(img) LATER   │  │(img) LATER   │       |
|  │ Main Show ◉  │  │   🔒         │  │   🔒         │       |
|  │ 👤x32 🏁🧩👑 │  │              │  │              │       |
|  └──────────────┘  └──────────────┘  └──────────────┘       |
```

```
|                        [ ▶️ Play ]                              |
+-----------------------------------------------------------------+
```

SC-03 Lobby

```
+-----------------------------------------------------------------+
| [✖️ Cancel]                                  👤 18 / 32          |
|                (img: beans gathering)                           |
|  🤖 "Some players may be computer players" (AD-17)              |
|      💡 Tip: (img) Dive to go fast!                             |
| [💬 Quick-chat]   party: (bean)(bean)                           |
+-----------------------------------------------------------------+
```

SC-04 Round intro

```
+-----------------------------------------------------------------+
|                    ROUND 2 of 5                                 |
|        🏁 (type icon)   "Hill Hop"  (name placeholder)          |
|           (img: animated objective — reach the trophy)          |
|                ✓ 20 beans go through                            |
+-----------------------------------------------------------------+
```

SC-05 HUD: see §4.4.

SC-06 Qualified / Out

```
+-----------------------------------------------------------------+
|                                                                 |
|            ✓  QUALIFIED!   (big, bouncy)                        |
|                (img: bean cheering)                             |
+-----------------------------------------------------------------+
+-----------------------------------------------------------------+
|        ★  Out this time - great try!                            |
|               (img: bean dizzy-happy)                           |
|        [ ◉  Watch    ]        [    Home ]                       |
+-----------------------------------------------------------------+
```

SC-07 Spectate

```
+-----------------------------------------------------------------+
| [🏠 Leave]  👁️ Watching: SunnyPickle07 ⭐party   ⏱️1:12  ✓14/20 |
|                                                                 |
|   [◀️]          (3D: followed bean)              [▶️]           |
|                                                                 |
| [▣ Player card]                          [💬 Quick-chat]        |
+-----------------------------------------------------------------+
```

SC-08 Results

```
+-----------------------------------------------------------------+
|              🏆 WINNER: ZippyNoodle55                           |
|             (img: winner celebration)                           |
|   You: Round 4 of 5  ⭐                                         |
|   ⭐ XP  +120 ▓▓▓
```

```
|  [🔒 Parent controls]   [🔒 Delete my data on this device]     |
+-----------------------------------------------------------+
Easy-play tab: Text size [A][A+][A++] · Colour help [Off][Red-green P][Red-green D][Blue-yellow]
· Grab [Hold][Toggle] · Mash [Tap][Hold] · Sound pictures [On] · Shake [On/Low/Off]
· Flashes [On/Low/Off] · Speed lines [On/Low/Off] · Motion blur [Off] · FOV 60°–85° · [▶️ Replay
tutorial]
```

SC-11 Parent gate

```
+-----------------------------------------------------------+
| [◀️]          For grown-ups                                |
|   What is  thirty-four  plus  twenty-seven ?              |
|                                                           |
|            | 6 1 |       [1][2][3]                       |
|                            [4][5][6]                      |
|   attempts ●●○             [7][8][9]                      |
|                            [⌫][0][✓]                      |
+-----------------------------------------------------------+
```

SC-11 Parent dashboard

```
+-----------------------------------------------------------+
| Parent Account ▾ (child: BouncyMuffin42 ▾)   Data as of 10:05 |
| [Activity][Friends (2)][Safety][Controls][Purchases][Data]    |
|  Playtime last 30 days  ▁▃▅▇ ...                          |
|  Friends pending: SunnyPickle07  [Approve] [Decline]      |
|  Controls: Quick-chat [On]  Builder [Off]  Sharing [Off]  |
|            Break reminder [60 min ▾]   Daily limit [Off ▾]|
|            ⓘ Changes apply from the next show              |
|  Data: [View] [Export JSON] [Delete profile]              |
+-----------------------------------------------------------+
```

SC-13 Report (≤ 2 interactions from SC-16)

```
SC-16 Player card                    SC-13 Report reasons (1 tap sends)
+----------------------------+      +----------------------------------+
| (bean) SunnyPickle07       |      | What happened?                   |
| [🔇 Mute]                  |  →   | [😠 Being mean] [🃏 Cheating]    |
| [   Block]  [   Report]    |      | [   Bad name or picture] [   Something upset me] |
+----------------------------+      +----------------------------------+
Confirmation: "Thank you! You did the right thing." · "Tell a grown-up you trust." · auto-block
notice · [✓ OK]
```

SC-14 Error

```
+-----------------------------------------------------------+
|                  (img: bean holding unplugged cable)      |
|                No internet right now.                     |
|     [ 🔁 Try again ◉ ]      [ 🎯 Practice ]               |
|                                       code: NET-01        |
+-----------------------------------------------------------+
```

SC-15 Practice Playground: same HUD as SC-05, with [🏠 Leave] and a "Practice: no stars or coins" badge (FR-01-12 AC2). **SC-17 Friends & Party**

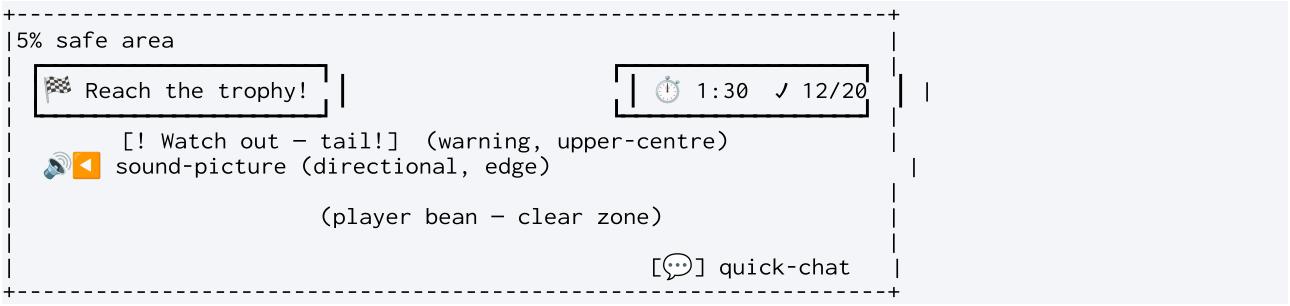
```
+-----------------------------------------------------------+
| [◀️] Friends                                               |
|  (bean) SunnyPickle07   🟢 online   [➕ Invite to party]  |
|  (bean) ZippyNoodle55   ⚪ away                            |
|  ⏳ Waiting for grown-up OK: 1                             |
|  [👥 Add friend → shows friend code; parent approves]      |
|  (no consent yet → padlock card: "Ask a grown-up to turn on |
|   friends")                                               |
+-----------------------------------------------------------+
```

SC-18 Season Path: horizontal track of reward nodes (img), the current step highlighted, ◀️ ▶️ scrolling, and node detail. No expiry countdowns. The season end date appears in Settings → Help only (FR-08-18, 04-UX-I6).
SC-19 Break / limit

```
+-----------------------------------------------------------+
|        (img: bean stretching)   Time for a little break!  |
|                 [   OK   ]                                |
| (limit version: "That's playtime for today. See you soon!") |
+-----------------------------------------------------------+
```

SC-20 Reconnecting: dimmed overlay, spinner bean, "Getting you back in…". After 61 s it shows SC-14 with "Disconnected – you can play again any time" (FR-07-16 AC2). **SC-22 Kids' privacy explainer**: 3 picture cards: "We keep your game safe", "No chatting with strangers", "A grown-up says yes to friends". Shown just in time before adding a friend (FR-11-13).

4.4 HUD layout (SC-05) — FR-08-01, FR-08-02, FR-08-12 AC4



- **Top-left:** objective (icon + ≤ 4 words). **Top-right:** timer and/or qualified count. Both are in the top 50%, inside the 5% safe area, and are tested at 16:9, 21:9 and 4:3 (FR-08-01 AC1). On 21:9 the HUD anchors to a 16:9 centre column.
- **Bean clear zone:** the central 40% × 50% has no HUD.
- **Warning cue:** "!" plus enlarged text, appearing on the next frame and flashing no more than 3 times per second (FR-08-02, NFR-10-19).
- **Team rounds:** a team badge top-centre with colour + shape + pattern (▲ ● ■ ★) (FR-08-11).
- **Sound pictures:** edge-anchored icons for gameplay-relevant audio such as a falling tile or a horn (FR-08-12 AC4, FR-08-08 AC2).
- HUD text is at least 32 px at 1080p (FR-08-13).

4.5 Kid readability rules

#	Rule	Source
R1	**Icon-first:** every action button = icon + word. Icon on the left (mirrored for RTL later).	FR-08-19
R2	**Minimal text:** ≤ 8 words for 95% of strings, with a goal of ≤ 4 on buttons. Year 2 / Grade 1–2 reading level.	FR-08-19 AC1
R3	**Font sizes at 1080p:** HUD ≥ 32 px, body ≥ 28 px, secondary ≥ 24 px, headings 48–72 px. Scale linearly with vertical resolution.	FR-08-13, AD-49
R4	**Typeface:** rounded humanist sans with single-storey a/g, large x-height and wide tracking. No all-caps body text (caps allowed for 1–2 word celebrations). Must support Latin-ext, Cyrillic and CJK fallback. Final family per §5.6 (REC-17).	FR-08-14
R5	**Target sizes:** PC (MVP) clickable targets ≥ 64 × 64 px at 1080p, primary actions ≥ 96 px tall, ≥ 16 px gap. Touch (Later) ≥ 48 × 48 dp, primary ≥ 64 dp. Controller: focus ring ≥ 4 px, 3:1 contrast.	04-UX-I4 (AD-51)
R6	**One primary action per screen**, focused by default and always the protective or safe choice.	FR-08-18 AC3
R7	**Positive wording only;** banned-word lint applies.	FR-08-03
R8	**No urgency:** no countdowns on choices or offers, no "hurry"/"last chance".	FR-08-18
R9	**Pictures explain rules** (round intro, report reasons, privacy).	FR-08-19, FR-11-06, FR-11-13
R10	**Numbers:** use digits and icons (🪙 40) instead of words.	FR-08-19

4.6 Accessibility

Area	Design	Source
Contrast	Text ≥ 4.5:1; large text, icons, hazard outlines and walkable edges ≥ 3:1. Tokens are checked in CI.	FR-08-06

Area	Design	Source
Colour-blind	Never hue alone: team = colour + shape + pattern, drop tiles = colour + crack pattern + icon, rarity = colour + gem shape + word. 3 presets (protan, deutan, tritan) that remap tokens.	FR-08-11
Text size	100 / 125 / 150%. Layouts use auto-layout with wrap, never fixed-width text boxes. 0 clipping at 150% (and at +40% pseudo-loc combined, see 04-UX-I5).	FR-08-12 AC3, FR-08-14 AC2
Reduced motion	Motion blur off by default. Shake, flashes and speed lines each On/Low/Off. FOV 60–85°. "Reduce motion" master switch also shortens UI tweens to fades of 150 ms or less. Flashing never exceeds 3 Hz.	FR-08-20, NFR-10-19
Audio → visual	Sound pictures, captions for UI sounds, mute per channel.	FR-08-12 AC4, FR-08-09
Motor	Grab toggle, hold-to-mash, full rebinding (swap on conflict), no timed UI inputs.	FR-08-12, FR-04-12
Cognitive	Consistent layout, one primary action, replayable tutorial, Practice Playground.	FR-08-17, FR-01-12
Mid-show	Settings overlay without pausing or disconnecting.	FR-08-12 AC2
Screen reader	Not required in MVP. All components expose a label string for future OS TTS (see 04-UX-I7).	—

4.7 Localisation rules (FR-08-14, AD-27)

- **L1:** MVP ships en-GB and en-US. US/UK differences use separate keys (Colour/Color, Favourite/Favorite).
- **L2:** No hard-coded strings. Every key has a context note, max length, screenshot reference and audience (child or adult).
- **L3:** Design for +40% expansion. Buttons grow horizontally first, then wrap to 2 lines, then shrink to the minimum font size, and never truncate. Pseudo-loc build uses accents, brackets and +40%.
- **L4:** Fonts cover Latin-ext, Cyrillic and CJK (glyph fallback chain). Leave line-height headroom for diacritics (≥ 1.3).
- **L5:** Layouts are mirror-ready (start/end, not left/right) for possible RTL later. Directional icons have mirrored variants.
- **L6:** Dates and times show in local time; resets are stored in UTC (AD-33). Use locale number formats (1,250 / 1 250).
- **L7:** The banned-word list, urgency list, quick-chat phrases and name word lists are maintained per language and signed off (FR-08-03, FR-08-18, FR-07-14, FR-11-05).
- **L8:** No text baked into images. Placeholder art must not carry words.
- **L9:** No string concatenation. Use ICU MessageFormat with plurals and gender-neutral wording.

4.8 UI component library spec

4.8.1 Design tokens (original palette, AD-26; final values come once the asset repo lands)

Token	Placeholder	Use
`color.bg.sky`	#E8F6FF	Screen background
`color.surface`	#FFFDF7 (Cloud, §5.2)	Cards and panels
`color.ink`	#1F2140 (Night Ink, §5.2)	Text (≥ 12:1 on surface; 15.3:1 checked)
`color.primary`	#E8336B (Berry, §5.2)	Primary button
`color.secondary`	#6B4BD8 (violet)	Secondary button
`color.accent`	#FFB627 (Sun, §5.2)	Rewards, Coins
`color.success`	#2E9E44 + ✓	Qualified
`color.warn`	#D9480F + !	Warnings
`color.team.{a,b,c,d}`	Coral #FF6B5B ▲, Ocean #2D7DD2 ●, Leaf #3DBB6B ■, Gold #F4C430 ★ (§5.2)	Team identity
`radius.{s,m,l}`	12 / 20 / 32 px	Corners
`space.unit`	8 px	Spacing grid

Token	Placeholder	Use
`type.{hud,body,secondary,h1}`	32 / 28 / 24 / 64 px	Text sizes
`motion.{fast,base}`	150 / 250 ms; reduced = fade only	Animation
`focus.ring`	4 px `color.ink` + 2 px white halo	Focus

Shared colour roles follow the art-direction palette in §5.2 (REC-16); the other tokens are UX placeholders until the asset repository lands.

4.8.2 Components

Component	Variants / states	Rules
BigButton	primary, secondary, quiet; default, focus, pressed, disabled (with reason icon)	Icon + label; ≥ 96 px high; pressed squish 0.95 + click SFX
IconButton	round 64 px	Label shown as tooltip on focus; must be in label table
Tile/Card	show card, item tile (owned ✓, equipped ★, locked 🔒), reward node	Ownership state shown with an icon, never colour alone
TabBar	icon tabs, LB/RB or Q/E to switch	Max 9 tabs; active tab = underline + fill
Slider	0–100, mute toggle	Step 5 (fine step 1 with modifier), applies within 100 ms
Segmented / Toggle	2–4 options	Default = most protective
Counter/Badge	👤 18/32, 🪙, ⭐ level	Digits + icon
HUD panel	objective, progress, warning, team badge, sound picture	Anchors per §4.4
Toast	info only, ≤ 3 s, never during a round	No reward pop-ups during play (FR-08-18)
Modal	confirm, error (SC-14 pattern), break	One primary action + optional Back; reference code slot
PlayerCard	overlay	Mute / Block / Report (no Add friend on a stranger's card, AD-04); ≤ 2 interactions to report or block
QuickChatWheel	8 slots, up to 16 phrases	Icon + phrase; rate-limited; hidden from blocked players
NumberPad	parent gate only	Digits only, not a text keyboard (04-UX-I2)
BeanPreview	idle, emote, celebration, rotate	360° rotation with mouse, stick or keys
ProgressTrack	XP bar, season path	No expiry timers
EmptyState	icon + 1 line + 1 action	e.g. Purchases "Spending is off"
FocusRing	global	Always visible on controller and keyboard

Every component ships with: a token-only style, an a11y label key, 150% text and +40% loc stories, colour-blind preset stories, and reduced-motion behaviour.

5 Art Direction and Asset Pipeline

Source: `05-art-direction-assets.md` (Proposal v0, framework only (23 Oct 2026)). **Authority:** Visual direction and asset pipeline; yields to Section 3 on physical scale.

Scope rule (UD-3): no specific characters are designed here. All character/costume imagery comes from the user's GitHub asset repository. This section defines the *frame* those assets must fit: pillars, rules, slots, names, budgets, rig and import checks. Final values are ratified at M2 against the delivered repository (FR-12-10 AC1). Hex values are proposals.

5.1 Visual pillars

#	Pillar	Meaning	Trace
P1	**Soft & safe**	Everything round, squishy, bouncy; nothing sharp, bloody or threatening. Failure = comic, never pain.	UD-4, AD-24
P2	**Read in 0.5 s**	A 6-year-old must tell safe / hazard / goal at a glance, at 720p, in a 32-bean crowd.	FR-08-11, AD-05
P3	**Toybox physical**	Worlds look like giant play-sets: plastic, foam, rubber, felt. Materials explain physics (bouncy looks bouncy).	FR-05-01
P4	**Characters own the colour**	Environments are calmer than beans so repository costumes always pop.	UD-3
P5	**Distinctly ours**	No tracing, recolouring or layout-copying of the reference game; own palette, shapes, UI.	UD-2, AD-26, FR-12-08

5.2 Colour palette rules (hex = proposals)

Value hierarchy (hard rule): goal/UI > hazards > beans > interactive safe surfaces > background. Background saturation ≤ 40%, beans/hazards ≥ 60% (HSV).

Role	Name	Hex	Notes
Safe surface base	Oat	`#F2E6C9`	Warm, low-sat walkable
Safe surface alt	Mint Foam	`#BFE8D2`	Tiles / platforms
Safe edge trim	Cocoa	`#6B4A3A`	Platform edge outline band
Hazard primary	Zing Orange	`#FF7A1A`	Always + stripe pattern
Hazard secondary	Plum Pop	`#7A2E8E`	Always + dot pattern (CVD-safe vs orange)
Warning / about-to-drop	Lemon	`#FFD93B`	+ "!" icon / wobble anim
Goal / finish / checkpoint	Sky Cyan	`#22C3E6`	+ chevron/flag icon
Bouncy / helpful	Bubblegum Lime	`#9BE15D`	Jump pads, boosts
Slime / out-of-bounds	Goo Teal	`#2F8F8A`	Low value, glossy
UI ink	Night Ink	`#1F2140`	Text & outlines (never pure black)
UI paper	Cloud	`#FFFDF7`	Panels
UI accent	Berry	`#E8336B`	Primary buttons
UI accent 2	Sun	`#FFB627`	Rewards/coins
Team A–D	Coral / Ocean / Leaf / Gold	`#FF6B5B` `#2D7DD2` `#3DBB6B` `#F4C430`	Each paired with pattern + icon (FR-08-11)

Rules:

1. **Never hue alone** — every hazard, team, tile state and rarity has colour + pattern + icon/shape (FR-08-11). CVD presets (protan/deutan/tritan) remap these roles only.
2. Reserve Zing Orange + stripes **exclusively** for hazards; no costume-trim of environment may use hazard stripe pattern.
3. Contrast: UI text ≥ 4.5:1; hazard vs adjacent safe surface $\Delta L^* \geq 30$.

4. Rarity labels use text + frame shape, colours: Common `#B8C0CC`, Uncommon `#3DBB6B`, Rare `#2D7DD2`, Epic `#9B5DE5`, Legendary `#FFB627`.
5. AD-26: palette deliberately avoids the reference game's signature pink/blue/yellow UI combo; Berry+Night Ink+Cloud is the brand trio.

5.3 Shape language

Shape	Meaning	Use
Circle / pill / blob	Friendly, safe, player	Beans, platforms, UI buttons, bumpers
Rounded square	Stable, standable	Floors, tiles, crates
Rounded triangle / wedge	Motion, direction, caution	Ramps, chevrons, spinners, hazard flags
Wavy line	Slippery / slime / water	Goo, ice, conveyors

Rules: min bevel radius 5 cm (no hard corners in collision-visible geometry); spikes forbidden — "threat" shown via rotating/punching foam shapes; silhouettes testable as black fills (hazards identifiable by silhouette + motion alone).

5.4 Lighting & mood per world theme (original themes — not reference seasons)

World theme	Key light	Ambient / fog	Mood	Hazard read note
Toybox Meadow (default/MVP)	Warm sun 5500K, 40°	Sky blue #BFE6FF, light fog	Sunny Saturday	Baseline palette
Candy Workshop	Soft 4500K + pastel rim	Peach #FFE1CC	Sweet, busy	Keep hazards orange; candy props pastel only
Frosty Fun-Park	Cool 7000K, low sun	Lavender-white #E9ECFF	Crisp, sparkly	Ice = Goo Teal sheen + wavy decal
Starlight Playground (night)	Moon 8000K + emissive props	Deep #2A2F5C	Cosy glow, never scary	Hazards emissive; min scene exposure EV kept ≥ day −1.5
Splash Jungle	5000K dappled	Green-blue #CDEFE3	Lively	Foliage desaturated ≤ 35%

Rules: one key + sky ambient + baked GI; no horror lighting, no red full-screen tints; light flicker/flash ≤ 3/s (NFR-10-19); bean rim light always on for crowd separation.

5.5 Readability rules (hazards vs safe)

1. Safe = matte, low-sat, rounded-square, edge trim band in Cocoa.
2. Hazard = saturated Zing/Plum + stripe/dot pattern + motion tell (wind-up ≥ 0.4 s).
3. Warning state (tile about to drop) = Lemon + "!" decal + wobble, ≤ 3 flashes/s (NFR-10-19, FR-08-11).
4. Goal = Sky Cyan + chevrons pointing direction of travel.
5. Out-of-bounds = Goo Teal, low value, soft caustics.
6. Beans must never be shaded like a hazard; repository costume review rejects hazard-stripe dominant designs (FR-12-08).
7. Test: greyscale + 3 CVD sims at 720p, 32 beans — testers ID safe/hazard 100% of 20 screenshots (mirrors FR-08-11 AC1).
8. Camera accessibility: shake/flash/speed-lines toggleable; motion blur off by default (FR-08-12).

5.6 UI art style

- Chunky, rounded "sticker" panels: Cloud fill, 4 px Night Ink outline @1080p, 6 px soft drop shadow.
- Typeface: rounded geometric sans, open licence (e.g. Nunito/Baloo family — final pick pending licence check, FR-12-07); min 28 px body @1080p for 6-year-olds; icon+text for all critical actions (pre-readers).
- Icons: 2-tone flat, 128 px grid, 8 px corner radius, Night Ink outline.
- Motion: squash/stretch on press (100–150 ms), no fast flashing.

- Source PNG/SVG at 2× 1080p (FR-12-03).

5.7 VFX style

- Stylised, chunky, hand-shaped particles (stars, puffs, confetti, bubbles); no blood, smoke-as-fire, or explosions with debris (AD-24).
- Elimination = "poof" cloud + confetti; impact = cartoon star ring.
- Max 3 flashes/s (NFR-10-19); full-screen effects ≤ 20% coverage; reduce-effects toggle (FR-08-12).
- Budget: ≤ 64 particles per effect, ≤ 2,000 live particles per scene, ≤ 1 overdraw layer >50% screen.

5.8 Asset manifest / slot list

All entries live in `assets/manifest.json` (FR-12-01). Placeholders = grey bean + "PH" texture (FR-12-06).

Category (FR-12-02)	Slot	Part(s)	Count MVP (proposal)	Source	Notes
`body`	Base bean	`full`	1	Pipeline	Shared rig owner
`upper`	Costume top	`top,thumb`	20	**User repo**	Pelvis & above
`lower`	Costume bottom	`bottom,thumb`	20	**User repo**	Pelvis & below
`face`	Faceplate	`mask,thumb`	10	User repo	`hidesFace` from upper (FR-05-05)
`pattern`	Body pattern	`mask,thumb`	15	User repo (2D OK)	Tileable
`colour`	Colour swatch	`thumb`	24	Data + icon	
`emote/celeb`	Animations	`full`	8 / 4	Anim on base rig	
`banner`	Nameplate art	`full`	10	User repo (2D OK)	
`obstacle`	Spinner, bumper, door, tile, pad, conveyor…	`full,lod`	~25	Pipeline	Hazard palette rules
`level`	Env kits per theme	`full`	5 themes	Pipeline	
`icon/ui`	HUD, menus, rarity, team	`full`	~120	Pipeline	
`vfx`	Effects	`full`	~30	Pipeline	
`sfx/music`	Audio	—	(see Section 6)	—	

Naming (FR-12-02 + proposed extensions)

`wb_<category>_<setid>_<part>_<variant>_v<nn>.<ext>` — lowercase ASCII, `[a-z0-9]`, `setid` = `[a-z]+[0-9]{2}`. Proposed suffix extensions (Issue 05-I-2): textures `…_v03_alb.png` / `_nrm` / `_msk` (R=AO, G=rough, B=pattern mask, A=emissive); LODs `…_v03_lod1.glb`. Examples: `wb_upper_set01_top_a_v01.glb`, `wb_lower_set01_bottom_a_v01.glb`, `wb_upper_set01_thumb_a_v01.png`, `wb_obstacle_spin01_full_std_v02.glb`.

Formats (FR-12-03)

3D `.glb` (FBX source OK); textures PNG (sRGB albedo, linear data); UI PNG-alpha/SVG at 2×; anim glTF/FBX on base rig; audio WAV 48k/24 → OGG q6. 2D-only repo images: direct use for UI/banners/patterns/icons; costumes require 3D modelling with the 2D as reference sheet (Q-D2, DEP-11).

Budgets

Asset	LOD0 / LOD1 / LOD2 tris	Textures	Bones	Trace
Costume part (upper or lower)	6,000 / 2,500 / 800	1× 1024^2 set (alb, nrm, msk)	≤ 8 accessory leaf	FR-12-05
Base bean	4,000 / 1,800 / 600	1024^2	base rig	proposal
Faceplate	500	512^2	0 (blend shapes)	proposal

Asset	LOD0 / LOD1 / LOD2 tris	Textures	Bones	Trace
Obstacle (large)	15,000 / 6,000 / 2,000	2048^2 source, 1024^2 runtime	≤ 16	proposal
Obstacle (small)	3,000 / 1,200 / 400	512^2	≤ 4	proposal
Level scene total	≤ 1.5 M visible tris incl. 32 beans	streaming ≤ 600 MB VRAM	—	NFR-10-04
UI icon	—	256^2 source (2×)	—	FR-12-03

Worst case beans: 32 × 2 parts × 6k = 384k tris → must pass 32-bean test at 60 fps 1080p on GTX 960 class (FR-12-05 AC2, NFR-10-04). Distance LODs: LOD1 > 8 m, LOD2 > 20 m (proposal).

Rig / skeleton requirements (FR-12-04)

- Units metres, Y-up, +Z forward, pivot at feet centre. Scale: the bean is authored to the collision capsule of gapfill-A (≈ 1.6 m tall, r 0.5 m; §3.3.1) — physics authority, REC-10; this supersedes the 1.00 m proposal of Issue 05-I-4.
- Bones (layer masks and ragdoll bodies: §3.3.1): `root > pelvis > spine_01 > spine_02 > head`; arms `clavicle/upperarm/lowerarm/hand_{l,r}`; legs `thigh/calf/foot_{l,r}`. Names fixed, case-exact.
- Face: blend shapes `eye_blink_l/r`, `eye_happy`, `eye_sad`, `mouth_open`, `mouth_smile` (decision for Issue 05-I-5).
- Waist seam: fixed edge loop from `seam_template.glb`; seam verts match position & weights exactly; ≤ 2 mm gap in 5 test poses.
- Upper skinned pelvis+above; lower pelvis+below; ≤ 4 influences/vertex; accessory bones prefixed `acc_`, leaf only, ≤ 8/part.
- No change to collision capsule (FR-05-01, FR-12-04 AC3).

5.9 Import checklist for the user's GitHub repo

1. ☐ Answer Q-D2: repo is 2D-only, 3D, or mixed? (drives modelling schedule, DEP-11)
2. ☐ Folder per category mirroring FR-12-02; files renamed to pattern; lint = 0 errors.
3. ☐ Manifest entry per asset with `id, category, set_id, part, variant, version, files[] (+sha256), rarity_label, hidesFace, traits, budget, provenance, review, placeholder_of`.
4. ☐ Provenance: source path/commit SHA, author, licence/ownership, AI tool use + which, date (FR-12-07). Unclear rights → blocked.
5. ☐ Formats valid (FR-12-03); textures power-of-two, correct colour space.
6. ☐ Budgets pass (tris per LOD, texture size, bones) (FR-12-05).
7. ☐ Rig: base skeleton unchanged, seam test passes all upper×lower pairs (FR-12-04).
8. ☐ Every top has a compatible bottom (full outfits delivered as pairs).
9. ☐ Readability: no hazard stripe/orange dominance; greyscale & CVD check (FR-08-11).
10. ☐ Content review: PEGI 3/ESRB E, UD-4 tone & inclusivity, originality vs reference game (FR-12-08); AI assets carry no-likeness note.
11. ☐ Thumbnails generated (`thumb` part, 512^2 PNG alpha).
12. ☐ Run `wb-asset-validate` (FR-12-09): 100 assets < 5 min, report with rule IDs; fix and re-run.
13. ☐ Release gate: no shipped slot has `placeholder_of` (FR-12-06 AC2). Suggested repo layout: `/characters/{upper,lower,face,pattern}/`, `/banners/`, `/icons/`, `/source/` (FBX/PSD refs), `LICENSE_ASSETS.md`.

6 Audio Design

Source: `06-audio.md` (Draft v1 (2026-10-23)). **Authority:** Audio, including the authoritative audio event list (FR-08-08 AC1).

Fall Guys is a genre reference only (UD-2, AD-25, AD-38). No reference music, SFX, stems or melodies are used, imitated or sampled.

6.1 Music direction (FR-08-07, UD-4)

- **Genre:** original, upbeat, toy-like electro-pop. **No lyrics, no sung words, no vocal chops that sound like words.**
- **Palette:** toy xylophone, kazoo, ukulele, squeaky synth leads, bouncy sub bass, hand-claps, slide whistle, boing springs, toy-box and household percussion (recorded new by us). Major and Mixolydian keys; 120–140 BPM for rounds, 95–110 BPM for menus.
- **Never:** minor-key tension horror, dissonant stingers, sirens, alarms, gun/explosion hits, war drums, laughter aimed at a player, anything "scary" (AD-24).
- **Originality guardrails:** no reference tracks in briefs except as written genre words; composers sign an originality statement; every track gets a melody-similarity check against the reference OST before sign-off (Issue 06-A-09). Every cue has a provenance record (FR-12-07) and content review (FR-12-08). AI-assisted music is allowed only with the AI-use flag and full rights.
- **Structure:** every round track is built from **horizontal sections + vertical stems** (drums, bass, harmony, lead, fun-layer) so the adaptive system in §6.2 can add or strip layers without hard cuts.

6.2 Adaptive music states per show phase (FR-01, FR-08-04, FR-08-07 AC1)

6.2.1 State machine

State	Screen	Music	Transition in	Notes
M0 Menu	SC-01, SC-02, SC-09, SC-10	`mus_menu` loop, light (drums+bass+harmony)	2 s crossfade	Dressing Room adds "fun-layer". Settings ducks music −6 dB. No urgency, no countdown music in the shop (UD-1 no-dark-patterns).
M1 Lobby	SC-03	`mus_lobby` loop, 100 BPM	on Play pressed, bar-synced	Layers add as lobby fills (25/50/75/100%). Never speeds up to pressure the child.
M2 Round intro	SC-04	`stg_roundintro_<type>` (Race/Survival/Team/Hunt/Final) ≤4 s	hard on bar 1	Type stinger = auditory twin of the round-type icon.
M3 Countdown	SC-05 pre-start	3-2-1-GO beats locked to the 1.0 s beat	music on downbeat of GO	Audio sync per countdown table (3.0 s).
M4 Round – play	SC-05	`mus_round_<theme>` intensity **L1**	on GO	Layers L1→L3 driven by intensity params below.
M5 Round – final seconds	SC-05	intensity **L3** + `fx_lastseconds` layer	at 15 s solo / 30 s team	Tempo is **not** raised; urgency comes from extra percussion only.
M6 Player Qualified	SC-06	`stg_qualified` (≤3 s), round music ducks −9 dB	instant	Player then hears round music filtered (low-pass 2 kHz) while others play.
M7 Player Out	SC-06 → SC-07	`stg_out_greattry` (≤3 s, gentle, major)	instant	Then M8.
M8 Spectate	SC-07	round music, low-pass + −6 dB	1 s	Follows spectate target's round; no "sad" music.
M9 Final	SC-05 (Final round)	`mus_final_<theme>`, distinct track	stinger `stg_final`	FR-08-07 AC1 "Final" cue.
M10 Victory / results	SC-08	`mus_victory` (winner) or `mus_results` (everyone else, equally cheerful)	0.5 s	Rewards jingles layer over (§6.3.5). Returns to M0 only when player presses "Play again" or Back (FR-08-04 AC2).

State	Screen	Music	Transition in	Notes
M11 Tutorial / Practice	SC-12, SC-15	`mus_practice` loop, calm	2 s	Offline; all assets local.
M12 Error / offline	SC-14	music continues at −6 dB, one soft `ui_error`	—	Never a harsh buzzer.
M13 Parent area	SC-11	music −12 dB	—	Adult area; neutral.

6.2.2 Intensity parameters (round tracks)

Param	Source	Effect
`round_progress` 0–1	timer or qualified count	L1 → L2 at 0.5
`qualify_pressure`	slots left ≤ 20%	L2 → L3
`last_seconds`	FR-08 final-seconds banner	L3 + `fx_lastseconds`
`slime_height`	FR-03-14	adds bubbly bass layer (fun, not scary)
`team_lead`	FR-01-06 (R2)	adds team-colour motif on own team lead

Rules: stems change only on bar lines; max one layer change per 2 bars; no tempo change mid-round.

6.2.3 Builder music (FR-09 level settings)

Builder levels choose a track from the **Wobblebeans original music library** (all FR-08-07 compliant). See Issue 06-A-03.

6.3 Audio event list (FR-08-08 AC1 — this is the authoritative list)

Columns: **ID** · trigger · sound brief · bus · 3D? · variations · **GR** = gameplay-relevant (needs visual indicator, FR-08-12 AC4) · visual indicator.

6.3.1 Character states (FR-04-01…08, FR-03-03, FR-03-13)

ID	Trigger	Brief	Bus	3D	Var	GR	Visual
`sfx_char_footstep_<surface>`	foot contact (default, metal, wood, slime, ice R2, water R2)	soft squishy toe-taps	SFX_Char	3D	6	–	–
`sfx_char_jump`	FR-04-02 take-off	rubbery "bwip"	SFX_Char	3D	4	–	animation
`sfx_char_land_soft/hard`	landing by fall speed	squash "pomf"	SFX_Char	3D	4	–	squash VFX
`sfx_char_dive`	FR-04-03	whoosh + belly slide	SFX_Char	3D	3	–	–
`sfx_char_dive_slide`	ground slide loop	rubber squeak loop	SFX_Char	3D	1 loop	–	–
`sfx_char_grab_start/hold/release`	FR-04-04	stretchy "gloop" / grip squeak	SFX_Char	3D	3	Y (being grabbed)	grab-link VFX + "hold" icon on grabbed player
`sfx_char_breakfree`	escape grab (incl. hold-to-mash alt)	pop	SFX_Char	3D	2	–	pop VFX
`sfx_char_bump`	FR-04-07 player collision	bonk on beanbag	SFX_Char	3D	5	–	–
`sfx_char_stumble`	FR-04-05 stumble	wobble-wobble	SFX_Char	3D	3	–	stars over head
`sfx_char_ragdoll`	ragdoll enter	jelly flop (no pain sounds)	SFX_Char	3D	4	–	animation
`sfx_char_recover`	get-up	"boing-up"	SFX_Char	3D	2	–	–
`sfx_char_fall_off map`	FR-03-13 fall-off	descending slide whistle	SFX_Char	2D own / 3D others	3	Y	fall vignette + respawn ring

ID	Trigger	Brief	Bus	3D	Var	GR	Visual
sfx_char_respawn	respawn + 1.0 s protection	sparkly "pop-in"	SFX_Char	3D	1	Y	shimmer outline during protection
sfx_char_oob_warn	soft OOB 3 s warning	gentle double-chime	UI_Gameplay	2D	1	Y	"!" banner (FR-08-02)
sfx_char_slime_out	slime elimination	bubbly "blub"	SFX_Char	3D	2	Y	slime splash + SC-06
sfx_char_emote_<id>	FR-05-06 emote	non-verbal toy sound per emote	SFX_Social	3D	1–3	–	emote animation
sfx_char_quickchat	FR-07-14	soft "pip"; no voice	SFX_Social	2D	1	–	quick-chat bubble
sfx_char_celeb_<id>	victory celebration (AD-35)	jingle	SFX_Social	2D	1	–	animation

6.3.2 Obstacles (FR-03; names from Appendix C)

⚑ *Merge note (MRG-02):* this table follows the full obstacle catalogue names and has no events yet for eight MVP kit pieces of §2.9 (Swing Ball, Spin Bar, Spin Mallet, Tilt Plank, Conveyor, Bumper, Pusher Block, Launch Arm). They must be added; none are invented here.

All obstacles: periodic hazards emit a **tell** sound ≥0.5 s before impact phase (FR-03-17 telegraphing), audible within 25 m, rolloff linear 5–25 m. Every tell is GR and has its visual twin (wind-up animation/glow).

ID	Obstacle (Req)	Sounds	GR / Visual
sfx_obs_turngate	Turn Gate (FR-03-02)	creaky toy-turnstile loop	–
sfx_obs_windmill	Windmill Paddles (FR-03-02)	whirring loop, paddle swoosh	–
sfx_obs_swinglog	Swinging Log (FR-03-02)	rope creak tell + whoomph (soft)	Y / wind-up highlight
sfx_obs_rolyroller	Roly Roller	rumble loop + bonk on hit	–
sfx_obs_boingglove	Boing Glove (FR-03-03)	spring-wind tell, "BOING" hit	Y / glove wobble 0.5 s
sfx_obs_flippad	Flip Pad (FR-03-08, R2)	click arm 0.25 s, fling	Y / pad glow
sfx_obs_button	Button (FR-03-08)	big toy click	Y / button light
sfx_obs_balllauncher	Ball Launcher (FR-03-09, R2)	"fwump" soft-ball pop (no cannon)	Y / muzzle puff + ground target ring
sfx_obs_fruit	Fruit projectile	squishy splat	–
sfx_obs_bouncepad	Bounce Pad (FR-03-10)	drum "bwoing" (pitch by force)	–
sfx_obs_zoomarch	Zoom Arch (FR-03-04)	rising sparkle	–
sfx_obs_faketile_reveal	Fake Tile (FR-03-06)	harmless "poof"	Y / tile puff
sfx_obs_trickdoor	Trick doors (FR-03-06)	real: "thunk"; fake: rubbery boing	–
sfx_obs_droptile_warn/drop	Drop tiles (FR-03-07)	wobble-rattle warn, "plop" drop	Y / tile shake + colour **and pattern** change (FR-08-11)
sfx_obs_fallingplatform	Falling Platform	creak warn, whoosh	Y / shake
sfx_obs_spinplate	Spin Plate	turntable hum loop	–
sfx_obs_hover	Hover Platform	soft hover hum	–
sfx_obs_bubblewall	Bubble Wall	bubble boop on contact	–
sfx_obs_puffballoon	Puff Balloon	inflate squeak tell, deflate	Y / inflate anim
sfx_obs_pinball	Pinball Paddles	flipper clack	Y / paddle glow
sfx_obs_lightbar	Light Bar	soft hum (no laser zap)	–
sfx_obs_windfan	Wind Fan / Floaty Zone (FR-03-11, R2)	breeze loop / dreamy pad	Y / wind particles
sfx_obs_whooshtube / sfx_obs_portal	Whoosh Tube / Portal	suck-whoosh / warble	–

ID	Obstacle (Req)	Sounds	GR / Visual
sfx_obs_gates_logs	Gates and Logs	wooden clonk	–
sfx_obs_ai_penguin/rhino	AI hazards	cute waddle / snort (no roar)	Y / footprints near player
sfx_obs_checkpoint	Checkpoint (FR-03-12 — sound cue required)	bright two-note chime	Y / flag flip + glow (already required)
sfx_obs_slime_rise	Rising slime (FR-03-14)	bubbly loop, step-up gurgle each rise	Y / slime line marker on HUD
amb_<theme>	Themes (FR-03-16)	crowd-free cheerful ambience beds	–

6.3.3 Round & show events (FR-01)

ID	Trigger	GR / Visual
stg_roundintro_<type>	SC-04	Y / type icon
sfx_round_count_3/2/1/go	3-2-1-GO	Y / on-screen numerals
sfx_round_lastseconds_tick	final 15/30 s, soft tick on each 5 s	Y / final-seconds banner
sfx_round_qualified_other	someone qualifies (quiet, throttled 1 per 0.5 s)	Y / qualified counter tick
sfx_round_end_whistle	round end	Y / "Round over" banner
sfx_team_score_<own/other>	team point (R2)	Y / score pop
sfx_hunt_tail_lost	tail taken (warning, FR-08-02)	Y / "!" icon

6.3.4 Warnings (FR-08-02)

sfx_ui_warning — friendly two-note "uh-oh" (pitched toy), max 1 per 1.5 s; paired visual "!" which never flashes >3/s (NFR-10-19).

6.3.5 Outcomes and rewards (FR-01-04, FR-01-09, FR-08-03)

ID	Rule
stg_qualified	bright fanfare ≤3 s
stg_out_greattry	**warm, major-key**, ≤3 s; no "wah-wah" trombone, no crowd boo or laugh; pairs with "Out this time – great try!"
stg_final, mus_victory, mus_results	results music is cheerful for every player
sfx_reward_xp/coins/item	sparkle jingles; **no slot-machine, coin-cascade or gambling sounds** (FR-11-18)

6.3.6 UI (SC-01…SC-15)

ui_nav_move, ui_confirm, ui_back, ui_toggle, ui_slider_tick (plays at new level so the child hears the change), ui_tab, ui_error (soft), ui_purchase_confirm (neutral, not celebratory — no reward-loop audio in the shop), ui_parentgate_open, ui_report_sent (reassuring), ui_mute_player. All UI sounds are 2D on the UI bus. **Block/mute (FR-07-15):** a blocked or muted player's emote, quick-chat and celebration sounds are not played for the blocker.

6.4 Mixing buses

```
Master (limiter -1 dBTP, kid-safe cap §6.6)
├── Music        (slider: Music)   ── stems, stingers
├── SFX          (slider: SFX)
│   ├── SFX_Char      own bean +2 dB priority
│   ├── SFX_Obstacle  tells never ducked
│   ├── SFX_Social    emotes/quick-chat, -3 dB, voice-limit 4
│   └── Ambience
└── UI           (slider: UI)
    ├── UI_Menu
    └── UI_Gameplay   countdown, warnings, qualified (never ducked)
```

Sidechain/ducking: UI_Gameplay and stingers duck Music −6 dB (attack 20 ms, release 400 ms). SFX_Obstacle tells duck SFX_Social −4 dB. Voice limits: 32 total real voices; per-obstacle-type 4; footsteps 8 (nearest). Priority:

own bean > tells > UI_Gameplay > others. Ambience follows the SFX slider (FR-08-09 lists only 4 sliders; Issue 06-A-05).

6.5 Loudness targets

Item	Target	Req
Music tracks (mastered)	**−16 LUFS ±1 integrated**, ≤ −1 dBTP	FR-08-07 AC3
Full in-game mix (all sliders 100)	−16 LUFS ±2 integrated over a 3-min round capture; short-term max −10 LUFS	added (06-A-02)
Stingers	−14 LUFS short-term peak, ≤ −1 dBTP	added
SFX one-shots (source)	peak −3 dBFS, normalised by category table in middleware	added
UI	−20 to −18 LUFS short-term	added
Dynamic range	loudness range (LRA) ≤ 10 LU for the full mix (TV + handheld friendly)	added
Low frequency	high-pass 30 Hz on master; no sub-bass hits > −12 dBFS	added

Measured with ITU-R BS.1770-4 / EBU R128 meter (T-FR-08-07).

6.6 Kid-safe volume (UD-1, FR-08-09)

- **Defaults:** Master 70, Music 60, SFX 80, UI 70 (persisted per profile; FR-08-09 AC1).
- **Hard cap:** master limiter at −1 dBTP and no single event over −10 LUFS short-term; no sudden level jumps >6 LU between consecutive 400 ms windows except the GO beat.
- **No startle:** no sounds louder than the music bed on fade-in of any screen; first launch fades audio in over 2 s.
- **Headphones mode (Settings toggle, default ON on handheld):** additional −6 dB master trim and gentle compressor (target ≤ 80 dB SPL guidance per WHO/ITU-T H.870 safe listening for children). See Issue 06-A-06.
- **Sliders:** 0–100, perceptual (dB) curve, 0 = full mute, effect ≤100 ms; slider tick sound plays at the new level.
- **Background/focus loss:** mute all on app background or controller disconnect (FR-04-15 pause screen).
- **Parent area (SC-11):** optional "max volume" setting that clamps Master (R2, Issue 06-A-06).

6.7 Accessibility — visual cues for audio (FR-08-12 AC4, AD-45, FR-08-08 AC2)

- **Definition of gameplay-relevant (GR):** any sound that tells the player about a threat, a timer, their own state, or a round outcome. All rows marked **Y** in §6.3 are GR (Issue 06-A-04).
- **Rule:** every GR sound has a visual twin that exists **whether or not** audio is on (not only at slider 0). An automated check in CI cross-references event IDs marked GR with a `visual_cue_id` in the manifest.
- **Directional indicator:** "Sound Ring" option (default off, on automatically when Master or SFX = 0, per the Ch. 8 §4 edge case): an arc at screen edge points to off-screen GR sounds (tells, projectiles, AI hazards) within 25 m. Icons + shape, never colour alone (FR-08-11). Pulses ≤3 Hz (NFR-10-19).
- **No captions needed for speech** (there is none, §6.8); a "sound icons" option shows small icons for emote sounds from others.
- **Haptics (where platform allows):** rumble on countdown GO, grab, qualified; reduce/off with motion-comfort settings (FR-08-20).
- **Mono audio** toggle and L/R balance for single-sided hearing.

6.8 VO policy — none / sound-only (AD-53, AD-02, FR-11-03, FR-08-17)

- **No voice-over anywhere:** no presenter, no narrator, no tutorial VO, no announcer, no spoken countdown.

- **No player voice** (chat or recorded) anywhere.
- **Bean vocalisations:** allowed only as **non-lexical toy sounds** (squeaks, hums, "hup!", giggles made with instruments or heavily processed so they carry no words in any language). Reviewed for accidental words in en-GB/en-US and target-launch languages (FR-08-14).
- No lyrics in music (FR-08-07). Licensed or UGC audio is not allowed in Builder levels (players pick from the library only).
- Benefit: no voice localisation cost (AD-53) and no voice moderation (AD-02).

6.9 Asset naming and format spec (FR-12-01/02/03/07/08)

6.9.1 Naming (FR-12-02 pattern, audio parts extension — Issue 06-A-07)

`wb_<category>_<setid>_<part>_<variant>_v<nn>.<ext>`

- `category`: `sfx` or `music` (existing FR-12-02 categories). Stingers and ambience use `music` and `sfx` respectively.
- `setid`: event group + index, e.g. `charjump01`, `obsboingglove01`, `roundforest01`.
- `part` (audio): `full` (one-shot / whole track, existing), plus proposed audio parts `loop`, `intro`, `outro`, `stem`.
- `variant`: `a`…`f` for random variations; for stems `drums|bass|harm|lead|fun`; for surfaces `default|metal|wood|slime|ice|water`.
- Examples: `wb_sfx_charjump01_full_a_v01.wav`, `wb_sfx_charfootstep01_full_slime_v02.wav`, `wb_music_roundforest01_stem_drums_v03.wav`, `wb_music_menu01_loop_main_v01.wav`.
- Runtime event IDs (middleware) use the short form in §6.3 (`sfx_char_jump`) mapped to files in the manifest.

6.9.2 Format (FR-12-03)

Item	Source	Runtime
All audio	WAV 48 kHz, 24-bit PCM	OGG Vorbis q6 (FR-12-03); platform-native codec allowed if the platform requires it (Issue 06-A-08)
Channels	SFX 3D: mono; UI/stingers: stereo; music: stereo stems	—
Loops	sample-accurate loop points, 10 ms zero-cross; BPM + bars in manifest	—
Head/tail	≤10 ms silence head; tails ≥ natural decay	—
Length caps	SFX ≤3 s (loops excepted), stingers ≤4 s, music loops 60–180 s	—
Memory	SFX bank ≤ 40 MB resident; music streamed	—

6.9.3 Manifest fields (FR-12-01 + audio)

`id, file, category, bus, is3d, rolloff_min/max, variations, loop, bpm, bars, lufs_measured, truepeak, gameplay_relevant, visual_cue_id, provenance (FR-12-07: author, licence, AI-use flag), review (FR-12-08), placeholder_of`. CI: name lint (FR-12-02 AC1), manifest schema (FR-12-01 AC1), loudness check (FR-08-07 AC3), event coverage (FR-08-08 AC1), GR→visual (FR-08-12 AC4).

6.10 Verification

Test	Covers
T-FR-08-07	loudness meter on all music; provenance audit; similarity check
T-FR-08-08	every event in §6.3 resolves to an asset; every GR has `visual_cue_id`
T-FR-08-09	slider range, ≤100 ms, persistence
T-FR-08-12	play a show with Master 0: all GR cues visible
T-AUD-01 (new)	full-mix capture −16 LUFS ±2, no window jump >6 LU
T-AUD-02 (new)	child-panel (6–8, n≥12): no sound rated "scary" or "sad" by >10%

Test	Covers
T-AUD-03 (new)	vocalisation word-check across launch languages

7 Progression, Rewards and Customisation

Source: `07-progression-rewards.md` (Systems/economy design (23 Oct 2026)). **Authority:** Economy and progression values; yields to Section 1 on game rules.

MVP is fully free: one earned currency (Coins), no real money, no premium currency, no loot boxes or chance rewards, no FOMO (AD-12…AD-16, FR-08-18, FR-11-09). Every number here is a **design estimate (not confirmed from Fall Guys)** and lives in config. Art for cosmetic slots comes from the user's asset repo later (FR-12-01/FR-12-04); this section only defines slots and data.

7.0 Design pillars

#	Pillar	Trace
P1	Everyone progresses: playing (not winning) moves you forward; a quit forfeits only the participation XP — per-round XP and Coins already earned are kept (01-ISS-01, REC-02)	FR-06-01, FR-01-09, FR-06-13 AC2
P2	Nothing is ever lost by not playing: banked dailies, rested XP, auto-granted season rewards	AD-15, FR-06-02 AC2, FR-06-13 AC3
P3	Deterministic & transparent: no chance anywhere, every item shows "How to get it"	AD-12, FR-05-09, FR-05-10, FR-05-17 AC1
P4	Cosmetic only, zero gameplay effect	FR-05-01
P5	No pressure UI: no countdowns, streaks, loss framing, push/marketing notifications	FR-08-18, FR-11-19, gapfill-C §13
P6	Healthy play: break reminders and parent limits end at show boundaries	FR-11-15, AD-23

7.1 XP & player level (FR-06-01, gapfill-C §1)

Source	Default XP	Range	Note
Show participation (not quit)	100	50–150	Eliminated players keep it (FR-01-09 AC1)
Per round qualified	150	100–250	
Reached Final	+200	100–300	
Win (Trophy)	+500	300–800	Never required by any challenge
Team-round win (R2)	+50	0–100	
Rested XP	+1,000 banked per full UTC day away, cap 3,000; doubles show XP until used	0–1,500/ day	Welcome-back, not a streak (AD-15)
Daily / weekly / season challenges	1,500 / 4,000 / 10,000	see §7.6	50–65% of path XP
Private Shows / Practice Playground	0	—	Anti-boost (AD-19, FR-01 offline AC2)
Level Hop (Could, Later)	80/level, soft cap 2,400/day then 50%		≤35% of total XP
XP boost event (Could, R2)	×2 announced with dates, no countdown		FR-09-21

Player level (account, non-seasonal, purely a badge): level = floor(lifetime XP / 25,000) + 1, uncapped; every 5 levels grants a banner/title from the "level" source (FR-05-10). Never resets, never decays.

7.2 Free Season Path (FR-06-02, AD-11, AD-14, gapfill-C §2)

- 8-week season, **70 tiers**, all free. Curve: T1–T10 ramp 1,500→5,000 XP (cum. 32,500), T11–70 flat 5,000 (cum. **332,500**). Flat cost tunable 4,000–6,500.
- Overflow after T70: +100 Coins per 10,000 XP, max 20 per season.
- Rewards auto-claim at earn time (claim button optional); anything unclaimed is granted at season end with neutral copy ("Your Season Path rewards are in your wardrobe!").
- UI shows **estimated playtime to next reward** (AC3) and never a % "behind" indicator.

7.2.1 Reward table (cadence: a reward every tier; a "big" reward every 5 tiers)

Tiers	Every tier (small)	Every 5th tier (big)
1–10	alternating: colour, pattern, banner, 100 Coins	T5 emote #1, T10 upper costume A
11–20	title / faceplate / 100 Coins / pattern	T15 lower costume A, T20 **500 Coins**
21–35	as above	T25 celebration, T30 upper B, T35 emote #2
36–50	as above	T40 lower B, T45 **500 Coins**, T50 banner (animated)
51–70	as above	T55 title, T60 faceplate set, T65 banner, T70 **500 Coins** + season title

Free-track minimums met: 3×500 Coins, 4 costume halves (A/B upper+lower), 2 emotes, 1 celebration, ≥2 banners, ≥2 titles (gapfill-C §2). Costume halves land by T40 so a **casual player (≈T51) gets both full season costumes**. Small-tier Coins: 100 × ~14 tiers ≈ 1,400 Coins across the path.

7.3 Coins: earn, sinks, shop (FR-06-05, FR-06-06, FR-05-11, gapfill-C §6–8)

Resolved retune (see 07-ISS-01): gapfill-C §6 rates over-pay ≈2× its own gapfill-C §12 targets. Proposed defaults:

Source	gapfill-C v3	This doc	Range
End of show	20 + 10/round + 50/win (≈35)	**10 + 5/round + 25/win (≈18)**	config
Daily challenge	50	**25**	25–50
Show-Coin soft cap	600/day then 25%	**300/day then 25%**	200–600
Season Path	3×500	3×500 + small-tier 100s	
Trophy Rank, overflow	unchanged	unchanged	

- Wallet = append-only ledger, idempotency key per grant (show ID / challenge ID / tier ID) (FR-06-05 AC2, FR-01-09 AC2).
- Sinks: Coin Shop only, prices per gapfill-C §7 (Common full costume 2,000 … Special 5,000). Weekly rotation Mon 10:00 UTC, 12 items + 2 sets, **no countdown**, every item returns ≤12 weeks; confirm + 24 h undo if unused (FR-05-11 AC3). Coins never shown next to a real-money value.
- Rarity is a label only, never a drop rate (FR-05-09).

7.4 Cosmetic slot system (FR-05-01…08, FR-05-02, FR-05-03)

User-facing names map to v3 categories:

Slot (brief)	v3 category	Equip count	Default owned	Rule
Top	Upper costume	1	1 basic	Any top + any bottom (FR-05-02); waist seam ≤2 mm (FR-12-04)
Bottom	Lower costume	1	1 basic	Missing half → default part shown
Colour	Colour	1	12	Colour Copy costumes recolour (FR-05-04)
Pattern	Pattern	1	6	
Face	Faceplate	1	6	Hidden if costume `hidesFace=full` (FR-05-05)
Emote	Emote	4 (wheel)	4 (Wave, Thumbs Up, Happy Dance, Cheer)	Friendly only, FR-12-08 review (FR-05-06)
(also) Celebration	Celebration	1	1	FR-05-07
(also) Banner / Title	Banner, Title	1 each	1 each	Titles from curated list only (FR-11-05)

Layer order (render): colour → pattern → faceplate → lower → upper → effects. Occlusion flags per item: `hidesFace (none|eyes|full)`, `hidesPattern (bool)`. **Starter kit:** every new profile can build ≥ 12×6×6 = 432 looks on day 1 with no play.

7.5 Dressing room logic (FR-05-17)

1. Tabs: Top, Bottom, Colour, Pattern, Face, Emotes, Celebration, Banner, Title (+ "Outfits" presets: 3 saved slots, Should).
2. Each tab lists owned first, then unowned greyed with **"How to get it"**: "Season Path tier 30", "Coin Shop (when it returns)", "Weekly challenge" — never a buy button or shop link (AC5).
3. Equip = instant live preview ≤100 ms (FR-05-03 AC1); 360° rotate; emote/celebration previews play (AC2).
4. Validation on equip: item owned (server check), slot matches, occlusion rules applied; unowned items can be *previewed* but not saved.
5. Save: signed-in child → server, follows across devices; guest → device; guest-to-child migration once (AD-22, gapfill-C §11).
6. "New" badge clears on first view (AC4); no badge counts on main menu buttons beyond a single dot (avoid nagging).
7. "Surprise me" randomises from **owned** items only (no chance reward—purely a styling helper).

7.6 Challenge design (FR-06-13, gapfill-C §9, AD-33)

Type	Slots	Reward	Refresh	Banking
Daily	3	1,500 XP + 25 Coins	10:00 UTC	Up to 7 days (21 slots); 1 free swap/day
Weekly	5	4,000 XP	Mon 10:00 UTC	Stay open all season
Season goals	5 stages	10,000 XP; cosmetic at stage 5	Season	Play-count based
Event (R2)	5–8	Event cosmetic track	7–14 days	All reachable ≤4 h play (FR-09-23)

Rules: ≤40 chars en-GB, one icon, literal (AC1); 0 require a win; 0 reward grabbing/knocking players (AC2); no "only N hours left". **Template pool (examples):** Play 3 shows · Qualify from 4 rounds · Finish 2 races · Reach 1 Final · Collect 10 fruit in Fruit Count · Try a team round · Do an emote in the lobby 3 times · Play with a friend's party (R2). Difficulty mix per day: 1 "play" (any outcome), 1 "do" (action), 1 "qualify" (achievable by ~70% of players in ≤3 shows).

7.7 Unlock cadence (what a child sees)

Moment	Unlock
First show	XP bar fills, 1–3 Season Path tiers (T1 = 1,500 XP ≈ 1 show + 1 daily)
Session 1 (≈1 h)	~4–6 tiers, emote #1 at T5
Week 1 casual	~T8, top costume A at T10 next session
Every ~1 h of play mid-season	~3–5 tiers ≈ 1 big reward per 1–1.5 h
Each week	enough Coins for ≥1 small shop item (regular)
Each season (regular)	full path + 2–3 shop costumes

7.8 Healthy play (FR-11-15, AD-23, FR-11-08, FR-11-19)

- Break reminder after 60 min continuous play (parent can set 30–120; only parent can disable). Shown at **next show end** (AC1), friendly copy + "Stretch break!" art, single "OK" button.
- Optional parent daily limit (off by default); when reached, play ends after the current show (AC2); remaining rewards are banked, not lost.
- No streaks, no daily-login rewards, rested XP rewards *absence*. No push/marketing notifications. Results screen (SC-08) never prompts purchase or "one more game" autoplay (FR-08-18: no auto re-queue).
- Progression telemetry (pseudonymous, aggregate, AD-21) flags if >10% of p90 sessions exceed 2 h → design review, not engagement push.

7.9 Data model (spreadsheet-style; tables = sheets)

items

item_id	slot	name_key	rarity_label	acquisition_source[]	season_id	coin_price	hides_face	hides_pattern	traits[]	asset_ref	review_status
top_choc_01	upper	item.top.choc	common	[seasonpath_free]	S1	null	none	false	[colour_copy]	repo:/costumes/choc_top	approved
face_star_02	faceplate	item.face.star	uncommon	[coins]	null	1000	none	false	[]	repo:/faces/star	approved

season_path_tiers: season_id | tier | xp_to_next | cum_xp | reward_type (item/coins) | reward_ref | reward_qty | is_big **challenges_def**: challenge_id | type (daily/weekly/season/event) | text_key | icon | metric (shows_played, rounds_qualified, …) | target | xp | coins | item_ref | enabled | min_age_ok **challenge_state**: profile_id | challenge_id | slot_date | progress | completed_at | swapped **ledger** (append-only): entry_id | profile_id | ts_utc | kind (grant/spend/undo) | currency (coins/xp) | amount | source (show/challenge/tier/trophy/overflow/admin) | idempotency_key | balance_after **inventory**: profile_id | item_id | acquired_ts | source | seen (bool) | used_in_match (bool, for 24 h undo) **loadout**: profile_id | preset_no | upper | lower | colour | pattern | faceplate | emote_1..4 | celebration | banner | title | updated_ts **shop_rotation**: week_start_utc | slot_no | item_id/set_id | price **wellbeing_settings**: profile_id | break_min (30–120) | break_enabled (parent-only) | daily_limit_min (null) | set_by_parent_id **economy_config**: key | value | min | max | owner (all §7.1–3 numbers) **(Later) entitlements**: profile_id | sku | platform_txn_id | parent_approval_id | price_minor | currency_iso | ts | refunded

7.10 Progression simulation (8 weeks; `sim.py` in the design folder reproduces it — re-run during the merge, Appendix E)

Weekly XP = show XP + daily slots×1,500 + weeklies×4,000 + rested.

Segment	Pattern	XP/week	Tier wk2	wk4	wk6	wk8	Shows	Wins	Coins/season v3 rates	Coins/season retuned
Casual	3 d × 1 h @2,000/h; 6 dailies, 3 weeklies, 3k rested	30,000	15	27	39	**51 (73%)**	120	4	~8,000	**~4,800**
Regular	5 d × 1.2 h @2,400/h; 15 dailies, 5 weeklies, 2k rested	58,900	27	50	70 (wk 5.7)	70 + 13 overflow	240	8	~18,000	**~10,900**
Hardcore*	7 d × 3 h @2,800/h; 21 dailies, 5 weeklies	110,300	47	70 (wk 3.1)	70+	70 + 20 overflow	840	30	~43,800	**~25,300**

*Above the 60-min break default; not a design target. Retuned regular = 10.9k matches gapfill-C target 9–11k (≈2–3 costumes + small items). Tier values reproduce gapfill-C §12 exactly. Telemetry gate: regular cohort >10 tiers off at week 4 → retune flat tier cost first, then challenge XP.

7.11 Later: paid cosmetics with parent approval (Q-B1; Could, post-MVP only)

Only if the user approves monetisation. Constraints (FR-11-09, FR-05-16, FR-06-02, AD-12):

- **Direct real-money SKUs** via platform store (e.g. a cosmetic set or an optional premium Season Path track), price shown in £/$; **never currency, never chance, never tier skips/XP boosts**.
- Each purchase routes to the Parent Account for per-transaction approval; spending defaults to £0; parent dashboard shows history (FR-11-10); refund per platform policy.
- Fits the model via `acquisition_source = seasonpath_premium | paid_sku` and the `entitlements` sheet; dressing-room "How to get it" would say "Ask a grown-up in the Parent Account" — still no buy

button inside the child dressing room (FR-05-17 AC5 stays). Premium track unlocks retroactively and shows estimated playtime to complete; free track remains complete and unchanged.

- Gameplay parity unaffected (FR-05-01 AC1 bit-exact test).

8 Technical Architecture

Source: `08-technical-architecture.md` (23 Oct 2026 (no version stated)). **Authority:** Systems (highest authority for systems).

*Every number here is a **design estimate, not confirmed from Fall Guys**, unless it cites a requirement. Where requirements fix a value, that value is used and not changed here.*

8.0 Scope and fixed decisions inherited from the requirements

Fixed input	Source
Kids 6–12; COPPA, UK Children's Code and GDPR-K apply; no free text or voice; parent consent	UD-1, FR-11-01..19, NFR-10-08/09
MVP platform: Windows PC; other platforms later	AD-51, NFR-10-01
32-player Main Show (config 8–40)	AD-05, NFR-10-03, D-12
One shared 50 Hz simulation step on client and server	AD-06, D-01
Server-authoritative results, client-predicted own movement	D-03, D-04, NFR-10-05
Server step P99 ≤ 9 ms, 2 shows per vCPU	AD-32, NFR-10-04
Default planning CCU F = 10,000; load test to 10×F	D-16, Q-C3, NFR-10-06
No real-money purchases in MVP; Coins only; no premium currency	AD-12, FR-05-14, FR-06-05
All tuning values are data, rollback ≤ 5 min	NFR-10-20
First-party, pseudonymous telemetry only	NFR-10-12, AD-21
"MiniMax" meaning unresolved	Q-A1, NFR-10-02

8.1 Engine choice (NFR-10-02)

Criterion	Unity 6	Unreal Engine 5	Godot 4
Networked physics for 32 ragdolls	PhysX via Rigidbody; NGO server-authoritative rigidbodies; mature third-party netcode (Photon Fusion, Fish-Net); the reference game shipped on Unity + modified UNet	Chaos physics; good replication framework; networked physics prediction is newer and harder to tune; physics-driven characters are heavier	Jolt/Godot Physics; high-level multiplayer API is basic; we would build prediction and rollback ourselves
Headless dedicated server	Linux dedicated-server build target; mature	Dedicated server needs a source build; heavier process (memory)	Headless export is easy and light
Cartoon art at 60 fps on GTX 960 class (NFR-10-04)	URP fits well	Possible with scaled-down settings; Nanite/Lumen not usable at min spec	Fine, smaller rendering ecosystem
Later ports (console, Switch, mobile; AD-51)	Strongest mobile and Switch story	Good on console, weaker on low-end mobile/Switch	Console ports go through third parties
Team hiring, kid-studio tooling, cost	Large talent pool; runtime fee was cancelled (check the current licence)	5% royalty above threshold; C++ hiring	MIT licence, smallest talent pool
Fit to gapfill-A/D (Unity-style fixedDeltaTime)	Direct	Needs translation	Needs translation

Recommendation: Unity 6 LTS (URP, Linux headless server build) with a **netcode decision spike at M0** between Unity Netcode for GameObjects (NGO) and Photon Fusion 2 (server mode) or a similar library. Pick the one that passes the ragdoll prototype (§8.13, R-01). Godot is the fallback if licence risk dominates. Unreal is not recommended for this scope. **"MiniMax" (Q-A1):** still unresolved. Plausible meanings: (a) an AI vendor (MiniMax) used for content, voice or asset generation; (b) a typo or internal name for a tool or engine; (c) the minimax algorithm (game AI). None changes the runtime architecture. If it means (a), it falls under the asset-provenance rules (FR-12-07) and must **never** receive child data (FR-11-14, NFR-10-12). **Action:** product owner answers Q-A1 before the M0 engine decision record is signed off (NFR-10-02 AC1).

8.2 System context and client/server architecture (FR-07-12)

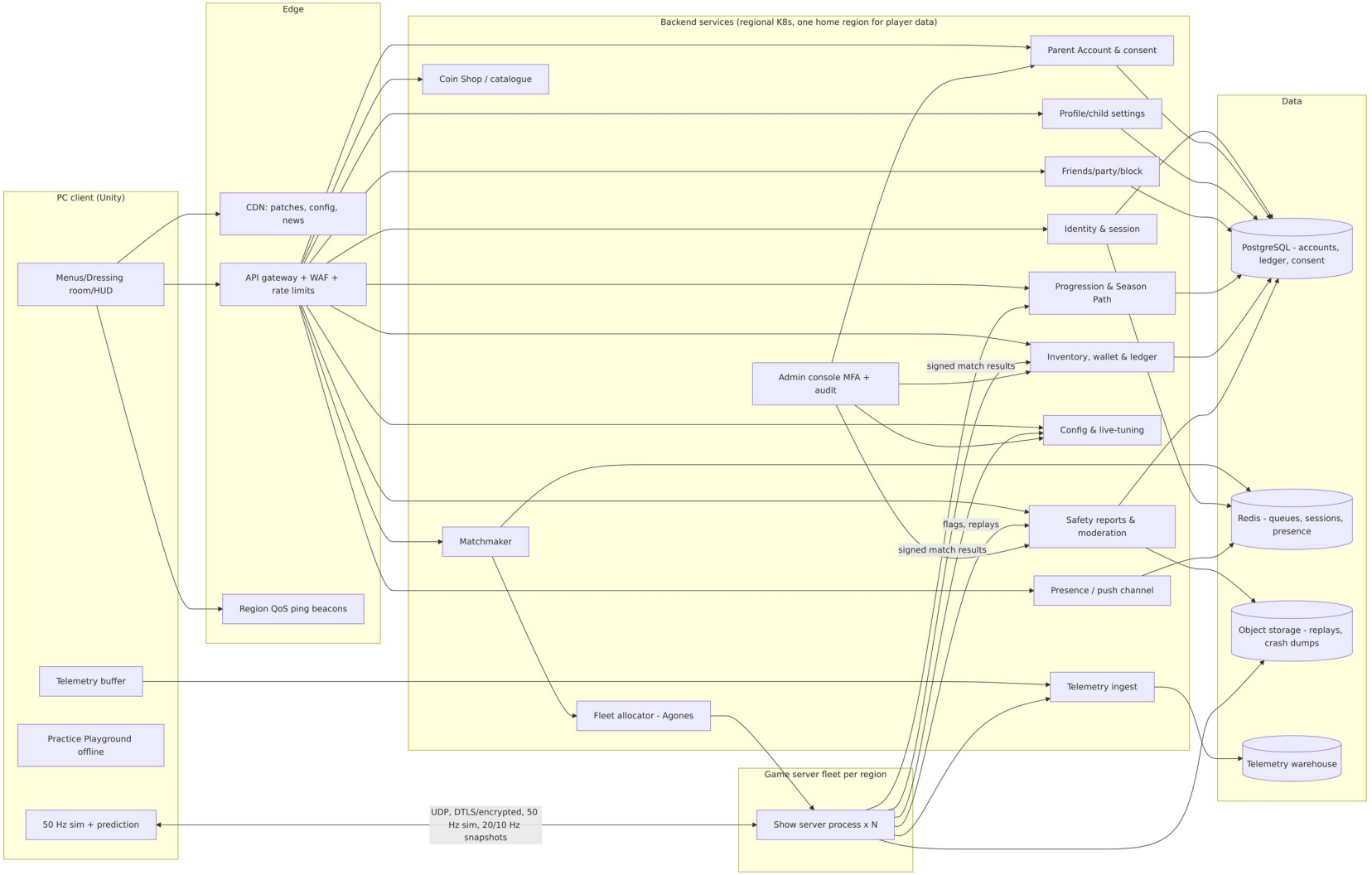


Figure 8.1 — System context and client/server architecture

Principles:

1. **Dedicated, server-authoritative show servers.** One headless process runs one show (D-15).
2. **Stateless HTTP/gRPC microservices behind one gateway.** Keep the service count small at MVP (a modular monolith with the listed modules is acceptable; split later).
3. **Player PII lives only in the Parent Account/consent store.** Every other service uses a pseudonymous `player_id` (FR-11-11, NFR-10-12).
4. **Game servers never write player data directly.** They post a signed `MatchResult` that Progression and Inventory apply idempotently.

8.3 Netcode model (D-01..D-10, NFR-10-05, FR-03-02, FR-03-07, FR-04-xx)

Topic	Design	Trace
Simulation	Fixed 50 Hz step on both sides; render decoupled. PhysX is not deterministic → **snapshot state sync**, not lockstep	AD-06, D-01
Authority	Server owns qualification, timer, scores, grabs, obstacles, tiles, respawn, fake-path layouts. Own bean is predicted and validated by the server	D-03, FR-02-02 AC4, NFR-10-05 AC3
Input	C→S at 25 Hz; each packet carries the last 6 steps (redundant), sequence plus client tick	D-02
Snapshots	S→C 20 Hz near/on-screen, 10 Hz far (interest management); delta-compressed against last acked; 16-bit quantised positions, smallest-three quaternions, ragdoll as root + 3–5 key bones (not every bone)	D-02, D-07
Prediction	Own move/jump/dive predicted; 1 s ring buffer; >0.3 m → rewind and replay; 0.05–0.3 m → smoothed over 100–200 ms; grabs/knockbacks from others are **not** predicted	D-04
Remote entities	Interpolated at 100 ms (adaptive 80–200 ms); extrapolate ≤150 ms then freeze	D-05
Obstacles	Kinematic movers driven by **server time** (synced clock, NTP-style offset estimate); clients sample the same timeline → phase error ≤50 ms P95	FR-03-02 AC1
Events	Reliable ordered channel for tile removal, eliminations, round state; tile hidden ≤100 ms after event	FR-03-07 AC2
Lag comp	Rewind other beans' hitboxes for grab/dive-hit start only, cap 200 ms	D-06
Ties	Sub-tick timestamp from the server-side contact solver; equal → all win	FR-01-05 AC2, AD-10
Bandwidth	≤64 kbps avg / 128 peak down, ≤16 kbps up; ≤2 Mbps per show	D-07
Latency	Target ≤80 ms, playable ≤150 ms, MM excludes >200 ms	D-08, D-14
Reconnect	60 s, max 2 per show, reconnect token bound to `player_id` + `show_id`	D-09, FR-07-16
Transport	UDP with encryption + authentication (DTLS or the netcode library's AEAD); connect token issued by the allocator (signed, 30 s TTL)	NFR-10-16
Spectating	Eliminated players receive a spectator snapshot stream (one followed bean at 20 Hz, others at 10 Hz)	FR-07-11

8.4 Game server lifecycle (FR-01-02, FR-01-05, D-09..D-12, AD-17)

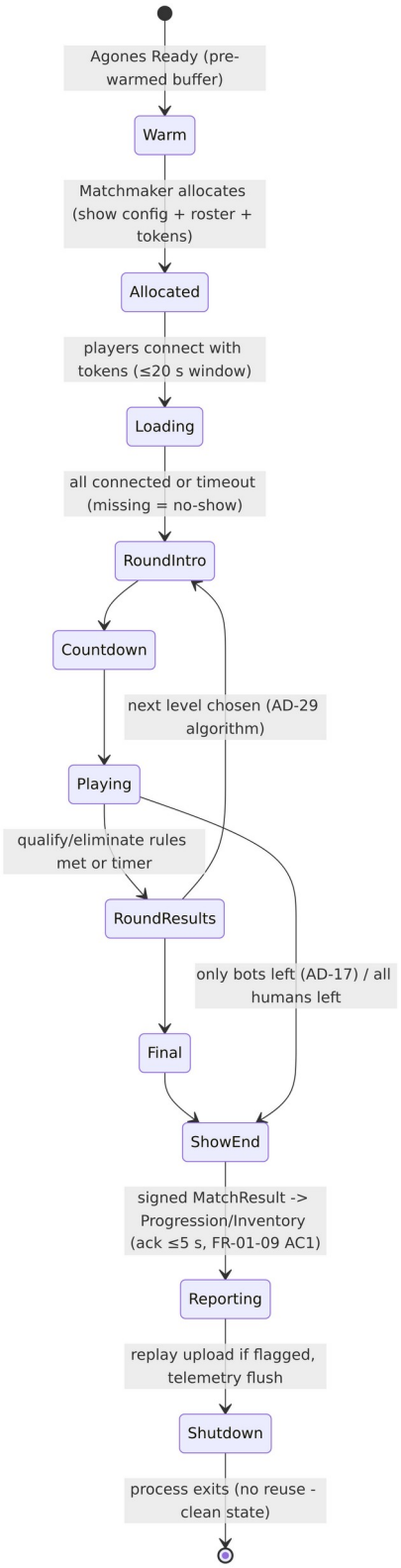


Figure 8.2 — Game server lifecycle

- **Pre-warm buffer:** 10% of peak plus 50 Ready servers per region; scale-up from 10% to 100% in ≤15 min (D-16).
- **Config pinning:** a show reads the config version that was live when it was allocated. A level disabled ≥60 s before show start is never chosen (FR-01-03 AC5).

- **Crash handling:** if a server crashes, a watchdog reports `ShowAborted`. Players get XP/Coins for the rounds they finished (rounds are checkpointed to Redis after each round) and return to the menu.
- **Bots:** run in-process, ≤12 per show, casual only, ~0.1 ms/bot/step (D-11).
- **Server-side drain:** on deploy, servers in `Allocated` state finish; new allocations go to the new fleet version.

8.5 Backend services

Service	Responsibilities	Key requirements
Identity & session	Device/platform login (Steam or Epic on PC, Q-C1), session JWT (15 min) + refresh, reconnect tokens, login queue at >80% capacity, kill switch on account creation	FR-07-06, D-16, NFR-10-06
Parent Account & consent	Parent email + verifiable parental consent (VPC, e.g. card check or ID check through a certified vendor such as KWS/PRIVO/Veratad; vendor choice open); parent gate; controls (play-time limits, friends on/off, quick-chat on/off, spending lock); dashboard; data export/deletion; consent record kept for life of account +3 years	FR-11-02, 07, 08, 09, 10, 12, NFR-10-21
Profile	Child profile with generated name (from a curated word list; not searchable), cosmetics loadout, settings (private by default)	FR-11-04, FR-11-11, FR-05-17
Social	Friend codes (parent-approved), party (≤4), block/mute, presence, quick-chat preset IDs only	FR-07-03, 07, 14, 15, FR-11-03
Matchmaker	Queue tickets, party atomicity, D-12 fill steps, region QoS, loose MMR bands, bot backfill decision, ≥2,000 players/s	FR-01-02, D-11..D-14, FR-07-02
Fleet allocator	Agones (or GameLift/Edgegap) allocation, autoscaler, multi-cluster per region	D-15, D-16
Progression	XP and level, Season Path (free track), challenges, Trophies; applies MatchResult idempotently	FR-06-01, 02, 07, 13, FR-09-19
Inventory & wallet	Append-only Coin ledger, entitlements, Coin Shop purchases, grants/rollback by admin with audit	FR-06-05, 06, 19, FR-05-10, 11
Config & live-tuning	Versioned, schema-validated config bundles; shows, levels, weights, physics values, kill switches, News Board	NFR-10-20, FR-01-03 AC5, FR-09-26
Telemetry	Event ingest (batched HTTPS), schema registry with PII lint, warehouse, dashboards and alerts	NFR-10-12, D-19
Moderation	Child-friendly reports (category picker, no free text), report queue, sanctions, anti-cheat flags, replay review, SLA tracking	FR-11-06, FR-11-17, NFR-10-13
Admin console	SSO + MFA, RBAC, immutable audit log	NFR-10-16, FR-06-19
Data lifecycle jobs	Daily retention deletion, export packages, account deletion cascade	NFR-10-21, FR-11-12

8.6 Data model (core entities)

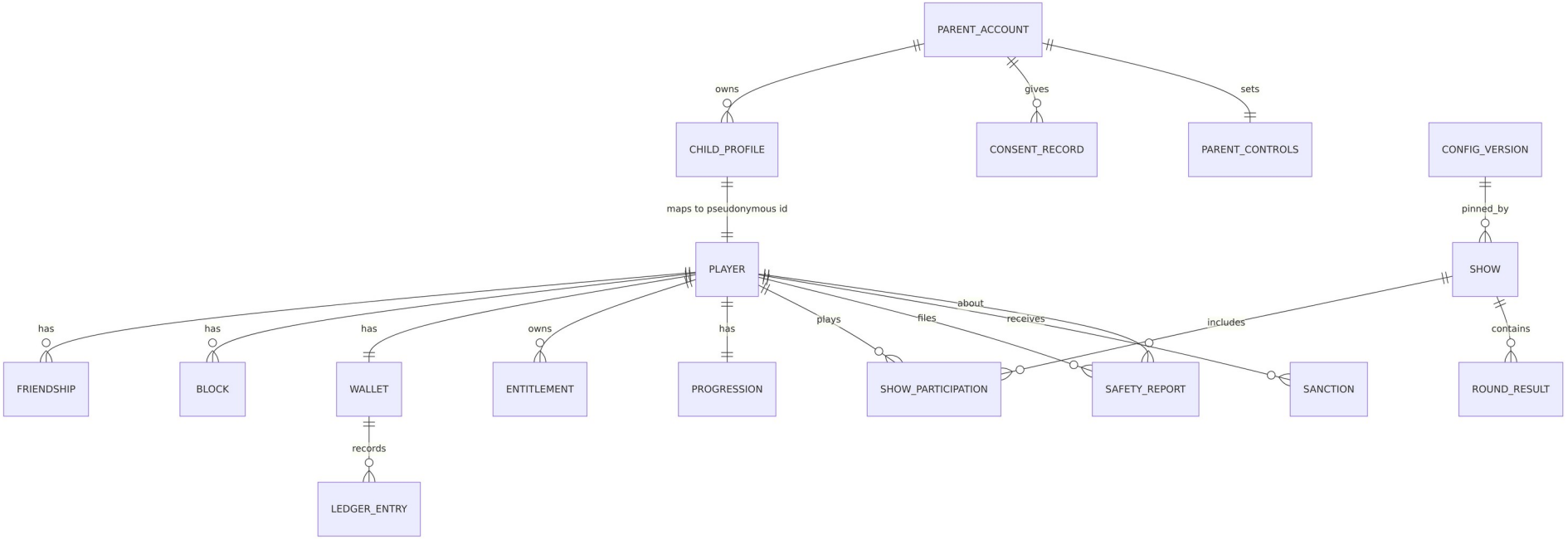


Figure 8.3 — Core data model

Entity	Key fields	Notes
parent_account	id, email (encrypted), country, created_at, vpc_method, status	Only table with direct PII; field-level encryption; separate schema and keys
consent_record	id, parent_id, child_id, scope, method, vendor_ref, granted_at, revoked_at	Append-only; kept life + 3 years
parent_controls	parent_id, child_id, daily_minutes, friends_enabled, quickchat_enabled, coin_shop_enabled	FR-11-08/09/15
child_profile	id, parent_id, birth_year (not date), generated_name_id, locale	Minimum data (FR-11-01)
player	player_id (UUIDv7), child_profile_id, platform_link_hash, mmr_mu, mmr_sigma, created_at	The pseudonymous id used everywhere else
friendship / block	player_a, player_b, state, approved_by_parent_at	
wallet / ledger_entry	ledger: id, player_id, delta, reason, source_ref (show_id/admin_id), idempotency_key UNIQUE, balance_after	Balance = sum of the ledger; never updated in place
entitlement	player_id, item_id, source, granted_at, revoked_at	
progression	player_id, season_id, xp, level, path_tier, trophies, challenge_state JSONB	
show / round_result	show_id, config_version, region, server_build, started/ended; per-round qualified/eliminated, reason	Results kept as aggregates
safety_report / sanction	reporter, subject, category, show_id, replay_ref, state, SLA timestamps	Retained 1 year
config_version	id, bundle hash, author, approved_by, schema_version, activated_at	Immutable
telemetry_event (warehouse)	event_name, player_pid (rotating salted hash), session_id, ts, payload	90 days raw, 2 years aggregates; no IP stored (coarse region only)

8.7 API list (gateway, HTTPS/JSON or gRPC-web; all need a session token unless marked)

Area	Endpoints
Auth	`POST /v1/auth/platform-login` · `POST /v1/auth/refresh` · `POST /v1/auth/logout` · `GET /v1/auth/queue-status` (no auth)
Parent (parent token, parent gate)	`POST /v1/parent/register` · `POST /v1/parent/consent/start` · `POST /v1/parent/consent/callback` (vendor, signed) · `GET/PUT /v1/parent/children/{id}/controls` · `GET /v1/parent/dashboard` · `POST /v1/parent/children/{id}/export` · `DELETE /v1/parent/children/{id}` · `POST /v1/parent/consent/revoke`
Profile	`GET /v1/me` · `PUT /v1/me/loadout` · `POST /v1/me/name/reroll` · `GET /v1/names/options`
Social	`GET /v1/friends` · `POST /v1/friends/code` · `POST /v1/friends/requests` (parent-approved) · `POST /v1/blocks` · `DELETE /v1/blocks/{id}` · `POST /v1/party` · `POST /v1/party/{id}/invite` · `POST /v1/party/{id}/leave`
Matchmaking	`GET /v1/shows` (available shows) · `POST /v1/mm/tickets` · `DELETE /v1/mm/tickets/{id}` · `GET /v1/mm/tickets/{id}` (or push) → returns `{host, port, connect_token}` · `POST /v1/mm/reconnect`
Progression	`GET /v1/progression` · `GET /v1/season/current` · `POST /v1/challenges/{id}/claim`
Inventory/shop	`GET /v1/wallet` · `GET /v1/inventory` · `GET /v1/shop/catalogue` · `POST /v1/shop/purchase` (Coins only, idempotency key)
Config	`GET /v1/config/client?version=` (CDN-cached, signed) · `GET /v1/news`
Safety	`POST /v1/reports` (category enum + show_id + subject) · `GET /v1/reports/mine`
Telemetry	`POST /v1/telemetry/batch` (schema-validated)
Internal (mTLS, service accounts)	`POST /internal/match-results` (signed by server key) · `POST /internal/anticheat/flags` · `POST /internal/fleet/allocate` · `GET /internal/config/server/{version}`
Admin (SSO+MFA, audited)	`POST /admin/config/activate` · `POST /admin/config/rollback` · `POST /admin/killswitch` · `POST /admin/wallet/grant` · `POST /admin/wallet/rollback` · `GET/POST /admin/moderation/queue` · `POST /admin/sanctions`

Area	Endpoints
Game protocol (UDP)	`Connect(token)·Input[seq, ticks×6]·Snapshot[delta]· ReliableEvent{RoundState, Eliminated, TileRemoved, Score, QuickChat(id), Emote(id)}·Ack·Ping`

8.8 Hosting and scaling (default F = 10,000 CCU; D-15, D-16, NFR-10-06)

Item	Sizing
Shows at 10k CCU	≈313 shows of 32 → ≈157 vCPU at 2 shows/vCPU → ~10 × 16-vCPU compute-optimised hosts across 4 regions (EU-W, EU-C, NA-E, NA-W; AD-39), plus a 30% buffer → ~14 hosts
Memory	~400 MB per process × 32 per host ≈ 13 GB → use 32 GB hosts
Egress	≈0.63 Gbps (≈$14/h) — larger than compute (≈$6.3/h); see issue 08-I-04
Load test target	10×F = 100k CCU for 1 h (NFR-10-06 AC1) → ~3,130 shows, ~1,570 vCPU burst; uses cloud burst capacity
Fleet	Agones on managed Kubernetes (GKE/EKS) per region, game-server node pools without autoscaler eviction; optional bare-metal baseline + cloud burst once the forecast (Q-C3) exists
Backend	Stateless pods with HPA; PostgreSQL primary in one home region (EU, for UK/GDPR) with a read replica and PITR (RPO ≤15 min, RTO ≤4 h; NFR-10-17); Redis cluster per region for queues and sessions
Edge	CDN for patches/config; WAF + rate limit at the gateway; login queue at >80% auth capacity
Availability	99.5% monthly (NFR-10-17); multi-AZ; game regions fail independently; MM re-routes to the next region ≤150 ms
Data residency	Child PII stored in one region (EU/UK); US players' PII handling is set by the DPIA (NFR-11-01, Q-C2)

8.9 Security and anti-cheat (NFR-10-13, NFR-10-16, D-18, AD-09)

Security baseline:

- TLS 1.2+ everywhere and an encrypted, authenticated UDP game protocol.
- Secrets in a managed vault (no secrets in the client).
- OWASP ASVS L2.
- Admin MFA and an immutable audit log (WORM storage).
- Pen test before launch covering the Parent Account, consent and friend codes.
- Signed config and patch bundles.
- Server keys rotated per fleet version.
- Service-to-service mTLS.
- Rate limits on friend-code and name-reroll endpoints (stop enumeration).

Anti-cheat layers:

1. **Authority:** the client sends only inputs and a predicted position. The server owns every result, so a client cannot set qualification (NFR-10-05 AC3).
2. **Server validation:**
    - Speed check: 1.15 × max(9, 11, 13, recent server Δv) over 0.5 s (AD-09).
    - Airtime and vertical-acceleration check.
    - Teleport check: >1 m delta outside a server event.
    - Collision/wall-clip check with server raycasts.
    - Finish/checkpoint order check.
    - Input-rate sanity check (≤ 50 steps/s).
    - Correction goes back within 2 ticks; repeat offences raise a flag score.
3. **Response:** at the threshold, remove the player from the show (AC3); save a 14-day replay; route to the moderation queue; apply escalating sanctions.

4. **Client-side:** obfuscation (IL2CPP), integrity checks, encrypted packets. A vendor kernel anti-cheat (e.g. EAC) is **optional and gated by the DPIA**, because of children's device data (NFR-10-13, Q-C3). Recommendation: launch without kernel anti-cheat; rely on server authority.
5. **Economy:** ledger idempotency keys; server-signed match results; anomaly alerts on Coin earn rate per player-hour.

Child safety:

- There is no free-text path (NFR-10-09 AC1 static check in CI).
- Quick-chat and emotes travel as enum IDs only.
- Names come from a curated list (FR-11-04/05).
- Reports use a category enum (FR-11-06).

8.10 Config and live-tuning (NFR-10-20, FR-01-03 AC5, FR-09-19/26)

- **Format:** Git repo of YAML/JSON bundles (`shows/`, `levels/`, `obstacles/`, `physics/`, `economy/`, `season/`, `killswitch/`, `news/`), each with a JSON Schema carrying the Chapter 13 ranges. Examples: conveyor ≤40% of run speed (AD-08); a ragdoll tag is required (FR-03-01 AC2).
- **Pipeline:**
    1. Pull request.
    2. CI schema and range validation, plus a simulation smoke test (headless 32-bot show on changed levels).
    3. Two-person approval.
    4. Signed bundle.
    5. Activate in staging.
    6. Canary in one region at 5% of new shows.
    7. Activate globally.
- **Runtime:** servers pin the version at allocation. Clients fetch the signed client subset from the CDN at login and on Show Picker open (FR-01-01 AC1). Scheduled activations are supported (level enable/disable, limited-time shows).
- **Rollback:** a one-click re-activation of the previous version. Target ≤5 min end to end; tested monthly.
- **Kill switches:** account creation, each show, Coin Shop, friend requests, and the Builder later (NFR-10-06 AC2).
- **Physics values:** shared by client and server. A show only admits clients whose physics hash matches the server's pinned version. A mismatch forces a config refresh before queueing.

8.11 CI/CD and environments

Env	Purpose	Data
dev (per-branch ephemeral)	Feature work; local headless server + docker-compose backend	Synthetic
int	Nightly integration; bot soak tests	Synthetic
staging	Release candidates; load tests (10×F); pen tests; restore drills	Synthetic, prod-shaped
prod (4 regions)	Live	Real; PII only in the home region

Pipeline:

1. Monorepo with trunk-based development.
2. Unity Build Automation or self-hosted runners build the Windows client and the Linux server.
3. Unit tests and PlayMode tests run.
4. Netcode tests run with network emulation (150 ms RTT / 2% loss) against FR-03-02, FR-03-07 and NFR-10-05.

5. A 32-bot headless perf test checks step P99 ≤9 ms.
6. A static check confirms no microphone permission and no text input reachable by a child (NFR-10-09).
7. A dependency scan confirms no ad or analytics SDKs (NFR-10-12 AC1).
8. Telemetry schema PII lint (NFR-10-12 AC2).
9. Flash analysis on level captures (NFR-10-19).
10. Container images are signed and released.
11. Backend uses blue/green deployment.
12. The game fleet uses versioned fleets with drain.
13. The client is patched through the store (Steam/Epic branches for beta).

Infrastructure is Terraform. Secrets are in Vault or the cloud KMS. Releases are tagged `client x.y.z ⇄ server protocol N`, and the server accepts N and N−1 during rollout.

8.12 Performance budgets

Client (NFR-10-04, D-17; min spec GTX 960 / i5-4590, 8 GB): 60 fps at 1080p low, P95 frame ≤16.7 ms.

Budget	ms	Notes
Sim (50 Hz; ≤1 step per frame normally)	2.5	Own prediction; 32 beans as interpolated kinematic proxies, not full ragdoll sim
Rollback replay worst case	+2.0	≤10 steps of own bean only
Animation + procedural wobble	2.5	GPU skinning, LOD bones far away
Gameplay/UI/audio	2.0	
Render CPU (URP, SRP batcher, GPU instancing)	4.0	≤1,500 draw calls
GPU	≤14	32 beans ≤15k tris each LOD0, ≤4 LODs
Headroom	~3	

- Memory ≤ 3.5 GB RAM, ≤ 2 GB VRAM.
- Install ≤ 15 GB.
- Load times: cold start ≤20 s, menu to first round ≤30 s, and level loads inside the 8 s results screen (NFR-10-18). Addressables preload the next level during results.

Server (AD-32): step P99 ≤9 ms per show:

- Physics ≤5 ms (32 bodies + ragdoll subsets).
- Gameplay/validation ≤1.5 ms.
- Bots ≤1.2 ms.
- Serialisation/network ≤1.3 ms.
- Memory ≤400 MB.

Network: D-07 budgets.

Backend:

- p95 API latency ≤200 ms.
- MatchResult applied ≤5 s (FR-01-09).
- Matchmaking P50 ≤20 s, P95 ≤60 s at 1k/10k/100k CCU (D-12).

8.13 Risk register

ID	Risk	L	I	Mitigation / owner action	Trace
R-01	**Server-authoritative physics for 32 (up to 40/60) ragdolls at 50 Hz misses the 9 ms P99 budget, or prediction mismatches cause rubber-banding**	H	H	**M0 prototype (4–6 weeks):** headless Unity server, 60 active-ragdoll bots (stress = 1.5–2× the 40 cap) in a worst-case pile-up arena and a Tile Drop and swing-ball course. Measure step P99, explosions, snapshot size, correction rate at 150 ms/2%. Exit criteria: 32 at ≤9 ms, 40 at ≤12 ms, 60 runs stably (no NaN, ≤20 ms). Fallbacks in order: (a) simplified server ragdoll (capsule + 3 bones; full ragdoll cosmetic on the client); (b) sleep/islands and physics LOD for far or eliminated beans; (c) 1 show per vCPU (cost ×2); (d) owner-auth movement + strict D-18 validation	NFR-10-02 AC2, NFR-10-03 AC2, NFR-10-04, D-01, gapfill-D risks
R-02	Netcode library choice (NGO vs third-party) falls short	M	H	Decide at M0 using R-01's results; abstract the transport	NFR-10-02
R-03	VPC vendor delays or country gaps	M	H	Pick the vendor at M0; DPIA; fallback to card-check VPC	FR-11-02, NFR-11-01
R-04	Egress cost exceeds compute	H	M	Delta/quantisation, relevancy tiers, negotiated egress or bare metal; track $/player-hour	D-07, D-15
R-05	Launch spike above 10×F	M	H	Login queue, kill switches, pre-agreed cloud burst quota, 10×F test	D-16, NFR-10-06
R-06	Cheating without kernel anti-cheat	M	M	Server authority; replay flags; reconsider after the DPIA	NFR-10-13
R-07	Child PII leaks through telemetry/logs	L	H	Schema lint, log scrubbing, no IP retention, audit	NFR-10-12, FR-11-14
R-08	Config error breaks live shows	M	M	Schema + bot smoke test + canary + ≤5 min rollback	NFR-10-20
R-09	Engine licence/pricing changes	L	M	Licence review at M0; Godot fallback noted	NFR-10-02
R-10	"MiniMax" turns out to be a mandated engine/platform	L	H	Resolve Q-A1 before M0 sign-off	Q-A1
R-11	Low population makes queues long (kids' title, regional)	M	M	Bot backfill ≤40%, region widening at 90 s	D-11, D-12
R-12	Min-spec 60 fps with 32 beans + VFX	M	M	Perf budget gates in CI; LODs; particle caps	NFR-10-04

8.16 Milestone hooks

- **M0:** engine and netcode decision record; R-01 ragdoll prototype; VPC vendor choice; Q-A1/Q-C1/Q-C3 answered.
- **M1:** vertical slice with one region and 32 players.
- **M2:** full backend plus config pipeline.
- **M3 beta:** DPIA signed; pen test; 10×F load test; restore drill.

9 Multiplayer, Social and Child Safety

Source: `09-multiplayer-safety.md` (Design v1 (23 Oct 2026)). **Authority:** Child safety (highest authority for child safety).

Rule: every design element below cites the requirement(s) it realises. Anything not backed by a requirement is marked **[Design]** and is listed in Appendix A for sign-off. Legal notes are design summaries, not legal advice (DEP-10).

9.1 Design pillars (traceability anchors)

#	Pillar	Requirements
P1	Everyone is a child; no adult/teen path in-game	UD-1, AD-01, FR-11-01
P2	No player-authored content between players: no text, no voice, generated names	AD-02, AD-03, FR-07-09/10 (Won't), FR-11-03, FR-11-04
P3	Strangers can play together but never contact each other	AD-02, AD-04, AD-19, FR-07-07, FR-07-08
P4	Parents decide; most protective defaults; child can't loosen	AD-22, FR-11-02, FR-11-08, FR-11-11
P5	No pressure: no leaver penalty, no countdowns, limits only at show end	AD-15, AD-23, AD-37, FR-11-15
P6	Safety in ≤ 2 interactions, picture-led, no typing	AD-43, FR-07-14/15, FR-11-06
P7	Minimise, truncate, delete	AD-21, FR-11-12, FR-11-16, NFR-10-21

9.2 Identity model

Identity	Who	Social features	Data	Req
Guest	Child, no account	Matchmaking, quick-chat (receive/send preset), emotes, block/mute, report. **No** friends, parties, Private Shows, online status	Random device-scoped ID, device class, gameplay events	FR-07-06 AC1, FR-11-01 AC2, AD-22
Child profile	Child under a Parent Account (max 4/parent)	Above + friends, parties, Private Shows (R2), only for purposes with consent	Pseudonymous profile ID, progress, friend graph, blocks, reports made	FR-07-06, FR-11-02 AC1
Parent Account	Verified adult (VPC)	Never appears in-game; dashboard + controls only	Email, consent records (method, time, purposes, notice version)	FR-11-02 AC3, FR-11-10
Bot	Server-side	None; cannot be friended, partied or reported into moderation	—	AD-17, D-11

Display names: Adjective + Noun + 2 digits, ≤ 16 chars, 5 re-rolls/day, curated per FR-11-05 (FR-11-04, AD-03). Platform account names never shown (FR-11-04).

9.3 Matchmaking & lobby UX

9.3.1 Flow

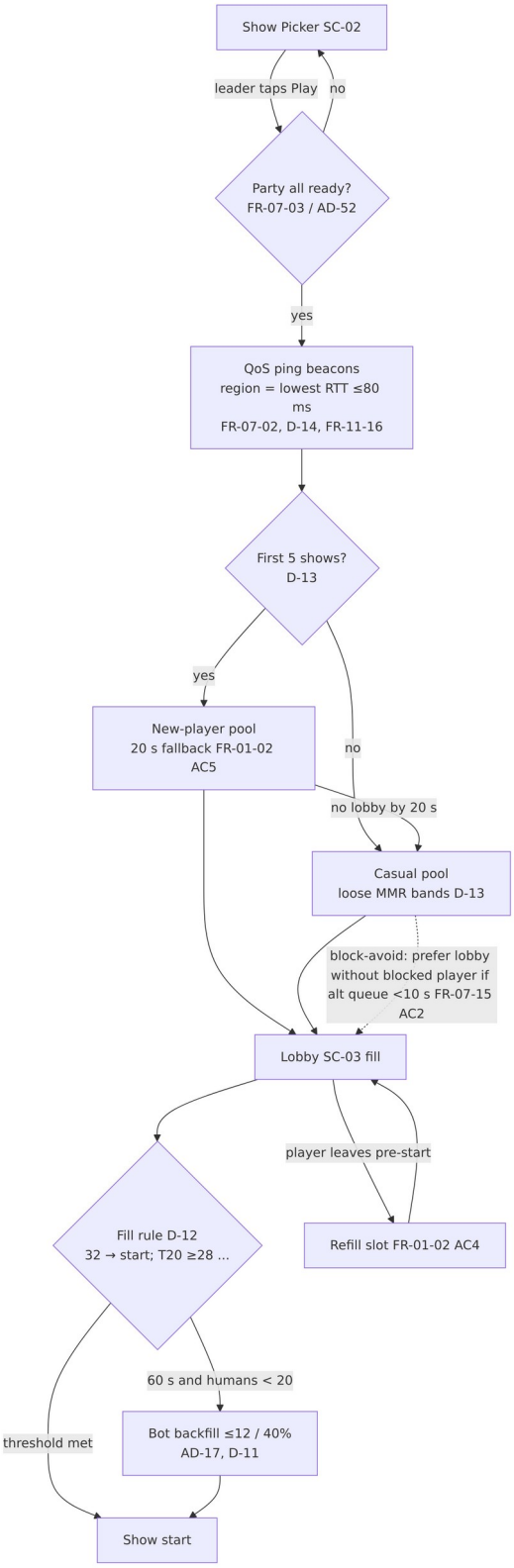


Figure 9.1 — Matchmaking and lobby flow

9.3.2 Rules

- Parties never split; same team in team shows (FR-01-02 AC2, AD-52). Team lobbies filled by whole parties (D-12).

- Region widens to ≤ 150 ms after 90 s; never > 200 ms; no kick for latency (D-08, D-14). Region from ping, never location (FR-11-16).
- Lobby screen SC-03 shows: party members (top), lobby fill as a friendly bean-counter (no ticking countdown — AD-15), "Some players may be computer players" (AD-17), player cards (generated name, costume, banner, title — FR-05 player card). **No** platform names, levels of spend, or country flags (FR-11-11, FR-11-16).
- Player card actions (lobby, spectate, results, friends list): **Mute · Block · Report** each reachable in ≤ 2 interactions (FR-07-14 AC2, FR-07-15 AC1, FR-11-06 AC1). There is **no** "Add friend" on a stranger's card (AD-04).
- Leaving a lobby or show: one confirm, no penalty (AD-37).

9.3.3 Bot backfill rules (consolidated)

Rule	Value	Req
Modes	Casual only; disabled in Private Shows and (n/a) Ranked	AD-17, D-11, AD-16
Trigger / start	< min_real (20) humans at 60 s step; the show starts if humans ≥ 8 and humans + bots ≥ 20, with bots = min(12, 32 − humans) (01-ISS-06, REC-09)	D-11, D-12
Cap	≤ 12 bots / ≤ 40 % of lobby — **also applies to new-player lobbies** (Issue 09-I-07)	AD-17
Final	Bots never reach a Final; rank below humans before the Final	AD-17, FR-01-02 AC7
Disclosure	SC-03 line + FAQ; parent privacy notice	AD-17, FR-11-13
All-bots	Show ends at once, no winner recorded	Ch.1 edge case 9
Names	Drawn from same curated generator **[Design]** so bots are indistinguishable in play, disclosure is at lobby level	AD-03, AD-17
Interaction	Bots never send quick-chat or emotes **[Design]** (avoids fake social pressure, AD-15 spirit); reports on bots are acknowledged to the child normally and discarded server-side	FR-11-06

9.4 Friends & friend codes

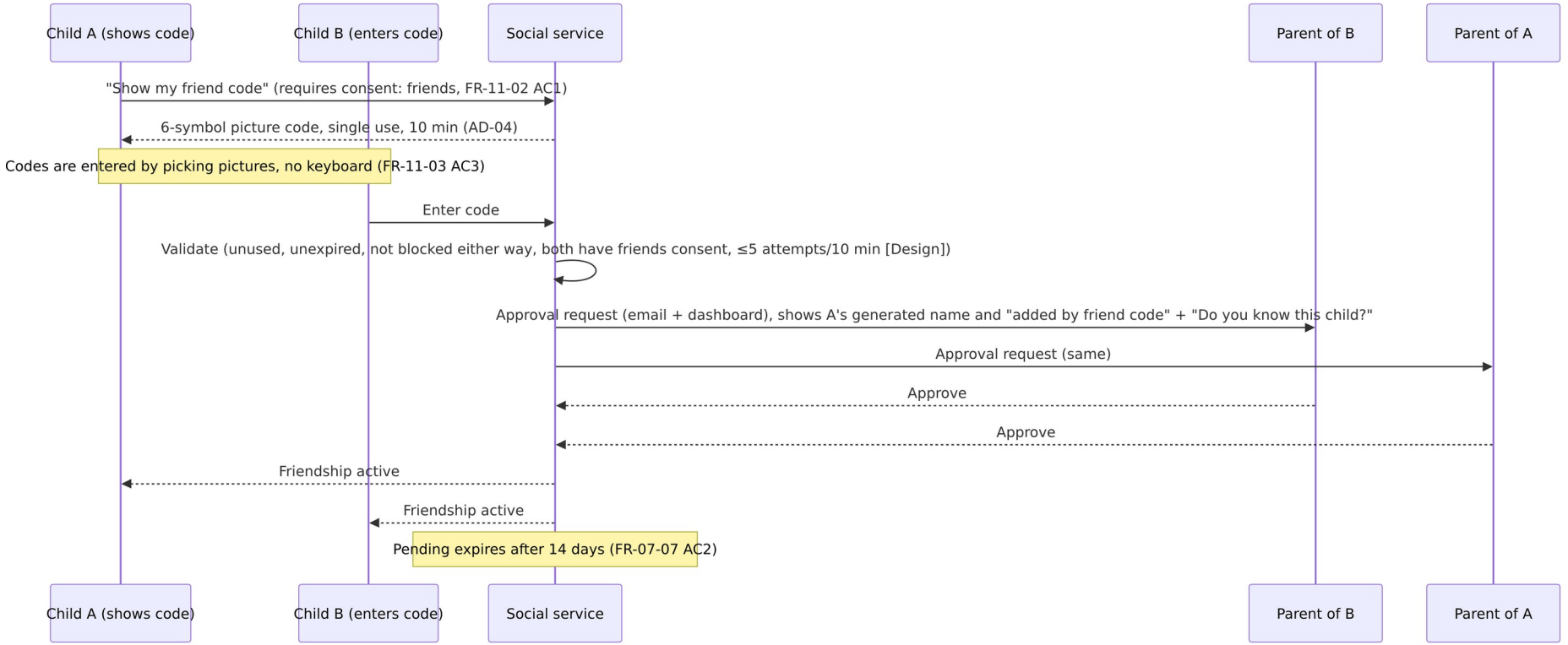


Figure 9.2 — Friend-code flow with parent approval

- Both parents approve **[Design, resolves 09-I-04]**; FR-07-07 says "every new friend needs approval in the Parent Account" — interpreted as each side's parent.
- Friend codes use a picture alphabet (e.g., 6 of 16 icons) so 6-year-olds can enter them without text fields (FR-11-03 AC3) **[Design]**.
- Platform-friend import: parent-only, behind parent gate, each import still per-friend approved (AD-04, FR-11-07).
- No name search API (FR-07-07 AC1). Online status visible to approved friends only (FR-07-07 AC4, FR-11-11).
- Remove/block friend ≤ 2 interactions (FR-07-07 AC3). Removal is silent to the other child **[Design]**.
- Friend requests are "off by default" (FR-07-07): the "Show my friend code" button is hidden until the parent grants the friends consent purpose.

9.5 Parties

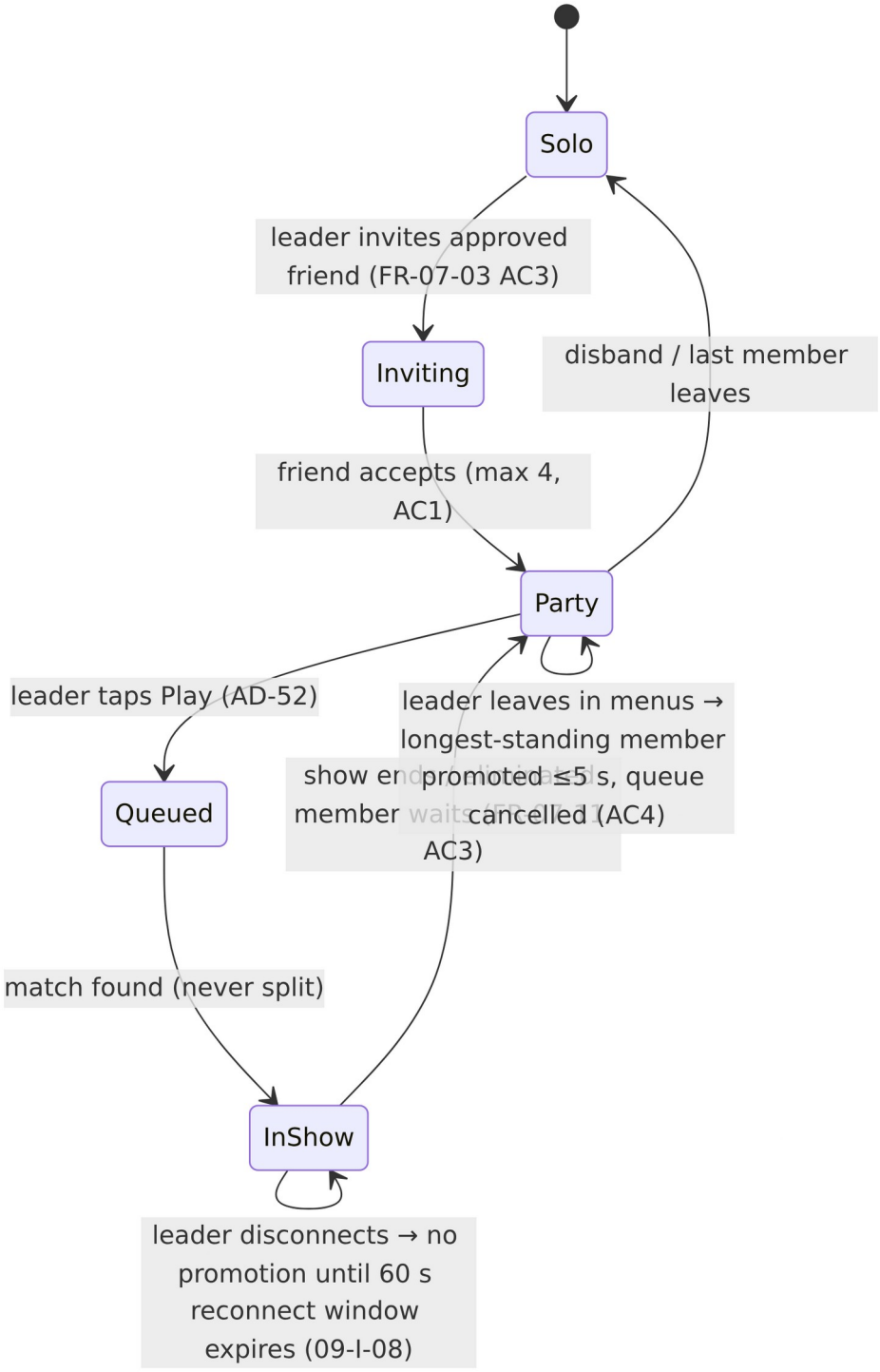


Figure 9.3 — Party state machine

- Invites only to approved friends; blocked players can't invite or join (FR-07-15 AC2).
- Leader changing show while members in Level Hop is blocked with message (Ch.1 edge 7).
- Parties are the only place a child sees "who they know"; party members are listed first in spectate (FR-07-11 AC2).

9.6 Communication: quick-chat & emotes only

Aspect	Design	Req
Channel	Lobby, rounds, results. Payload = phrase ID (uint8); client renders localised string	FR-07-14 AC3, FR-11-03 AC2
Set	≤ 16 positive/neutral phrases (launch 8 listed) + icons; FR-11-05 review per language	FR-07-14, FR-11-05
Input	Radial wheel, ≤ 2 inputs	FR-07-14 AC1
Rate	1 msg / 2 s per player; excess dropped silently	FR-07-14 AC1
Emotes	4 equipped, cancel on any input, not while airborne/grabbing/ragdolled; rate 1 / 2 s **[Design]**	FR-05-06, AD-34
Audience	Whole lobby in lobby/results; in rounds, only beans within view **[Design]**; spectators' quick-chat goes to party only **[Design, 09-I-06]**	FR-07-11
Mute	Mute all / mute one player ≤ 2 interactions; parent can disable quick-chat entirely (on by default)	FR-07-14 AC2, FR-11-08
Repetition abuse	Same phrase to same target ×5 in 30 s → sender's quick-chat soft-throttled to 1 / 10 s for 60 s **[Design]**; a phrase list with no negative phrases keeps worst case low	FR-07-14

Explicit **Won'ts** (negative tests): voice (no mic permission, FR-07-09, FR-11-03 AC1), text chat (FR-07-10), custom names (FR-11-04 AC2), free-text report descriptions (FR-11-03).

9.7 Block, mute, report

9.7.1 Child flow

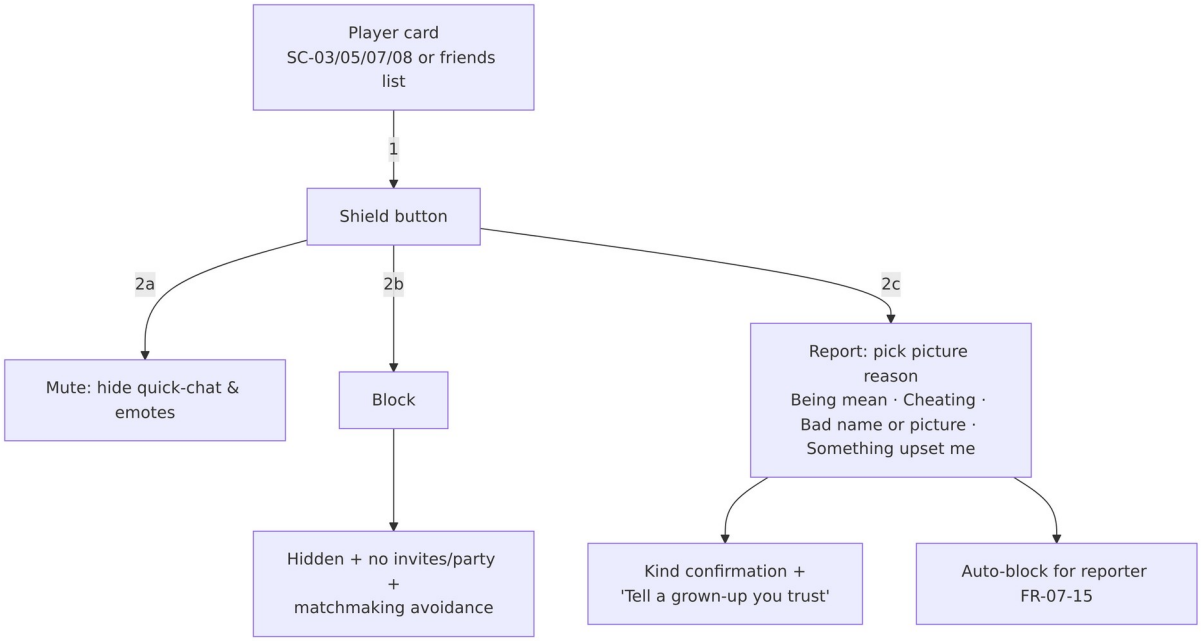


Figure 9.4 — Block, mute and report: child flow

"Shield" = one tap opens the three choices; the choice is tap 2 (≤ 2 interactions: FR-07-15 AC1, FR-11-06 AC1). Report = choose reason as the second tap (the reason icons are shown directly in the shield sheet) **[Design]**.

9.7.2 Report context (auto, no typing — FR-11-06)

show ID, round, reporter & reported pseudonymous IDs, phrase-ID/emote log for last 60 s, 30 s server replay snippet (server-side state, no client capture), reason code. Level reports add level ID & version.

9.7.3 Moderation pipeline

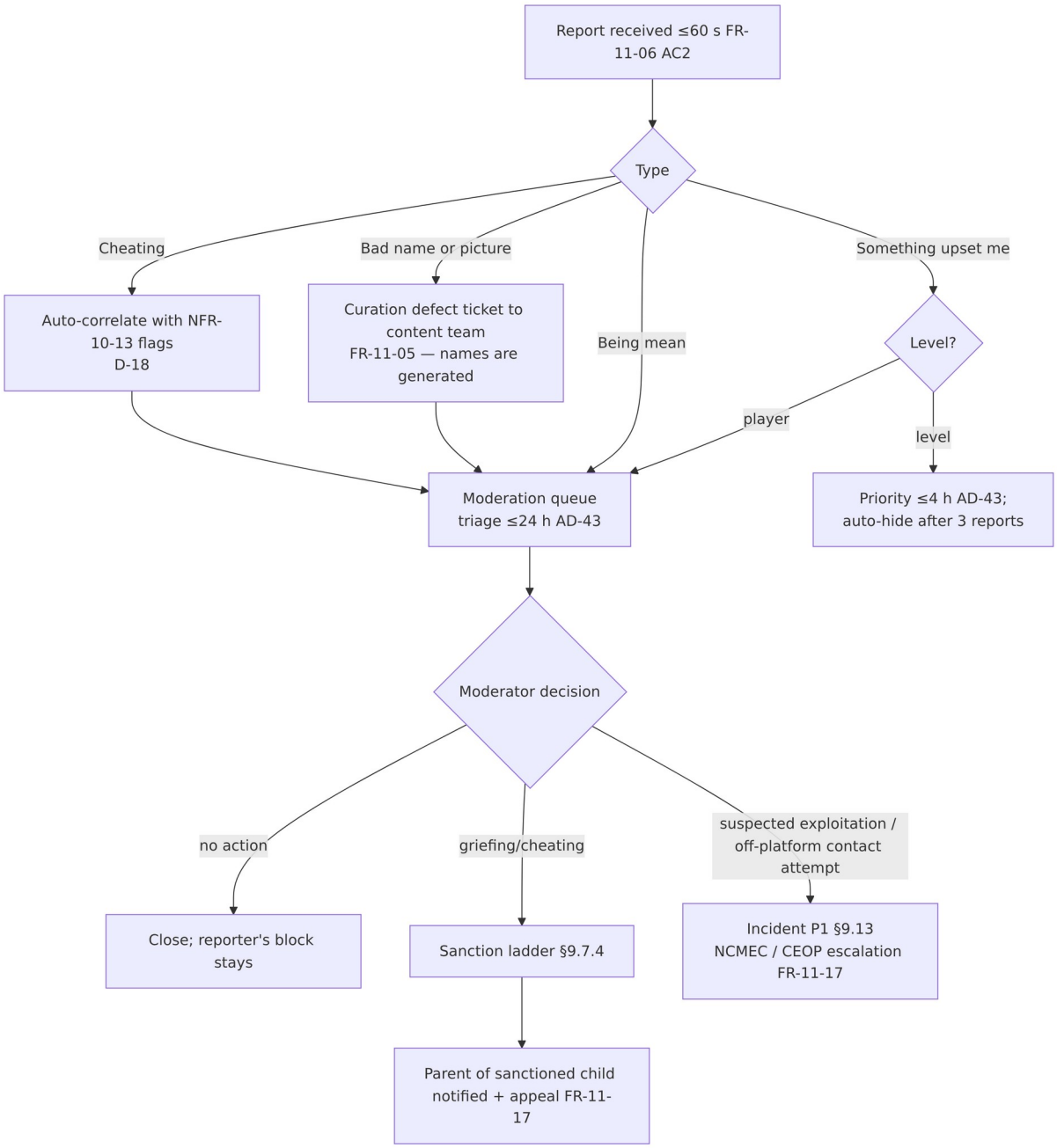


Figure 9.5 — Moderation pipeline

9.7.4 Sanction ladder [Design] (bound by FR-11-17, AD-37)

No automated sanctions from report counts alone for players (avoids brigading; AD-43's auto-hide applies to levels). Server-detected cheats are removed from the show immediately (NFR-10-13 AC3).

1. Warning card (picture-led, kind wording) + parent email.
2. 24 h online suspension (offline Practice Playground still available, AD-18).
3. 7-day suspension. 4. Profile online ban, parent appeal. Name issues → forced re-roll (doesn't count against 5/day).

9.8 Private Shows (R2)

FR-07-08, AD-19: host invites approved friends only; no join codes; 2–32 players; 0 XP / 0 Coins; host migrates to longest-standing; no bots (D-11). Min players: 2 solo / 2 full teams (AC1). Gated on friends consent. Block rules apply (blocked player can't be invited). Quick-chat same as public.

9.9 Spectating

FR-07-11, AD-36: on elimination → "Watch" or "Leave"; cycle players, party first, auto-switch 2 s after target finishes/eliminated; HUD shows timer + qualified count. Spectator view shows generated names only; player card actions (mute/block/report) available. Spectators cannot be seen by players and their emotes are not broadcast to the lobby **[Design]**. Leaving returns to menu; rejoin party at show end (AC3).

9.10 Disconnect / reconnect UX

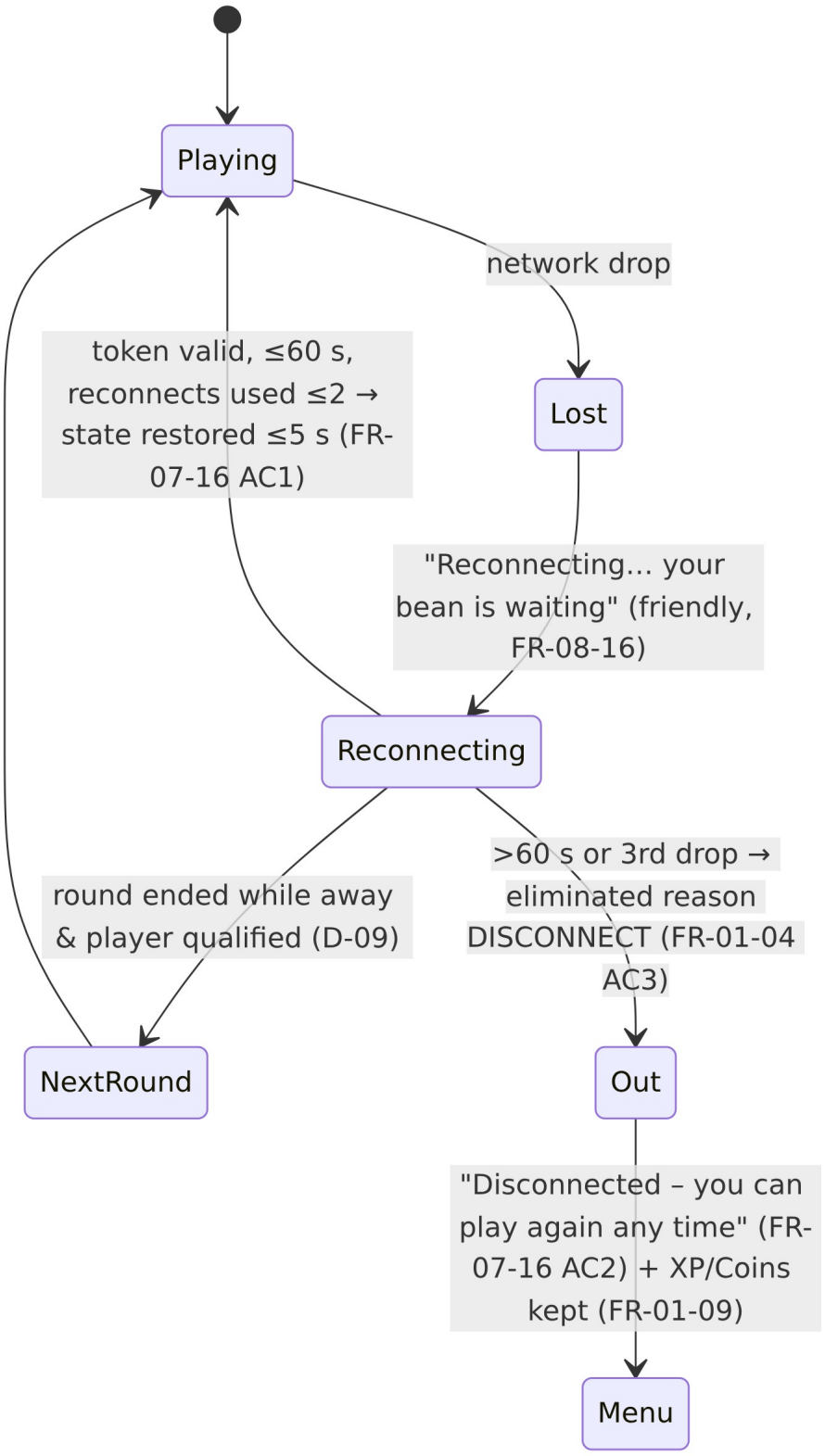


Figure 9.6 — Disconnect and reconnect states

- Bean frozen (race: can be knocked; survival/team: counts present); team points frozen (D-09, FR-07-16 AC3). Quota not recomputed (FR-01-04 AC3).
- Party members are not blocked; they continue (D-09). No leaver penalty (AD-37). Trophy idempotent on reconnect (FR-06 AC1).
- Crash-restart: client offers "Jump back in?" on launch within window (D-09 AT).

- Playtime limit reached → applied at show end, never mid-round (FR-11-15 AC2).

9.11 Parental dashboard & controls

Feature	Default	Effect timing	Req
Consent per purpose: online play, friends, Builder sharing	Off until VPC	Withdraw ≤ 1 h + deletion starts	FR-11-02
Approve/remove friends, pending list	—	≤ 5 min / next show	FR-11-08, FR-11-10
Quick-chat	On (preset only)	next show, ≤ 5 min	FR-11-08
Builder / sharing	Builder per scope; sharing Off	—	FR-11-08, AD-20
Break reminder	60 min (30–120), only parent can disable	next show end	AD-23, FR-11-15
Daily limit	Off	end of show	FR-11-15
Cross-play (Later)	On when shipped, parent can disable	—	FR-07-05, AD-51
Activity: playtime/day 30 d, friends, reports made, blocks, Trophies, Season Path	—	data ≤ 15 min old	FR-11-10
Moderation actions on my child + appeal	—	on action	FR-11-17 **(09-I-10)**
Purchases	Off, cannot enable in MVP; history empty	—	FR-11-09, AD-12
Data view / export JSON (≤72 h) / delete (≤30 d)	—	—	FR-11-12
Weekly summary email	Opt-in	weekly	FR-11-19

Parent gate (sum-in-words, 3 fails → 5 min lock, back doesn't bypass) + Parent Account sign-in for any change (FR-11-07). No settings loosening from the child profile (FR-11-11).

9.12 Consent & data flows

9.12.1 Onboarding / consent

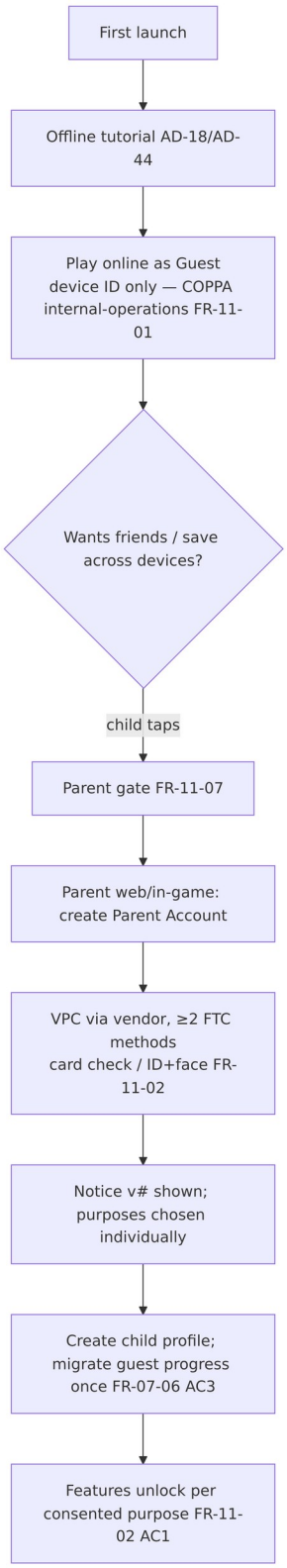


Figure 9.7 — Onboarding and consent data flow

9.12.2 Data inventory (social/safety slice — feeds FR-11-01 AC1)

Element	Purpose	Basis	Retention	Consent?
Device/profile pseudonymous ID	run game, security, progress	COPPA internal ops / legitimate interest + contract (GDPR-K)	life of profile; guest local	No
Truncated IP (/24, /48)	connection, security	internal ops	raw ≤ 24 h	No (FR-11-16)
QoS RTT per region	region choice	internal ops	90 d raw (AD-21)	No
Hidden MMR	matchmaking	internal ops (FR-11-14)	profile life	No
Quick-chat/emote ID logs	moderation context	safety	90 d raw; attached to report per §9.12.3	No
Friend graph, blocks	social features	consent "friends"	until removed / deletion	Yes (friends)
Reports made / about	safety	legal obligation / safety	§9.12.3	No
Replay snippets (flagged)	moderation	safety	14 d (AD-21) unless legal hold	No
Parent email, consent records	VPC, notices	legal obligation	retained as legally required (NFR-10-21)	—
Anti-cheat device data (if vendor)	security	must be in DPIA	per DPIA	NFR-10-13, NFR-11-01

9.12.3 Retention resolution for safety evidence

Default flagged-replay retention 14 d (AD-21). **Legal hold** extends retention for cases escalated to NCMEC/CEOP/law enforcement per statutory preservation duties (US 18 U.S.C. §2258A preservation, currently 1 year) — covered by the "legally required safety records" carve-out in FR-11-12 AC2 / NFR-10-21 (09-I-05). Counsel sign-off (DEP-10).

9.12.4 Regulatory mapping (summary)

COPPA: child-directed, VPC before non-internal collection, parent review/delete (FR-11-01/02/12). UK Children's Code: Std 2 DPIA (NFR-11-01), 7 default settings (FR-11-11), 8 minimisation (AD-21), 10 geolocation off (FR-11-16), 11 parental controls transparency — child sees a small "a grown-up can see your play time" icon **[Design]**, 12 no profiling (FR-11-14), 13 no nudges (AD-15), 15 online tools (FR-07-14/15, FR-11-06). GDPR Art 8: parental consent for consent-based processing; UK age 13, EU member-state 13–16 — we treat all as children, so VPC is used regardless (AD-01). OSA 2023: children's risk assessment (NFR-11-01), reporting + complaints (FR-11-06/17).

9.13 Incident response

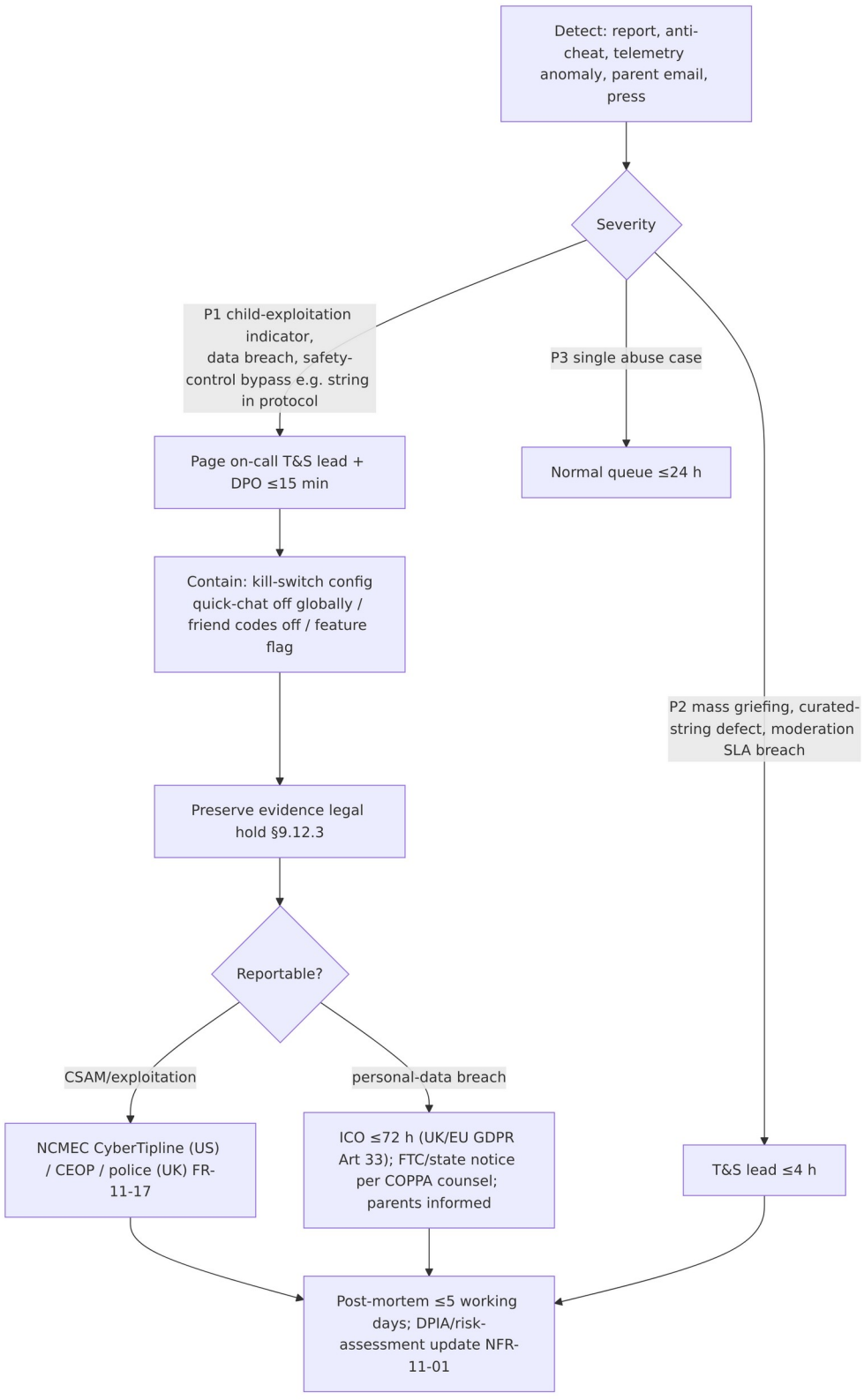


Figure 9.8 — Incident response

Kill switches **[Design]** (config-driven, NFR config req): `quickchat.enabled`, `friendcode.enabled`, `privateshows.enabled`, `phrase.<id>.enabled`, `namegen.word.<id>.enabled` (forces re-roll for affected names). Runbooks + contacts approved pre-launch (FR-11-17 AC1); monthly SLA report ≥ 95 % (AC2). Admin tools MFA + audit logs (AD-41).

9.15 Verification hooks

T-FR-01-02, T-FR-07-02/03/06/07/08/11/14/15/16, T-FR-11-01…19, T-NFR-11-01, T-NFR-10-13/21. Added design tests: picture friend-code entry by 6–8 year olds (≥80 % unaided); both-parent approval E2E; spectator chat scope; leader-disconnect-in-show; bot-report discard; kill-switch drill (each flag effective ≤ 5 min); legal-hold retention audit.

10 Builder (Creative) and Live Ops

Source: `10-creative-liveops.md` (23 Oct 2026 (no version stated)). **Authority:** Builder and live ops; yields to Section 8 on systems and Section 9 on child safety.

Label rule (v3): every number here is a **Design estimate** unless marked **Ref** (Fall Guys community value) or quoted from a requirement. The Fall Guys research file (src 09-creative-liveops.md) is reference only, not requirements.

10.0 Release map (respecting v3 priorities)

Area	Req IDs	Priority	Release
Season container, News Board, level disable/rotation config, Season Path, Challenges, telemetry	FR-09-19, FR-09-26, FR-01-03 AC5, FR-06-02, FR-06-13, NFR-10-12	Must/Should	**MVP**
XP boost events, limited-time shows/events	FR-09-21 (Could), FR-09-23 (Should)	Could/Should	**R2**
Builder editor, publishing, codes, likes, discovery, Level Hop	FR-09-01…17, FR-01-08	Could	**Later**
Builder moderation	FR-09-18	Must *when Builder ships*	**Later** (gate for Builder launch)
Builder SURVIVAL / POINTS modes	FR-09-09	Could	**Later+1** (after COURSE)
Mobile/console play of shared levels (no building)	FR-09-01 AC1	Could	**Later**

Retired IDs not reused: FR-09-20→FR-06-02, FR-09-22→FR-06-13, FR-09-24→FR-01-03, FR-09-25→FR-01-07 (Won't).

10.A Part A — Builder (Release: Later)

10.A1 Design pillars

1. **Build in 60 seconds** — a child places a start, a few platforms and a finish and plays it within one minute (templates, not blank canvas).
2. **Can't make it wrong** — invalid actions are prevented with pictures, not error text (FR-08-19 friendly messages).
3. **Nothing to type** — no free text anywhere (AD-02, FR-09-15); stickers curated (FR-09-07).
4. **Safe by construction** — private by default; sharing is a parent-granted permission (FR-09-01, FR-11-08).
5. **Proud, not pressured** — feedback is likes only, no dislikes, no public counts used as rankings for kids (FR-09-16).

10.A2 Access & permissions (FR-09-01, FR-11-08)

- Entry: Main menu tile "Build!" — hidden entirely when parent has Builder off (FR-09-01 AC1). PC/Windows only.
- Permission states per child profile:

State	Build	Playtest	Share to approved friends by code	Submit for public (Builder Picks/Level Hop)
Builder off	–	–	–	–
Builder on, sharing off (**default**)	✔	✔	–	–
Builder on, sharing on	✔	✔	✔ (after automated checks)	✔ (goes to human pre-moderation)

- When a child taps Share with sharing off: friendly card "Ask a grown-up to turn on sharing" → sends parent approval request email (allowed category in FR-11-19). No nag repeat within 7 days.

10.A3 Editor UX (kid-friendly)

Layout (1920×1080 baseline, scales to 1280×720):

- Left: **Object Drawer** — 7 category tabs as big icons (FR-09-02): Platforms, Blocks, Moving, Obstacles, Decorations, Interactions, Effects. Each tile: picture, cost pip, cap badge ("3/10").
- Top: **Budget Bean Jar** meter (FR-09-03 AC1) — a jar filling with beans + "1,240 / 2,500". Turns amber at 80 %, red at 95 %.
- Right: **Properties panel** (only when an object is selected): colour swatches (curated palette, FR-09-07), material, visible/collide toggles, physics, destructible, speed/direction sliders.
- Bottom bar: Undo / Redo (50 steps, FR-09-05 AC3), **Play** ▶️ (FR-09-11), Level Settings ⚙️, Save, Share.
- Camera: orbit (RMB), pan (MMB / WASD), zoom (wheel); "Bean-eye view" button drops to player height.

Modes: *Beginner* (default for profiles aged 6–8 by parent-set age band) hides Logic, numeric entry and physics/destructible toggles; *Builder Pro* reveals all. Same data, filtered UI — no separate requirement needed (Design addition, traces FR-09-05/07/08).

Tools (FR-09-05/06):

Tool	Behaviour	Kid UX
Move	Gizmo arrows + grid snap (0.5 m default, toggle 0.25/1/off)	Snap on by default
Rotate	15° snap rings	
Scale	Uniform or limited; hidden for non-scalable (AC1)	Handles absent, tooltip picture "This one can't stretch"
Precise entry	Numeric fields clamped (AC2)	Pro mode only
Copy/paste/duplicate	Respects caps (FR-09-04 AC1) and budget	Refused with cap badge shake
Overlap check	Red ghost + 🚫 icon; cannot confirm (FR-09-06 AC1)	
Link (Logic)	Drag "string" from Button/Plate/Zone to Controller; lines drawn (FR-09-08 AC1)	Pro mode; max 10 controllers

Starter templates (Design, Later): "Straight Dash", "Spinny Island", "Button Bridge", "Blank". Each a valid COURSE with Start + Finish so that publishing gates are met early. Template cost counts toward budget.

Level settings (FR-09-10): timer 1–10 whole minutes (default 3), max players up to 32 (AD-05), qualify count, sky, music (from the Wobblebeans original music library — 06-A-03, REC-23), theme. Out-of-range input rejected (AC1).

Game modes (FR-09-09): COURSE only at Builder launch; required objects (Start Point/Line, Finish Line) shown as a checklist "To share you need: ✔️ Start ✖️ Finish". SURVIVAL/POINTS Later+1.

10.A4 Object budget & caps (FR-09-02/03/04/08)

- Budget **2,500 units** (Ref value, configurable server-side). First Start Line and first Finish Line free (Ref).
- Physics/destructible toggles add cost (FR-09-03). Controllers: 5 + 15 per linked object, max 10 (FR-09-08).
- Catalogue is data-driven (FR-09-02 AC1): `object_id, category, cost, max_count, overlappable, scalable(uniform|limited|none), physics_cost, destructible_cost, review_id (FR-12-08), object_version, mode_mask`.

Example object	Cost (Design)	Cap
Basic platform 4×4	5	–
Tilt Plank	25	10
Spinning plate	20	50 (Ref)
Turnstile	30	10 (Ref)
Slime floor tile	4	200 (Ref)
Start Point	2	40 (Ref)

Example object	Cost (Design)	Cap
Group Start Zone	15	4 (Ref)
Effects (each)	10	50 total (FR-09-04)
Controller	5+15/link	10
Sticker decal	3	100

- Server re-computes cost on publish; tampered files exceeding budget/caps are rejected (FR-09-18 AC1).
- Rebalance policy: published levels grandfathered at published cost; removed objects become inert decorations and trigger re-moderation (Ch.9 §D edge cases). Each level pins `catalogue_version`.
- Performance guard (Design addition → **10-ISS-10**): server also checks a *perf score* (dynamic objects × players) against a ceiling so levels run on minimum PC spec at 32 players.

10.A5 Testing & publishing pipeline

```
Draft (local+cloud save) → Playtest (FR-09-11, <3 s switch)
 → Verification run: creator must qualify/finish on CURRENT version (FR-09-12 AC1)
 → Automated checks (FR-09-18 stage 1): budget, caps, bounds, required objects,
   sticker/arrangement classifier, perf score, schema (no free text — FR-09-15 AC1)
 → [sharing on?] → SHARED-FRIENDS: 4-4-4 code issued (FR-09-13)
 → [creator taps "Ask to be a Builder Pick"] → Human pre-moderation queue (stage 2)
 → APPROVED-PUBLIC: eligible for Builder Picks / Level Hop (FR-09-17, FR-01-08)
 → Reports (stage 3, FR-11-06) → Hidden-pending-review (3 reports/1 h) → Takedown or Restore
(stage 4)
```

- **Level states:** Draft · Verified · Shared-Friends · In-Review · Approved-Public · Hidden · Removed. Every transition logged with actor + reason.
- **Republish (FR-09-14):** same code; new version re-runs verification + automated checks; Approved-Public levels drop to In-Review and the *last approved version* stays live until the new one passes (resolves 10-ISS-07). Lobbies keep their loaded version (AC1).
- **Names/tags (FR-09-15):** name picker = 3 wheels (Adjective + Noun + optional Number 1–99), e.g. "Wobbly Pudding 7". Word lists reviewed (FR-12-08); a combination denylist blocks unlucky pairs. Up to 3 tags from fixed list (Easy, Tricky, Spinny, Jumpy, Slimy, Logic, Short, Long, Silly, Team-play).
- **Level codes (FR-09-13):** 12 chars from 30-symbol alphabet without 0/O/1/I/L → $30^{12} \approx 5.3\times10^{17} \geq 10^{15}$ ✔️. Random, non-sequential; 10 entries/min/profile; friendly error for invalid/removed.

10.A6 Safe sharing & moderation of kid-made levels (FR-09-18, FR-11-06, FR-11-17, FR-11-18)

Threats specific to kid UGC & mitigations

Threat	Mitigation	Trace
Offensive symbols/words built from blocks or stickers	Classifier renders top-down + 4 side + spawn-view captures of whole level (not just stickers) → image classifier + text-OCR; positive → human queue	FR-09-18 stage 1 (10-ISS-05)
Free-text side channels	No text fields; protocol test rejects any	FR-09-15 AC1
Stranger contact via codes posted on YouTube/social	Shared-Friends levels playable only by approved friends of creator; public code play only for Approved-Public	AD-20, 10-ISS-03
Creator identity exposure	Level card shows creator's generated display name only; no profile link for non-friends	FR-11-11
Impossible/troll levels	Mandatory verification run; "Skip" 1 s hold in Level Hop	FR-09-12, FR-01-08
Modded level files	Server-side validation + signature	FR-09-18 AC1
Brute-force code scanning	Rate limit + large space	FR-09-13 AC2
Review-bombing	Likes only	FR-09-16
Upsetting content slipping through	Report in ≤2 taps ("Bad name or picture", "Something upset me"); hide at 3 independent reports within 1 h; human review ≤24 h	FR-11-06, FR-09-18 AC2
Grooming/exploitation signal	Escalation path (NCMEC / CEOP)	FR-11-17

Human pre-moderation checklist (per level): PEGI 3 / ESRB E + UD-4 tone (FR-11-18); no symbols/words/likeness of real people/brands; no Fall Guys recreations (originality, FR-12-08); playable & fair on

fly-through; name combo ok. SLA target (Design): 72 h for submission; reports 24 h (FR-09-18 AC2). **Moderator console:** queue sorted by report count then age; 3D fly-through, auto-captures, version diff, one-click Approve / Reject (picture-coded reason sent to child: "Not quite right for Builder Picks — try changing your decorations") / Takedown / Escalate. Rejections are kind, never shaming; no strikes visible to children; repeat offence (3 takedowns/30 days) → sharing auto-paused and parent notified.

10.A7 Discovery (FR-09-17, FR-01-08)

- **Builder Picks:** curated rows (≤6 rows × 10 levels) — "Staff Favourites", "This Season's Theme", "Great First Levels", "Logic Puzzles". Pinned server-side (AC1). Refresh weekly (Design).
- **Level Hop:** random playlist of Approved-Public levels supporting ≥20 players, lobbies up to 20 (FR-09-17 / FR-01-08 AC2). Selection weighted: 60 % curated pool, 30 % recent approved, 10 % newer creators with <3 approved levels (fairness, no personal profiling — FR-11-14).
- **My Levels / Friends' Levels:** grid of own and approved friends' shared levels.
- **Enter Code:** 4-4-4 entry with large on-screen keypad. ⚑ This conflicts with FR-11-03 AC3 (no on-screen keyboard or text field in a child profile); the entry method is open (REC-24, MRG-03).
- No leaderboards of creators, no follower counts. Creator sees own likes as a badge tier (10/50/250/1000) rather than raw counts publicly.
- Official playlists (Main Show) use first-party levels only (FR-09-17).

10.A8 Builder acceptance (design-level, maps to T-FR-09-xx)

- Child 8–10 builds & plays a template level in ≤60 s (moderated playtest, 80 % success).
- 0 shared levels without verification record; 0 Discovery levels without pre-mod approval (FR-09-17 AC2).

10.B Part B — LIVE OPS

10.B1 Cadence (FR-09-19, FR-06-02; AD-11)

- **Season = 8 weeks**, UTC start/end server-configured, no client patch needed (FR-09-19 AC2). Client shows end date, never countdown (AD-15, AC1).
- **Mid-season drop at week 4** (Design): 1–2 new/variant levels, cosmetics, News Board post.
- **Client patch cadence:** every second season, aligned to season start (content otherwise data-driven).
- Annual calendar (6.5 seasons/yr): S1 Launch (MVP) · S2 · S3 · … with themed events slotted in R2+: Autumn egg/leaf-collecting (Oct), Winter (Dec), Spring (Apr), Summer (Jul–Aug). Culturally neutral themes, no religious holidays as themes (UD-4).

10.B2 Season content budget (Design per season)

Item	MVP season	R2+ season
New rounds/variants	2 + 2 variants	3 + 3 variants
Season Path	70 tiers (FR-06-02)	70
Cosmetics	~25	~30
Challenges	Daily/weekly/season sets (FR-06-13)	+ event challenges
Limited-time shows	–	2 (FR-09-23)
XP boost windows	–	≤2 weekends (FR-09-21)
Builder Picks themed row	–	Later

10.B3 Event templates (R2; FR-09-21, FR-09-23)

All events: UTC schedule, appear/disappear without patch, no countdown, rewards earnable in ≤4 h total play (FR-09-23) and event window ≥7 days so ≤ ~35 min/day suffices (10-ISS-08). Missed rewards are never framed as loss.

Template	Config fields	Example
LTS – Limited-time show	show_id, playlist, slot rules, start/end, reward track (≤6 rewards), min/max players	"Egg Scramble Week"
XP Boost	multiplier (×2 default; cap ×2 for kids — 10-ISS-06), start/end, scope (all/show)	"Double XP this weekend (Sat 9 – Sun 10 Nov)"
Theme Week	level pool weight overrides (FR-01-03), skybox skin, News Board post	"Slime Week"
Community Build Challenge (Later)	theme tag, submission window, Builder Picks row	"Build a Bouncy Castle" — picks, no prizes ranked
Returning Favourite	re-enable retired LTS from vault	Winter show returns

Event schema: `event_id, type, title_loc_key, start_utc, end_utc, playlist_ref, reward_table_ref, xp_mult, news_post_ref, review_id, kill_switch`.

10.B4 Content pipeline

1. **Plan** (T-12 wk): theme brief, content list, originality brief (FR-12-08).
2. **Build** (T-12→T-5): levels in data (FR-02-01), cosmetics, loc keys.
3. **Review** (T-5): content & originality review records (FR-12-08 AC1), PEGI3/UD-4 (FR-11-18); child playtest for new levels (6–8 & 9–12).
4. **Cert/patch** (T-4) only if client code changes.
5. **Config staging** (T-2): season/event/rotation config in staging env; automated schema + time-zone tests (AD-33).
6. **Soft launch** (T-0): 10 % population canary 24 h for new levels (Design) → full.
7. **Operate**: dashboards, kill switch, News Board (FR-09-26, review record AC2).
8. **Retro** (T+8): KPI review feeds next season.

10.B5 Rotation rules (FR-01-03, FR-02-01)

- Selection per FR-01-03: slot-valid, N within min/max, divisible by team divisor; no repeat in a show (AC2); no same type twice in a row (AC3); fallbacks Tile Drop / Tile Return / Bumpy Parade (AC4).
- Live-ops levers: per-level `enabled`, `selection_weight`, schedule windows (AC5; effective for shows starting ≥60 s after change).
- Design rules: new levels get ×1.5 weight for first 2 weeks; levels with qualification rate outside 30–80 % (Design band) or crash/desync spikes are auto-flagged for weight reduction (human-approved, not auto-applied); every level stays in rotation at least 1 season before vaulting; never vault the fallbacks.
- Level Hop rotation: §10.A7.

10.B6 Roadmap

Phase	Live ops	Builder / UGC
MVP	Season container (FR-09-19), Season Path free track (FR-06-02), Challenges (FR-06-13), News Board (FR-09-26), rotation config + kill switch (FR-01-03 AC5), telemetry & Live-Ops dashboard v1 (NFR-10-12)	Internal-only level authoring tools (data-driven FR-02-01) — groundwork for Builder catalogue
R2	LTS & events (FR-09-23), XP boost (FR-09-21), event templates, canary rollout, Moderation ops (FR-11-17) scaled up	Builder tech spike + closed creator playtests with children (internal); moderation classifier training
Later	Community Build Challenges, themed Builder Picks rows	Builder PC COURSE (FR-09-01…18), Level Hop (FR-01-08), mobile/console play-only; then SURVIVAL/POINTS

Builder launch gate: FR-09-18 stages 1–4 operational, moderator staffing for 72 h/24 h SLAs, parent controls (FR-11-08) live, classifier precision/recall reviewed.

10.B7 KPIs & telemetry dashboard (NFR-10-12, FR-11-14)

Constraints: first-party, pseudonymous, no PII, no third-party SDKs; KPIs used for content health, **not** individual targeting or monetisation optimisation (FR-11-14). No KPI may reward longer child sessions per se — wellbeing guardrails are primary.

Dashboards

1. **Season Health** — DAU/WAU (aggregate), season retention D1/D7/D30, Season Path median tier by segment vs target (casual ≥T35 by wk 8, FR-06-02 AC4), challenge completion %, banked-slot usage.
2. **Round/Level Health** — per level: pick rate, qualification rate, avg duration, quit/disconnect during round, desync/crash rate, skip rate (Level Hop); traffic-light vs bands.
3. **Event Performance** (R2) — participation %, median minutes to full rewards (target ≤4 h, FR-09-23), reward claim %.
4. **Wellbeing & Safety** — median session length, % sessions hitting parent limit, break-reminder acceptance (FR-11-15), reports per 1k matches, report SLA compliance (FR-11-06/17), takedowns.
5. **Builder (Later)** — levels created/verified/shared/submitted, approval rate, time-in-queue, classifier hits & false-positive rate, Level Hop plays & skip rate, % creators with ≥1 approved level.
6. **Ops** — config change log, kill-switch events, canary comparisons.

Core telemetry events (all with pseudonymous id, build, platform, UTC ts): `season_start_seen, show_start, round_start{level_id,slot,N}, round_end{result,duration}, show_end, challenge_complete, path_tier_up, event_join, event_reward_claim, news_view, report_submit{type}, builder_open, builder_object_place{object_id}, builder_playtest, builder_verify{pass}, level_share, level_submit, level_moderation{decision}, levelhop_play{level_id,skip}, like_give`. Retention follows NFR-10-21 / AD-21 as set in §8.6: raw events 90 days, aggregates 2 years (REC-22); deletion propagates per FR-11-12.

Appendix A — Design Issues Register

This register merges the issue lists of all ten source documents. They held **126** entries. **10** were duplicates of another entry and were merged into the same row, leaving **116 unique issues** (DI-001 to DI-116). Source IDs are kept and prefixed with the source document number (for example 04-UX-I8). Where several sources raised the same issue, the row lists all source IDs, with the issue text and resolution of each. Issues that the merge itself raised are listed separately in A.2.

Follow-up status (derived from each resolution and from the sources' own open-item notes):

Follow-up	Issues
Resolution adopted in the design	84
Requirement change proposed (analyst)	12
Analyst sign-off before M0 lock	5
Change request to requirements owner	3
Owner/analyst: amend FR-06-01 AC3 (Appendix C, item 15)	1
Owner confirmation (Appendix C, item 13)	1
Analyst: amend FR-11-03 AC3 (Appendix C, item 14)	1
Owner decision Q-D2 (Appendix C, item 8)	1
Owner decision	1
Owner decision Q-C3 (Appendix C, item 6)	1
Owner decision Q-C2 (Appendix C, item 5)	1
Owner decision Q-C1 (Appendix C, item 4)	1
Counsel sign-off, DEP-10 (Appendix C, item 10)	1
Counsel sign-off, DEP-10 (Appendix C, item 11)	1
Ops sign-off (Appendix C, item 12)	1
Open: thresholds to be set with engineering	1

A.1 Register

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-001 (01-ISS-01)	1	Contradiction: FR-01-09 says quitting mid-show "forfeits only the participation bonus" (per-round XP kept); FR-06-01 says "Quitting a show grants 0 XP". Also tension with AD-37 (no leaver penalty) and FR-01-04 AC2 (eliminated keep earnings).	FR-01-09 AC4 vs FR-06-01 (body + AC3), gapfill-C §1	**Quit = keep all per-round XP/Coins already earned; forfeit only the +100 participation XP** (and Final/win bonuses not reached). Read FR-06-01 "0 XP for a quit" as "0 *participation* XP". Proposed edit: FR-06-01 AC3 → "0 participation XP is granted for a quit". Leaving *after* elimination is not a quit (participation kept).	Owner/analyst: amend FR-06-01 AC3 (Appendix C, item 15)
DI-002 (01-ISS-02)	1	Duplicate / overlap: Two features describe the same offline tutorial space without saying they are one.	FR-08-17 (tutorial "in an offline space") vs FR-01-12 (Practice Playground "contains … tutorial stations")	One offline space: the **tutorial is a guided path through the Practice Playground** (§1.8.2). FR-08-17 owns tutorial flow/ACs; FR-01-12 owns the free-play space.	Resolution adopted in the design
DI-003 (01-ISS-03, 04-UX-I8)	1, 4	Wrong cross-reference: Cite "tutorial stations of **FR-08-16**" — FR-08-16 is *Error, offline and empty states*; the tutorial is **FR-08-17**. **Also raised as 04-UX-I8:** FR-01-12 refers to "the tutorial stations of **FR-08-16**", but the tutorial is FR-08-17 (FR-08-16 is errors).	FR-01-12 body, US-01-7	Treated as FR-08-17. Fix the reference. **04-UX-I8:** Treated as a typo for **FR-08-17**. Fix in the docx. *Merge note:* A further instance was found in the merge: the assists table of Section 3 (§3.5) also cited FR-08-16 for the Practice Playground; corrected to FR-08-17.	Resolution adopted in the design
DI-004 (01-ISS-04, 02-I-8)	1, 2	Contradiction / ambiguity: FR-01-05 AC5: "If 2–10 players are left **before round 4**, the next round is a Final" — implies Final only from round 4. AD-29: Final whenever N ≤ 10 *or* round_idx = MAX_ROUNDS (5). Also FR-02-01/FR-01-03 list slots R1, R2, R3, F only, while gapfill-B §3 has R4 and MAX_ROUNDS = 5. **Also raised as 02-I-8:** FR-01-05 AC5 says "2–10 left **before round 4** → Final", but the slot rule makes any round a Final when N ≤ 10.	FR-01-05 AC5 vs AD-29, gapfill-B §3.2–3.4, FR-02-01, FR-01-03; FR-01-05 AC5, AD-29	**AD-29 governs**: Final at any round when 2 ≤ N ≤ 10, or at round 5. Slot set is {R1, R2, R3, R4, F}, with R4 sharing R3's weights. Proposed edit: FR-01-05 AC5 → "If 2–10 players remain after any round, the next round is a Final"; add R4 to FR-02-01. **02-I-8:** The AD-29 slot rule is followed (Final whenever N ≤ 10). AC5 should be reworded.	Requirement change proposed (analyst)
DI-005 (01-ISS-05, 02-I-1)	1, 2	Scope conflict: R2 and R3/R4 slot weights include Logic 10, but no Logic level exists in MVP — the weights can't be met and the gapfill-B §3.4 AT ("type mix within ±5% of weights") would fail. **Also raised as 02-I-1:** R1 type weights are Course 80 / Survival 10 / Points 10, but no MVP Survival or Points level is tagged R1.	gapfill-B §3.2 slot weights vs FR-02-05 (Logic = Could, R2); B §3.2, FR-02-11	MVP uses **renormalised weights without Logic** (§1.4.4). Original weights apply once Logic ships (R2). **02-I-1:** Weights renormalise over the valid candidates, so R1 is 100% Course from 2 candidates. This meets the FR-02-11 AC3 minimum of 2. Logged for R2 content.	Resolution adopted in the design
DI-006 (01-ISS-06)	1	Contradiction (numbers): D-11: fill "to 32" with bots if < 20 humans at 60 s, but cap is 12 bots (≤ 40%) — with < 20 humans, 32 is unreachable. D-11 AT expects a show to start with **10** humans (10 + 12 = 22). D-12: "T = 60 s ≥ 20 + bots" is ambiguous (20 humans, or 20 total?).	FR-01-02 AC7 / D-11 default vs D-11 AT, D-12	At 60 s: **start if humans ≥ 8 (Main Show min) and humans + bots ≥ 20, with bots = min(12, 32 − humans)**. A lobby of 10 humans starts with 22 beans. Slot/final selection uses the real N (FR-01-03 handles any 8–40).	Resolution adopted in the design

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-007 (01-ISS-07)	1	Ambiguity: "Skipped after the first completion" vs "Pressing Play straight away is allowed" on first run.	FR-08-17	Before first completion: a **"Play now"** button bypasses it (offered again next launch). After completion: tutorial no longer auto-runs, replayable from Settings. No "Skip" during a station is needed.	Resolution adopted in the design
DI-008 (01-ISS-08)	1	Terminology / consistency: Reference terms leak into normative chapter text; two different disclosure strings.	Ch.1 "Area" and core-loop text ("Crown wins", "Game Selector") vs glossary (Trophy, Show Picker); SC-03 inventory string "AI players may join" vs FR-01-02 AC7 "Some players may be computer players"	Use **Trophy** and **Show Picker** throughout; single disclosure string from FR-01-02 AC7 ("Some players may be computer players") — simpler for 6–8 (FR-08-19). *Merge note:* Further instances found in the merge ("Robot friends may join" in 04; reference terms in 04 wireframes) were corrected (REC-03, REC-08). *Merge note:* Further instances found in the merge ("Robot friends may join" in 04; reference terms in 04 wireframes) were corrected (REC-03, REC-08).	Resolution adopted in the design
DI-009 (01-ISS-09, 10-ISS-02)	1, 10	Wrong cross-reference: "passed moderation (FR-09-18, **FR-11-12**)" — FR-11-12 is *Data access, export and deletion*; moderation ops is **FR-11-17**. **Also raised as 10-ISS-02:** FR-01-08 AC2 cites FR-11-12 (data deletion) as a moderation source — wrong ref.	FR-01-08 AC2	Treated as FR-09-18 + FR-11-17 (Later scope). **10-ISS-02:** Should cite FR-11-06/FR-11-17. Flag as ID typo CR.	Change request to requirements owner
DI-010 (01-ISS-10, 02-I-5)	1, 2	Missing rule: Fallback for any empty slot is "Tile Drop", a 5:00 Final-type level; using it mid-show (non-final slot) would end in one winner or blow the show length. **Also raised as 02-I-5:** Tile Drop is the Main Show fallback for *any* slot (`FALLBACK[mode][slot]`), but as a last-standing Final it would end the show early in a non-final slot.	FR-01-03 AC4 / gapfill-B §3.3 fallback; FR-01-03 AC4, FR-01-05	Fallback in a non-final slot = **Tile Drop with a qualifier quota (60%, 1:45 limit)**; in the F slot = normal Tile Drop Final. Requires FR-02-01 per-slot overrides (FR-01-04 AC4 allows per-show limits). **02-I-5:** In a non-final slot the fallback runs Tile Drop under a **survival-quota rule** (it ends at the quota) with the same layer table. **Owner confirmation needed.**	Owner confirmation (Appendix C, item 13)
DI-011 (01-ISS-11)	1	Missing rule (fairness/child-safety): Points rounds include grab-to-steal tails; for R1 new players this makes another child the direct cause of elimination.	FR-02-04 (tail grabs) in MVP Points pool vs FR-06-13 AC2 spirit, FR-08-03	GDD content rule: **grab-steal Points levels are not allowed in R1**; MVP Points level (L6) uses zones/buttons.	Resolution adopted in the design
DI-012 (01-ISS-12)	1	Ambiguity: A trophy-grab final with respawns could time out with many "winners", diluting the Trophy.	FR-01-05 AC3 (trophy-grab time-out all win) + respawn finals	Accepted (AD-10, P2), but L8 tuned so time-out occurs in ≤ 5% of runs (gapfill-B §1.2 AT); telemetry gate.	Resolution adopted in the design
DI-013 (02-I-2)	2	Summit Grab uses the reference player range of 2–9, but the Final slot fires at N ≤ 10. At N = 10 only Tile Drop would be valid, which breaks FR-02-11 AC3 ("final slot has 2").	FR-02-11 AC3, AD-29	Summit Grab maxP is set to **10**, and its spawn plaza is sized for 10.	Resolution adopted in the design

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-014 (02-I-3)	2	FR-03-13 says "Finals eliminate a player who falls into slime", but a trophy-grab climb with launch arms needs respawn.	FR-03-13, FR-01-05 AC1	Summit Grab has no slime. Falls use kill-plane respawn at a checkpoint (FR-03-13 AC1 allows a fall rule per level). FR-03-13 wording should be amended.	Requirement change proposed (analyst)
DI-015 (02-I-4)	2	Wall Dodge needs moving walls (B §4.1 "Sweeper/moving wall"), but that is not in the MVP subset of 11 obstacles. The MVP table lists Pusher Block.	FR-03-01, FR-02-11	Walls are built as a **Pusher Block wall variant** (K7): linear travel of 3→5 m/s, Δv 10. No new obstacle type is added.	Resolution adopted in the design
DI-016 (02-I-6)	2	The MVP table lists "grab" as a Tail Chase obstacle. Grab is a player verb, and Tail Chase is POINTS with a 60% quota, while a tail is held or not held.	FR-02-04, FR-03-01	Tails are seeded so that the number of tails = quota, making qualification exact. Bumper is the only kit obstacle. A 1.0 s re-grab immunity was added.	Resolution adopted in the design
DI-017 (02-I-7)	2	Checkpoint spacing conflicts: "25–45 m" against "43–67 m" after re-basing on gapfill-A.	FR-03-12, B §6.2	The **12 s fall-cost rule governs**, and spacing is set per course in playtest (4–5 checkpoints in this set).	Resolution adopted in the design
DI-018 (02-I-9)	2	The brief asks for difficulty variants, but named variant sets (FR-03-15) are R2. Only speed_mult (FR-03-04, Should) is MVP.	FR-03-04, FR-03-15	The MVP ships with speed_mult and layer-count variants only. Named swap variants are designed here but disabled until R2.	Resolution adopted in the design
DI-019 (02-I-10)	2	Spin Bar Δv is 9 m/s, above the 8 m/s knockdown threshold, yet A §5.4 says "stumble if low beam".	FR-03-03, A §5.4	Ragdoll is tagged per contact state: stumble when grounded, knockdown in the air. The grounded hit is capped at Δv 7.9 so it stays a stumble.	Resolution adopted in the design
DI-020 (02-I-11)	2	Survival rounds can be "survive the timer" or a quota, and the catalogue marks Wall Dodge as "survivors of timer".	FR-02-03 AC1, 02 catalogue	Both survival levels use **quota 60% plus a timer fallback** (everyone still in qualifies at time-out), matching B §1.2's "ends early when quota reached".	Resolution adopted in the design
DI-021 (02-I-12)	2	Tile Drop's 4-layer band (13–20 players) can never occur in the Main Show, because Finals run at N ≤ 10.	B §4.3, AD-29	The band is kept for fallback and R2 or config use. Main Show Finals use 3 layers for 8–10 players and 2 for 7 or fewer.	Resolution adopted in the design
DI-022 (02-I-13)	2	Slime Rise is typed COURSE, which normally means respawn, but slime contact puts the player out. Checkpoints under the slime are ambiguous.	FR-02-02 AC2, FR-03-13	Falls respawn only if the checkpoint is above the slime; otherwise the player is out. Slime contact always puts the player out.	Resolution adopted in the design
DI-023 (03-I-01)	3	Jump+dive reach AT in gapfill-A §3.1 says **5.8–6.3 m**, but FR-04-03 AC1 (AD-31) says **5.4–5.9 m standing / 8.9–9.5 m running**.	A §3.1 vs FR-04-03	FR-04-03 AC1 governs (later decision, derived from the values). Update gapfill-A §3.1 AT text.	Analyst sign-off before M0 lock
DI-024 (03-I-02)	3	Two stumble durations (impact 0.30 s, A §5.2; hard landing 0.35 s, AD-30) but only `land.hardStumbleTime` is a config key; landing knockdown vy ≤ −20 and stumble control 40 % have no keys.	A §5.2, §3.8	Add keys `ragdoll.stumbleTime` 0.30, `ragdoll.stumbleControl` 0.4, `land.knockdownVy` −20. Keep both durations (different causes).	Analyst sign-off before M0 lock
DI-025 (03-I-03)	3	"One air dive per airborne period" — unclear whether a ground dive (which adds +3 m/s up and goes airborne) consumes it.	FR-04-03	A ground dive *is* the dive for that airborne period; no second dive until grounded. Prevents dive-chaining.	Resolution adopted in the design
DI-026 (03-I-04)	3	FR-04-01 AC3 slows the bean "while grab is held" — ambiguous if nothing is grabbed (grab toggle makes this worse: kids may leave it on).	FR-04-01 AC3, FR-04-04 AC4	×0.6 applies while grab input is active (matches AC3 literally); in **toggle** mode, grab auto-releases after 1.0 s with no target, so toggle cannot leave a child stuck slow.	Resolution adopted in the design

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-027 (03-I-05)	3	No requirement covers haptics/controller vibration or its setting, yet FR-08-12 asks for comfort toggles and FR-04-15 adds gamepads.	gap	Add vibration intensity slider (0–100 %, default 70 %) to FR-08-12 options; propose as new AC.	Analyst sign-off before M0 lock
DI-028 (03-I-06)	3	Mobile grab-lock (src 04 §1.2) "lets player jump while grabbing", contradicting gapfill-A §4.3 / edge case "no jump/dive while grabbing".	src 04 §1.2 vs FR-04-04	Do not adopt grab-lock's jump exception; on touch, grab toggle provides the comfort benefit. Re-decide when mobile (Later) is scoped.	Resolution adopted in the design
DI-029 (03-I-07)	3	Grab while airborne not specified (ledges need it; player grabs in air would be abusive).	FR-04-04	Airborne grab targets **ledges and objects only**; player grabs need grounded grabber.	Resolution adopted in the design
DI-030 (03-I-08)	3	Ledge climb (FR-04-04 AC1) has no parameters (max ledge height, mantle time). gapfill-A §2.1 says a 2.0 m ledge is unreachable without dive/bounce.	FR-04-04 AC1	Add `grab.ledgeMaxHeight` = ledge top ≤ 0.4 m above feet (apex 1.5 m + 0.4 → 1.9 m, so 2.0 m ledges stay gated; >0.45 would break gating) and `grab.ledgeMantleTime` 0.45 s; validate with level-gating test.	Analyst sign-off before M0 lock
DI-031 (03-I-09, 05-I-2, 06-A-07)	3, 5, 6	FR-12-02 naming categories have no slot for locomotion/action animations (`emote`, `celeb` only). **Also raised as 05-I-2:** FR-12-02 has no token for texture maps or LODs; `part` list lacks `lod`; `category+part` redundant (`upper…top`). **Also raised as 06-A-07:** FR-12-02 parts (full, top, bottom, mask, thumb) do not fit audio (loops, stems).	FR-12-02	Add category `anim` (and part `full/top/bottom` reused as layer masks). **05-I-2:** Add suffixes `_alb/_nrm/_msk`, `_lod<n>`; keep redundancy (cheap lint check). Needs FR-12-02 amendment. **06-A-07:** Extend parts with `loop, intro, outro, stem` for `sfx/music` only.	Analyst sign-off before M0 lock
DI-032 (03-I-10)	3	Dive default key: Ctrl vs right mouse is "Unverified" in src 04 §1.1; Left Shift + Left Ctrl adjacency is fine for adults, awkward for small hands.	src 04 §1.1	Primary Ctrl / Shift as reference; ship mouse alternates (RMB dive, LMB grab) bound by default as secondary bindings.	Resolution adopted in the design
DI-033 (03-I-11)	3	Hold-to-mash (1 input / 0.25 s, −0.25 s each) means holding halves a 3.0 s grab to 1.5 s, while fast mashing (~8/s) frees almost instantly — accessibility option is weaker than mashing.	FR-08-12, A §4.2	Accept (cap kept), but cap mash credit at 4 inputs/s for everyone so hold users are within 2× of mashers; flag for playtest.	Resolution adopted in the design
DI-034 (03-I-12, 08-I-06)	3, 8	Ragdoll replication "root + 3 key bones" doesn't name bones; must align with split rig. **Also raised as 08-I-06:** Server ragdoll fidelity vs bandwidth: full-bone ragdoll replication breaks the D-07 budget	A §7, FR-12-04	Root, pelvis, spine_02, head (§3.3.1). **08-I-06:** The server simulates a reduced ragdoll (root + key bones). The client adds cosmetic secondary motion. Gameplay outcomes depend only on server bodies *Merge note:* Authority: Section 3 / gapfill-A for ragdoll physics; Section 8 for how it is replicated.	Resolution adopted in the design
DI-035 (03-I-13)	3	Camera other-player occlusion behaviour unspecified; with 32 beans, collision pull-in on beans would pop constantly.	FR-04-08	Camera ignores bean colliders; beans near camera fade (§3.6).	Resolution adopted in the design
DI-036 (03-I-14)	3	Auto-recentre has no rate limit; fast swings can cause discomfort (FR-08-20).	FR-04-08	Recentre at ≤ 90 °/s ease-in-out.	Resolution adopted in the design
DI-037 (04-UX-I1)	4	The brief asks for an "age gate" flow, but AD-01 / FR-11-01 rule out any age gate: everyone is a child.		**No age gate.** Replaced by the "Grown-ups" button → parent gate → Parent Account (§4.2.2). The age question is never asked of the child.	Resolution adopted in the design

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-038 (04-UX-I2)	4	FR-11-07 example says the adult must "type the answer", but FR-11-03 AC3 says a child profile has **no on-screen keyboard or text field anywhere**. Also, the kids "counting answers ≤ 20" AC (in the docx) must not be confused with the parent gate.		The gate uses a **digits-only NumberPad component** (not a text field and no letters), and the sum is written in words with an answer of 21–99, harder for 6–8s. Propose amending FR-11-03 AC3 to "no free-text field; the parent-gate numeric pad is excepted". Parent Account sign-in and creation open in a separate adult context (web via QR, or an adult-mode screen after the gate).	Analyst: amend FR-11-03 AC3 (Appendix C, item 14)
DI-039 (04-UX-I3)	4	The screen inventory lacks screens the requirements need: player card, friends/party, Season Path, break/limit, reconnect, name picker, kids' privacy explainer. The parental dashboard is folded into SC-11.		Added **SC-16…SC-22** as UX IDs (§4.1.1). Propose adding them to src 08 §1.	Requirement change proposed (analyst)
DI-040 (04-UX-I4)	4	The brief asks for touch-target sizes, but MVP is PC-only (AD-51). There is no requirement for target size at all.		Defined PC targets (≥ 64 px, primary ≥ 96 px at 1080p) and touch targets for Later (≥ 48 dp) in R5. Propose a new NFR.	Requirement change proposed (analyst)
DI-041 (04-UX-I5)	4	FR-08-12 AC3 (150% text) and FR-08-14 AC2 (+40% pseudo-loc) are tested separately. Together they can break layouts, and the combination is untested.		Design and test the **combined worst case** (150% + 40%). Propose adding it to T-FR-08-14.	Requirement change proposed (analyst)
DI-042 (04-UX-I6)	4	FR-08-18 bans countdowns on offers, but the Coin Shop and Season Path imply rotation and season end. It is unclear whether "shop refreshes in 3 h" or "season ends in 5 days" is allowed.		No countdown in the Shop or Season Path. The season end date appears as a static date only in Help. Shop items do not show rotation timers. Propose stating this in FR-08-18.	Requirement change proposed (analyst)
DI-043 (04-UX-I7)	4	Screen-reader / TTS support is not mentioned. It matters for pre-readers (6-year-olds) and visually impaired players.		Out of MVP scope, but every component carries a label key so OS TTS or narrated UI can be added. Suggest an open question for R2 ("read-aloud button").	Resolution adopted in the design
DI-044 (04-UX-I9, 06-A-10)	4, 6	Round intro (SC-04) and lobby timers could read as "countdowns on choices". **Also raised as 06-A-10:** UD-1 bans "countdowns", but 3-2-1-GO and final-seconds countdowns are gameplay.		They are not choices. Countdowns appear only on non-interactive game-state screens. Choice screens (Watch/Home, Play again) never have timers. **06-A-10:** Clarify: ban applies to commercial/shop pressure; gameplay timers allowed; no countdown audio in SC-09.	Requirement change proposed (analyst)
DI-045 (04-UX-I10)	4	Adult-facing screens (consent, dashboard) have no stated readability or contrast standard. Child rules (≤ 8 words, 28 px) do not fit legal text.		Adult screens: WCAG 2.2 AA, body ≥ 24 px at 1080p, plain-English notices. Child rules exempted. Propose noting this in FR-08-19's scope.	Requirement change proposed (analyst)
DI-046 (05-I-1)	5	FR-12-10 says art direction is *derived from* the repo and pending; work can't wait to M2.		This doc = provisional framework (environment, UI, VFX, readability, pipeline); character-dependent values (accent palette tuning, final themes) ratified at M2. Request owner amend FR-12-10 to "framework v0 at M0, final at M2".	Requirement change proposed (analyst)
DI-047 (05-I-3)	5	Budgets exist only for costume parts; none for obstacles, levels, VFX, UI.		Proposed budgets in §5.8/§5.7; validate in NFR-10-04 prototype (NFR-10-02 AC2).	Requirement change proposed (analyst)
DI-048 (05-I-4)	5	Seam tolerance 2 mm has no defined unit/scale for the bean.		Fix units = metres, bean height 1.00 m, Y-up. Add to `rig_spec.md`.	Resolution adopted in the design

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-049 (05-I-5)	5	FR-12-04 allows face bones **or** blend shapes — two pipelines.		Choose blend shapes (cheaper, easier for repo contributors); face bones disallowed.	Resolution adopted in the design
DI-050 (05-I-6)	5	FR-12-03: 2048² max source vs FR-12-05 1024² per part — ambiguous.		2048² allowed as source only; runtime downscaled to 1024²; validator warns not fails.	Resolution adopted in the design
DI-051 (05-I-7)	5	Chapter 3 "Themes and seasons" lists reference-game legacy themes (medieval, neon 4041, etc.). Risk to UD-2/FR-12-08 originality.		Treat as research only; use original world themes in §5.4.	Resolution adopted in the design
DI-052 (05-I-8)	5	FR-08-11 covers hazards/teams/rarity but not art-side rule for costumes resembling hazards.		Added readability rule 6 + checklist item 9; propose adding to FR-12-08 criteria.	Requirement change proposed (analyst)
DI-053 (05-I-9)	5	32 × 3 × 1024² costume textures risk VRAM on GTX 960 (≈ 2 GB).		BC7/BC5 compression, mip streaming, LOD2 uses 256² mip; verify in FR-12-05 AC2 test.	Resolution adopted in the design
DI-054 (05-I-10)	5	Q-D2 unresolved: 2D-only repo means ~40 costume parts need modelling (DEP-11 schedule risk).		Default assumption: mixed; plan modelling capacity; escalate Q-D2 to owner now.	Owner decision Q-D2 (Appendix C, item 8)
DI-055 (05-I-11)	5	No typeface/licence requirement for UI fonts.		Open-licence rounded sans; record in provenance (FR-12-07).	Resolution adopted in the design
DI-056 (06-A-01)	6	FR-08-07 Music is **Should**, yet it holds Must-grade obligations (original-only/IP risk, provenance) and FR-12/Ch.16 rely on it.		Raise FR-08-07 to **Must**, or split: originality/provenance AC2 = Must; distinct cues AC1 = Should.	Resolution adopted in the design
DI-057 (06-A-02)	6	Only music has a loudness target (−16 LUFS); no full-mix, true-peak, SFX/UI or LRA targets.		Adopt §6.5 table; add T-AUD-01.	Resolution adopted in the design
DI-058 (06-A-03)	6	Builder level settings say "music track (from the **licensed** library)", conflicting with "Original music only" (Ch.12) and FR-08-07.		Reword to "from the Wobblebeans original music library"; no UGC audio.	Requirement change proposed (analyst)
DI-059 (06-A-04)	6	"Gameplay-relevant" (FR-08-12 AC4) and "audio event list" (FR-08-08 AC1) are referenced but undefined.		§6.3 is the event list; GR defined in §6.7; flag stored in manifest.	Resolution adopted in the design
DI-060 (06-A-05)	6	FR-08-09 has 4 sliders; ambience and social (emote) sounds have no home.		Ambience and Social sit under SFX; optional "Other players' sounds" toggle (R2).	Resolution adopted in the design
DI-061 (06-A-06)	6	UD-1 kid-safety has no volume/hearing-safety requirement.		Add NFR: defaults, limiter cap, headphones mode, startle rule (§6.6); parent max-volume R2.	Resolution adopted in the design
DI-062 (06-A-08)	6	FR-12-03 fixes runtime OGG q6; some consoles prefer native codecs; mono/stereo unspecified.		Allow platform-native runtime codec; channel rules in §6.9.2.	Resolution adopted in the design
DI-063 (06-A-09)	6	No check that "original" music is not derivative of the reference OST.		Composer originality statement + similarity check in T-FR-08-07.	Resolution adopted in the design
DI-064 (06-A-11)	6	FR-08-08 bans violent/scary sounds but says nothing on sad/mocking outcome sounds, against FR-08-03 tone.		Add AC: elimination audio is major-key, no mocking/laughter/boos (§6.3.5).	Resolution adopted in the design
DI-065 (06-A-12)	6	Reward audio could create gambling feel (FR-11-18 bans gambling imagery but not sound).		Extend FR-11-18 to sound: no slot/coin-cascade/jackpot sounds.	Resolution adopted in the design

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-066 (06-A-13)	6	AD-53/AD-02 forbid voice but do not say if non-verbal bean vocals are allowed.		Allow non-lexical toy vocals only, with word-check (§6.8, T-AUD-03).	Resolution adopted in the design
DI-067 (06-A-14)	6	FR-07-15 block/mute does not mention emote/quick-chat sounds.		Muted/blocked players' social sounds are suppressed (§6.3.6).	Resolution adopted in the design
DI-068 (06-A-15)	6	FR-08-12 AC4 edge case only covers "sliders at 0"; deaf/hard-of-hearing players may keep audio on.		Visual twins always shown; Sound Ring optional (§6.7).	Resolution adopted in the design
DI-069 (07-ISS-01)	7	gapfill-C §6 earn rates (≈35 Coins/show, 50/daily, 600 cap) yield ~18k (regular) / ~44k (hardcore) per season, contradicting the gapfill-C §12 table (~9.5k / ~18k) and the 9–11k target		Retune to 10+5/round+25/win, 25/daily, 300/day cap → 4.8k/10.9k/25.3k. Update gapfill-C §6/§7.12 and FR-06-06 text	Resolution adopted in the design
DI-070 (07-ISS-02)	7	Brief's slot names (top/bottom/face) differ from v3 (Upper/Lower/Faceplate)		Map 1:1 (§7.4); keep v3 IDs in data, kid-friendly labels in UI	Resolution adopted in the design
DI-071 (07-ISS-03)	7	Brief lists 6 slots; v3 also has Celebration, Banner, Title		Keep all v3 slots (Must per FR-05-07/08)	Resolution adopted in the design
DI-072 (07-ISS-04)	7	"Random Style" costume trait (FR-05-04) could read as chance		It's a visual variant of an owned item, not a reward; allowed. Never used for acquisition	Resolution adopted in the design
DI-073 (07-ISS-05)	7	"New" badges could become nagging/FOMO		Badge in dressing room only; one dot on menu, no counts, no expiring badges	Resolution adopted in the design
DI-074 (07-ISS-06)	7	Small-tier Coins on path (≈1,400) not in gapfill-C		Small addition; included in retuned sim; within source/sink 1.0–1.2 if shop grows by ~2 items/season	Resolution adopted in the design
DI-075 (07-ISS-07)	7	No account-level "player level" defined (FR-06-01 title mentions it)		Defined in §7.1 as non-decaying badge level	Resolution adopted in the design
DI-076 (07-ISS-08)	7	Hardcore cap: XP keeps flowing post-T70 with no wellbeing hook		Overflow capped at 20 tiers; break reminder default on; p90 >2 h telemetry review	Resolution adopted in the design
DI-077 (07-ISS-09)	7	FR-11-15 is Should, but healthy-play is core to brief		Recommend raising to Must for MVP (user decision)	Owner decision
DI-078 (07-ISS-10)	7	Paid future items must not add purchase entry in dressing room		Purchases only via Parent Account flow (§7.11)	Resolution adopted in the design
DI-079 (08-I-01)	8	"MiniMax" (Q-A1) is undefined, yet the brief names it		Architecture is engine-neutral at the service layer. Unity is recommended for the client and server. **Owner must answer Q-A1 before M0.**	Resolution adopted in the design
DI-080 (08-I-02)	8	NFR-10-03 AC2 asks for a 40-player load test, but AD-32's 9 ms budget is sized for 32. The brief asks for a 60-ragdoll prototype, beyond the 40 cap		Budget: 32 @ ≤9 ms (2 shows/vCPU); 40 @ ≤12 ms (accept 1.5 shows/vCPU for 40-player configs); 60 is a stress and stability test only, not a supported lobby size (R-01)	Resolution adopted in the design
DI-081 (08-I-03)	8	gapfill-A/D are written for Unity (fixedDeltaTime, NGO), while v3 claims engine independence		Keep values as engine-neutral numbers (50 Hz step, thresholds). Only the implementation notes are Unity-specific	Resolution adopted in the design

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-082 (08-I-04)	8	D-15 cost model: egress (~$14/h) is more than 2× compute (~$6.3/h) at 10k CCU; budget is open (Q-C3)		Treat egress as the main cost lever. Measure real kbps in the M0 prototype; target mean ≤40 kbps; negotiate egress. Product owner to give F and budget	Resolution adopted in the design
DI-083 (08-I-05, 09-I-16)	8, 9	Kernel anti-cheat vs children's data (NFR-10-13 vendor subject to the DPIA) **Also raised as 09-I-16:** Anti-cheat vendor may collect device data from children	NFR-10-13, Q-C3	Launch with server authority + server validation only; revisit after beta cheat telemetry **09-I-16:** Prefer server-side only for MVP; any vendor needs DPIA entry and parent notice before enabling *Merge note:* Authority: Section 8 (systems) for the vendor and budget; Section 9 (child safety) for any device data collected from children (DPIA, NFR-11-01).	Owner decision Q-C3 (Appendix C, item 6)
DI-084 (08-I-07)	8	Reconnect after a crash of the *server* (not the client) is not specified		Checkpoint per round; abort show; grant rewards for completed rounds; result reason SERVER_ABORT	Resolution adopted in the design
DI-085 (08-I-08)	8	Data residency for US vs UK/EU children is not specified (Q-C2)		Single EU home region for PII at MVP; the DPIA confirms; US-region PII store if counsel requires	Owner decision Q-C2 (Appendix C, item 5)
DI-086 (08-I-09)	8	Client/server physics config drift could break prediction		Physics config hash check at connect (§8.10)	Resolution adopted in the design
DI-087 (08-I-10)	8	Platform login on PC (Steam/Epic) depends on the storefront decision (Q-C1)		Identity service supports platform-agnostic linking; the first provider follows Q-C1	Owner decision Q-C1 (Appendix C, item 4)
DI-088 (08-I-11)	8	Telemetry pseudonymous ID vs deletion requests (FR-11-12)		Rotating salted hash, with the salt per player deleted on account deletion (crypto-shredding)	Resolution adopted in the design
DI-089 (09-I-01)	9	Guest can play a full online show without consent (FR-07-06 AC1), yet FR-11-02 lists "online play" as a consent purpose	FR-07-06 vs FR-11-02	Guest online play relies on COPPA internal-operations exception: no social graph, preset comms only, pseudonymous ID. "Online play" consent purpose governs persistent profile online features (cross-device sync, friends' visibility). Clarify FR-11-02 wording. **Counsel**	Counsel sign-off, DEP-10 (Appendix C, item 10)
DI-090 (09-I-02)	9	Quick-chat for guests not specified	FR-07-14 silent	Allowed (preset IDs, no personal data); parent can't control a guest, so guests get quick-chat **receive+send** but mute-all is one tap. Revisit in DPIA	Resolution adopted in the design
DI-091 (09-I-03)	9	Friend code entry requires a text field, but child profiles have none	FR-07-07 vs FR-11-03 AC3	Picture-alphabet codes (6 of 16 icons)	Resolution adopted in the design
DI-092 (09-I-04)	9	Ambiguous whose parent approves a friendship; a shared/screenshotted code could let a stranger in	FR-07-07, AD-04	Both parents approve; approval card shows "added by code" and "Do you know this child in real life?"; ≤5 code attempts per 10 min	Resolution adopted in the design
DI-093 (09-I-05)	9	Flagged replay retention 14 d (AD-21) shorter than evidence preservation needed for NCMEC/CEOP escalations	AD-21 vs FR-11-17	Legal-hold exception under NFR-10-21 "legally required safety records". **Counsel**	Counsel sign-off, DEP-10 (Appendix C, item 11)
DI-094 (09-I-06)	9	Spectator communication undefined (could message active players)	FR-07-11, FR-07-14	Spectator quick-chat to party only; spectators invisible to players	Resolution adopted in the design
DI-095 (09-I-07)	9	Bot share in new-player lobbies unbounded ("disclosed bots", D-13)	FR-01-02 AC5 vs AC7	40 % / 12-bot cap applies to all lobbies	Resolution adopted in the design

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-096 (09-I-08)	9	Leader-disconnect promotion in 5 s (FR-07-03) clashes with 60 s reconnect window mid-show (D-09)	FR-07-03 vs FR-07-16	5 s promotion in menus/queue only; in a show, promotion waits for window expiry	Resolution adopted in the design
DI-097 (09-I-09)	9	Disconnect wording differs: "Disconnected – you can play again any time" vs "eliminated – disconnected"	FR-07-16 AC2 vs D-09 AT	Use FR-07-16 wording (FR-08-16 friendly language); update D-09 AT	Resolution adopted in the design
DI-098 (09-I-10)	9	Parent dashboard shows reports the child *made*, but not moderation actions *against* the child, though FR-11-17 requires a parent appeal path	FR-11-10 vs FR-11-17	Add "Safety notices & appeals" section to dashboard + email	Resolution adopted in the design
DI-099 (09-I-11)	9	"Bad name or picture" report against generated names/curated cosmetics can't be the child's fault	FR-11-06 vs AD-03	Route to curation defect queue; offending word disabled via kill switch; no sanction on reported child	Resolution adopted in the design
DI-100 (09-I-12)	9	Report auto-hide after 3 reports (AD-43) — unclear if it applies to players; risk of brigading	AD-43	Levels only; players never auto-sanctioned by report count	Resolution adopted in the design
DI-101 (09-I-13)	9	Reports on bots would waste moderation and could reveal bot identity if handled differently	AD-17, FR-11-06	Same child-facing confirmation; silently discarded server-side	Resolution adopted in the design
DI-102 (09-I-14)	9	Emote spam/rate limit undefined	FR-05-06	1 emote / 2 s, mirroring quick-chat	Resolution adopted in the design
DI-103 (09-I-15)	9	Block avoidance only "when alternative lobby < 10 s"; a blocked player can still share a show	FR-07-15 AC2	Accept (pool health), since no contact channel exists; blocked player's emotes/chat hidden and their bean gets no player-card from blocker's view	Resolution adopted in the design
DI-104 (09-I-17)	9	Children's Code Std 11 (parental monitoring transparency) has no child-facing indicator	FR-11-13	Child sees a small icon/explainer that a grown-up can see playtime & friends	Resolution adopted in the design
DI-105 (10-ISS-01)	10	FR-01-08 says Course/Points levels have a "30-minute limit"; FR-09-10 caps timer at 1–10 min.		Level Hop uses the level's own timer (≤10 min). Propose CR to edit FR-01-08 to "the level timer (FR-09-10)".	Change request to requirements owner
DI-106 (10-ISS-03)	10	AD-20 "publish to friends by code" vs codes being leakable to strangers (FR-09-13 only rate-limits).		Unmoderated (Shared-Friends) levels playable only by creator's approved friends; any-player code play only for Approved-Public.	Resolution adopted in the design
DI-107 (10-ISS-04)	10	FR-01-08 mentions Survival/Points levels, but FR-09-09 launches COURSE only.		Level Hop launches COURSE-only; Survival/Points rules apply Later+1.	Change request to requirements owner
DI-108 (10-ISS-05)	10	FR-09-18 classifier scoped to "sticker arrangement"; offensive shapes can be built from blocks/platforms.		Classify whole-level renders (multi-view + OCR).	Resolution adopted in the design
DI-109 (10-ISS-06)	10	FR-09-21 allows ×3/×4 (Ref) boosts — pushes play pressure on kids (UD-1/AD-15 spirit).		Cap at ×2 by config; ×3/×4 disabled.	Resolution adopted in the design
DI-110 (10-ISS-07)	10	FR-09-14 republish keeps code and re-moderates, but undefined what plays during review.		Last approved version stays live until new version passes.	Resolution adopted in the design
DI-111 (10-ISS-08)	10	FR-09-23 "≤4 h" rewards with no minimum event length → could be short, intense windows.		Events ≥7 days.	Resolution adopted in the design
DI-112 (10-ISS-09)	10	FR-09-18 AC2 has 24 h report review but no pre-moderation SLA for submissions.		Design target 72 h; add to FR-11-17 staffing plan (open — needs Ops sign-off).	Ops sign-off (Appendix C, item 12)

ID / source IDs	Sec.	Issue	Requirements	Resolution adopted	Follow-up
DI-113 (10-ISS-10)	10	Budget limits memory but not runtime cost at 32 players (min-spec perf).		Add perf-score check in automated stage; values TBD with engineering.	Open: thresholds to be set with engineering
DI-114 (10-ISS-11)	10	Likes visible as public counts could create popularity pressure.		Public shows tier badges only; creator sees own count.	Resolution adopted in the design
DI-115 (10-ISS-12)	10	FR-09-01 is Could/Later while FR-09-18 is Must-when-shipped — risk of Builder launching with partial moderation.		Explicit launch gate (§10.B6).	Resolution adopted in the design
DI-116 (10-ISS-13)	10	Live-ops KPIs risk conflicting with FR-11-14 (no profiling).		KPIs aggregate/content-level only; no per-child targeting (§10.B7).	Resolution adopted in the design

A.2 Issues raised by the merge

These were not in any source issue list. They were found while merging and are counted separately from the register above.

ID	Where	Issue	Handling in this document	Follow-up
MRG-01	01 §6 vs FR-02-11	The GDD MVP level list did not match FR-02-11 / AD-40 and used a reference-game name ("Slime Climb", AD-25). No source doc logged this.	Resolved in the merge by REC-01 (FR-02-11 set used). GDD owner to confirm and update 01-core-gdd.md.	GDD owner confirmation
MRG-02	06 §6.3.2 vs 02 §2.9	The audio obstacle list follows the full catalogue names (Turn Gate, Windmill Paddles, Boing Glove …) and has no events for eight of the eleven MVP kit pieces: Swing Ball, Spin Bar, Spin Mallet, Tilt Plank, Conveyor, Bumper, Pusher Block and Launch Arm (Trick Door, Drop Tile and Slime are covered). FR-08-08 AC1 and FR-03-17 telegraph tells need them.	Audio owner to add event IDs, tells and visual twins for K1–K8 before M1. Not invented here.	Audio owner
MRG-03	10 §10.A7 vs FR-11-03 AC3	Level-code entry is described as an on-screen keypad for a 30-symbol code, which conflicts with "no on-screen keyboard or text field" in a child profile.	Open. Needs an entry design that is not a free-text keyboard (09 used a picture alphabet for friend codes) or an explicit exception like the parent-gate number pad (04-UX-I2). Builder is Later scope. See REC-24.	UX + Trust & Safety decision (Later)
MRG-04	05 §5.8 budgets vs 08 §8.12	05 budgets a costume part at 6,000 LOD0 tris (two parts per bean), a base bean at 4,000 and a faceplate at 500, and computes the worst case as 32 × 2 × 6k. 08 budgets "32 beans ≤ 15k tris each LOD0". If the base body renders under the costume, a bean can reach 16.5k.	Open. Art and engineering to confirm whether the base mesh renders under costumes and align the two budgets in the FR-12-05 AC2 test.	Art + engineering
MRG-05	03 §3.3 vs 05 §5.8 naming	The animation example "wb_anim_base_full_run_v01.fbx" (03-I-09) does not match the 05 setid pattern [a-z]+[0-9]{2} (it would need e.g. "base01").	Open (minor). Fix the example when the `anim` category is added to FR-12-02 (see the merged naming issue in A.1).	Art pipeline
MRG-06	06 §6.7 vs 03 §3.7	06 proposes rumble on the countdown GO; the 03 haptic list has no GO event.	Open (minor). Controls owner to accept or reject; see REC-14.	Controls owner

A.3 Source ID to register ID

Duplicates merged into one row (all other source IDs map one-to-one, in order):

Register ID	Merged source IDs
DI-003	01-ISS-03, 04-UX-I8
DI-004	01-ISS-04, 02-I-8
DI-010	01-ISS-10, 02-I-5
DI-005	01-ISS-05, 02-I-1
DI-009	01-ISS-09, 10-ISS-02
DI-083	08-I-05, 09-I-16
DI-044	04-UX-I9, 06-A-10
DI-031	03-I-09, 05-I-2, 06-A-07
DI-034	03-I-12, 08-I-06

Full map by source document:

Section 1: 01-ISS-01 → DI-001; 01-ISS-02 → DI-002; 01-ISS-03 → DI-003; 01-ISS-04 → DI-004; 01-ISS-05 → DI-005; 01-ISS-06 → DI-006; 01-ISS-07 → DI-007; 01-ISS-08 → DI-008; 01-ISS-09 → DI-009; 01-ISS-10 → DI-010; 01-ISS-11 → DI-011; 01-ISS-12 → DI-012.

Section 2: 02-I-1 → DI-005; 02-I-2 → DI-013; 02-I-3 → DI-014; 02-I-4 → DI-015; 02-I-5 → DI-010; 02-I-6 → DI-016; 02-I-7 → DI-017; 02-I-8 → DI-004; 02-I-9 → DI-018; 02-I-10 → DI-019; 02-I-11 → DI-020; 02-I-12 → DI-021; 02-I-13 → DI-022.

Section 3: 03-I-01 → DI-023; 03-I-02 → DI-024; 03-I-03 → DI-025; 03-I-04 → DI-026; 03-I-05 → DI-027; 03-I-06 → DI-028; 03-I-07 → DI-029; 03-I-08 → DI-030; 03-I-09 → DI-031; 03-I-10 → DI-032; 03-I-11 → DI-033; 03-I-12 → DI-034; 03-I-13 → DI-035; 03-I-14 → DI-036.

Section 4: 04-UX-I1 → DI-037; 04-UX-I2 → DI-038; 04-UX-I3 → DI-039; 04-UX-I4 → DI-040; 04-UX-I5 → DI-041; 04-UX-I6 → DI-042; 04-UX-I7 → DI-043; 04-UX-I8 → DI-003; 04-UX-I9 → DI-044; 04-UX-I10 → DI-045.

Section 5: 05-I-1 → DI-046; 05-I-2 → DI-031; 05-I-3 → DI-047; 05-I-4 → DI-048; 05-I-5 → DI-049; 05-I-6 → DI-050; 05-I-7 → DI-051; 05-I-8 → DI-052; 05-I-9 → DI-053; 05-I-10 → DI-054; 05-I-11 → DI-055.

Section 6: 06-A-01 → DI-056; 06-A-02 → DI-057; 06-A-03 → DI-058; 06-A-04 → DI-059; 06-A-05 → DI-060; 06-A-06 → DI-061; 06-A-07 → DI-031; 06-A-08 → DI-062; 06-A-09 → DI-063; 06-A-10 → DI-044; 06-A-11 → DI-064; 06-A-12 → DI-065; 06-A-13 → DI-066; 06-A-14 → DI-067; 06-A-15 → DI-068.

Section 7: 07-ISS-01 → DI-069; 07-ISS-02 → DI-070; 07-ISS-03 → DI-071; 07-ISS-04 → DI-072; 07-ISS-05 → DI-073; 07-ISS-06 → DI-074; 07-ISS-07 → DI-075; 07-ISS-08 → DI-076; 07-ISS-09 → DI-077; 07-ISS-10 → DI-078.

Section 8: 08-I-01 → DI-079; 08-I-02 → DI-080; 08-I-03 → DI-081; 08-I-04 → DI-082; 08-I-05 → DI-083; 08-I-06 → DI-034; 08-I-07 → DI-084; 08-I-08 → DI-085; 08-I-09 → DI-086; 08-I-10 → DI-087; 08-I-11 → DI-088.

Section 9: 09-I-01 → DI-089; 09-I-02 → DI-090; 09-I-03 → DI-091; 09-I-04 → DI-092; 09-I-05 → DI-093; 09-I-06 → DI-094; 09-I-07 → DI-095; 09-I-08 → DI-096; 09-I-09 → DI-097; 09-I-10 → DI-098; 09-I-11 → DI-099; 09-I-12 → DI-100; 09-I-13 → DI-101; 09-I-14 → DI-102; 09-I-15 → DI-103; 09-I-16 → DI-083; 09-I-17 → DI-104.

Section 10: 10-ISS-01 → DI-105; 10-ISS-02 → DI-009; 10-ISS-03 → DI-106; 10-ISS-04 → DI-107; 10-ISS-05 → DI-108; 10-ISS-06 → DI-109; 10-ISS-07 → DI-110; 10-ISS-08 → DI-111; 10-ISS-09 → DI-112; 10-ISS-10 → DI-113; 10-ISS-11 → DI-114; 10-ISS-12 → DI-115; 10-ISS-13 → DI-116.

Source notes kept from the issue sections:

- *Section 1:* No other duplicates/conflicts were found within this GDD's scope; the 34 duplicates already merged in v3 (e.g. DUP-02 FR-02-09→FR-01-03, DUP-04 FR-02-07→FR-01-05, DUP-08 FR-01-11→FR-07-08, DUP-09 FR-02-10→FR-03-15) are honoured and retired IDs are not used as design anchors.

- *Section 3:* None of these change a Must AC numerically except 03-I-01 (AT wording) and new keys/ACs in 03-I-02, 03-I-05, 03-I-08, 03-I-09 — all need analyst sign-off before M0 lock.
- *Section 4:* Open dependency: the final palette, icon set, bean and show imagery wait for the user's asset repository (UD-3). All `(img)` slots are placeholders.
- *Section 9:* Open for owner: Q-C2 (launch countries → regions & GDPR-K ages), Q-C3 (anti-cheat vendor), DEP-10 counsel sign-off on 09-I-01, 09-I-05.
- *Section 10:* Open (need decisions): 10-ISS-09 SLA & staffing; 10-ISS-10 perf thresholds; CRs for 10-ISS-01/02/04 to requirements owner.

Appendix B — Reconciliation log

B.1 Conflicts between source documents (26 reconciliations)

Each row records a conflict between two or more source documents, the authority used and how the master text was changed. Section references point to this master document.

ID	Topic	Sources in conflict	Conflict	Authority	Resolution in the master	Where
REC-01	MVP level set	01 §1.6 vs 02 (whole section) and FR-02-11 / AD-40	01-core-gdd §6 listed eight levels (Wobble Dash, Conveyor Canyon, Slime Climb (rising), Bumper Bowl, Plank Party, Bean Bell Hunt, Tile Drop (fallback), Summit Scramble) with their own slots and obstacle mixes. 02-rounds-levels specifies the eight FR-02-11 levels (Knock-Knock Run, Bumpy Parade, Slime Rise, Beam Hop, Wall Dodge, Tail Chase, Tile Drop, Summit Grab). Both claim to trace FR-02-11, but only the 02 list matches the requirement; the GDD list was not logged as a deviation, and "Slime Climb" is a reference-game round name (AD-25).	Core requirements (FR-02-11, AD-40, AD-25). The GDD authority covers game rules, not the content list fixed by a Must requirement.	The master uses the FR-02-11 set everywhere (§1.6 rewritten as a summary of Section 2). The GDD rules still apply to it: slot and selection rules (§1.4.4), fall-rule principles, and the R1 content rule of 01-ISS-11 (no grab-steal Points level in R1, which Tail Chase meets because it is tagged R2/R3 only). The GDD owner should confirm (see MRG-01).	§1.6, §2
REC-02	XP when a child quits a show	07 §7.0 (pillar P1) vs 01-ISS-01	07 pillar P1 said "quitting gives 0". 01-ISS-01 rules that a quit keeps all per-round XP and Coins already earned and forfeits only the +100 participation XP.	Core GDD (game rules).	07 P1 rewritten to the 01-ISS-01 rule. The proposed requirement edit (FR-06-01 AC3) stays open in Appendix C.	§7.0, §1.7.3
REC-03	Bot disclosure wording	04 §4.2.5 and SC-03 wireframe vs 01-ISS-08 (FR-01-02 AC7)	04 used "Robot friends may join"; 01-ISS-08 fixes a single string, "Some players may be computer players" (FR-01-02 AC7), which 09 §9.3.2 also uses.	Core GDD (game rules), matching FR-01-02 AC7.	All screens and flows use "Some players may be computer players".	§4.2.5, §4.3
REC-04	Round intro duration	04 §4.2.5 ("4 s") vs 01 §1.4.3	04 showed the round intro as 4 s. The GDD phase timeline is a 6.0 s intro fly-through with the picture card, then a 2.0 s spawn hold and a 3.0 s 3-2-1-GO.	Core GDD (game rules).	04 sequence diagram updated to "6 s intro + 2 s spawn hold". The 06 round-intro stinger (≤ 4 s) still fits inside the 6 s intro.	§4.2.5
REC-05	Results screen: placement number	04 §4.2.7 ("my placement") vs 01 §1.9.2	04 listed "my placement" on SC-08. The GDD shows a medal per round reached and hides the placement number for eliminated players (finalists only).	Core GDD (game rules, fail-forward pillar P2).	04 §4.2.7 now shows the furthest-round medal; a placement number is shown to finalists only.	§4.2.7
REC-06	Tutorial skip rule	04 §4.2.4 vs 01-ISS-07	04 said a Skip button appears only after the first completion. 01-ISS-07 rules: before the first completion a "Play now" button bypasses the tutorial (offered again next launch); after completion it no longer auto-runs and is replayable from Settings; no Skip inside a station.	Core GDD (game rules).	04 §4.2.4 now points to §1.8 and keeps only presentation rules.	§4.2.4
REC-07	Tutorial stations	04 §4.2.4 vs 01 §1.8.2	04 listed Move → Jump → Dive → Grab → Emote/Quick-chat → Finish arch. The GDD path is Move, Jump, Dive, Grab, Bump & tumble, Say hi (emote + quick-chat), then a mini-race against 3 practice bots, with assists after 3 tries / 20 s.	Core GDD (game rules).	The GDD station list governs; 04 keeps the UI rules (animated input picture, single-word label, ghost bean, gate, demo replay after 20 s).	§4.2.4, §1.8.2

ID	Topic	Sources in conflict	Conflict	Authority	Resolution in the master	Where
REC-08	Reference terminology in screens	04 SC-04, SC-05, SC-08 wireframes and 09 Figure (matchmaking flow) vs 01-ISS-08	Wireframes used "reach the crown" and a crown icon for the winner; the 09 matchmaking flow used "Game Selector". 01-ISS-08 requires Trophy and Show Picker throughout.	Core GDD (game rules / terminology).	Changed to "trophy" (🏆) and "Show Picker".	§4.3, §4.4, §9.3.1
REC-09	Bot backfill start rule at 60 s	09 §9.3.3 vs 01-ISS-06	09 gives the trigger "< 20 humans at the 60 s step" and the cap (≤ 12 bots / ≤ 40%) but not the start condition; 01-ISS-06 resolves the D-11/D-12 arithmetic: start if humans ≥ 8 and humans + bots ≥ 20, with bots = min(12, 32 − humans).	Core GDD (game rules); the child-safety cap from 09 (40% also in new-player lobbies, 09-I-07) is kept.	09 §9.3.3 trigger row extended with the GDD start rule.	§9.3.3
REC-10	Bean scale for the rig	05 §5.8 ("bean height 1.00 m", 05-I-4) vs gapfill-A conventions and 03 §3.3.1 (capsule 1.6 m × r 0.5 m)	05 proposed a 1.00 m bean height to give the 2 mm seam tolerance a unit. gapfill-A fixes the collision capsule at ≈ 1.6 m tall, 0.5 m radius, and all movement values (jump apex 1.5 m ≈ 0.95 bean heights, gap rules) are scaled to it.	Physics (03 character-controls and gapfill-A).	Units stay metres, Y-up; the mesh is authored to the 1.6 m capsule. 05-I-4 is kept only for "units = metres".	§5.8
REC-11	Ragdoll bones replicated over the network	03 §3.3.1 / 03-I-12 vs 08 §8.3 ("root + 3–5 key bones")	03 names the replicated bones (root, pelvis, spine_02, head). 08 allows "3–5 key bones" for bandwidth.	Physics (03) for which bones; systems (08) for the network budget.	Baseline is root + pelvis, spine_02, head; up to 5 key bones only if the D-07 bandwidth budget allows. Merged as one issue in Appendix A.	§3.2.6, §8.3
REC-12	Tile Drop used as a non-final fallback	02-I-5 vs 01-ISS-10	Both docs found that Tile Drop (a last-standing Final) would end a show early in a non-final slot. 02 proposed a "survival-quota rule" pending owner confirmation; the GDD fixed 60% qualifier quota and a 1:45 limit.	Core GDD (game rules).	Non-final fallback = Tile Drop with a 60% qualifier quota, 1:45 limit, same layer table; F slot = normal Tile Drop Final. Owner confirmation still open (Appendix C).	§1.4, §2.7
REC-13	Sound briefs for character events	03 §3.7 (SFX column) vs 06 §6.3	The two docs describe the same events with different sound briefs (e.g. jump "boing" vs "bwip").	Audio doc 06, which is the FR-08-08 AC1 event list (not covered by the user order of authority; decided by the requirement).	06 §6.3 is the authoritative event list; the 03 juice table keeps VFX, haptics and camera and its SFX column is read as a feel brief.	§3.7, §6.3
REC-14	Haptic events	06 §6.7 vs 03 §3.5 / §3.7	06 lists rumble on the countdown GO; the 03 juice table (vibration default 70%, events per action) has no GO pulse.	Controls doc 03 (input and feedback specification).	03 §3.7 is the haptic list; the GO pulse is logged as a proposal (MRG-06).	§3.7, §6.7
REC-15	Default volume levels	04 SC-10 wireframe (70/50/80/60) vs 06 §6.6 (Master 70, Music 60, SFX 80, UI 70)	The Settings wireframe showed different slider defaults from the audio spec.	Audio doc 06 (FR-08-09 defaults).	Wireframe updated to 70 / 60 / 80 / 70.	§4.3, §6.6
REC-16	UI colour tokens	04 §4.8.1 vs 05 §5.2	04 proposed teal primary, white surface, #1B2340 ink, tangerine accent and teal/tangerine/violet/lime teams; 05 sets the brand trio Berry, Night Ink and Cloud, Sun for rewards and Coral/Ocean/Leaf/Gold teams.	Art direction doc 05 for shared colour roles (both are placeholders until the asset repo, AD-26).	Shared roles use the 05 values; tokens 05 does not define (secondary, sky background, success, warning) keep the 04 placeholders. Night Ink on Cloud is 15.3:1, so the "≥ 12:1" rule still holds (checked during the merge).	§4.8.1

ID	Topic	Sources in conflict	Conflict	Authority	Resolution in the master	Where
REC-17	UI typeface	04 R4 (rounded humanist sans) vs 05 §5.6 (rounded geometric sans, e.g. Nunito/Baloo, licence pending)	Two different typeface descriptions.	Art direction doc 05 picks the family; the 04 legibility criteria (single-storey a/g, large x-height, glyph coverage) are constraints on that pick.	Cross-referenced; final family pending licence check (FR-12-07).	§4.5, §5.6
REC-18	Player card actions for strangers	04 SC-16, SC-13 wireframe and PlayerCard component ("Add friend") vs 09 §9.3.2 (AD-04)	04 put "Add friend" on every player card. 09 rules that there is no Add friend on a stranger's card; actions are Mute · Block · Report, and friends are added only by parent-approved friend code.	Multiplayer-safety doc 09 (child safety).	04 inventory, wireframe and component updated to Mute / Block / Report.	§4.1.1, §4.3, §4.8.2
REC-19	Report flow and reason labels	04 SC-13 vs 09 §9.7.1	04 opened a separate reasons screen from the player card; 09 shows the reasons directly in the shield sheet so the report is the second tap, with the labels Being mean · Cheating · Bad name or picture · Something upset me.	Multiplayer-safety doc 09 (child safety).	09 flow governs; the 04 wireframe labels aligned.	§4.3, §9.7.1
REC-20	Consent purposes	04 §4.2.3 vs 09 §9.11 / §9.12	04 listed the purposes as Online friends, Builder sharing and Weekly email. 09 lists online play, friends and Builder sharing as consent purposes, with the weekly summary email as a separate opt-in.	Multiplayer-safety doc 09 (child safety).	04 consent flow updated to the 09 purposes.	§4.2.3
REC-21	Parental dashboard content	04 §4.2.10 vs 09 §9.11 (and 09-I-10)	04 had its own list of controls and no place for moderation actions against the child; 09 adds "Safety notices & appeals" (FR-11-17).	Multiplayer-safety doc 09 (child safety).	Control content, defaults and timing are specified once in §9.11; 04 keeps the tab layout and its Safety tab gains notices and appeals.	§4.2.10, §9.11
REC-22	Raw telemetry retention	10 §10.B7 ("13 months (Design)") vs 08 §8.6 and NFR-10-21 / AD-21 (90 days raw, 2 years aggregates)	Different retention periods for the same raw events.	Technical architecture doc 08 (systems), which matches the Must requirement NFR-10-21.	10 §10.B7 now cites 90 days raw / 2 years aggregates.	§10.B7
REC-23	Builder level music library	10 §10.A3 ("licensed library") vs 06 §6.2.3 and 06-A-03	10 copied the requirement wording "licensed library"; 06 resolves it to the Wobblebeans original music library (original music only, FR-08-07, Ch. 12).	Audio doc 06 with FR-08-07 (not in the user order of authority).	10 updated to the original library; no UGC audio.	§10.A3
REC-24	Builder level-code entry	10 §10.A7 ("4-4-4 entry with large on-screen keypad") vs FR-11-03 AC3 and 09 friend-code approach	FR-11-03 AC3: no on-screen keyboard or text field anywhere in a child profile. 09 solved the same problem for friend codes with a picture alphabet (09-I-03). The 30-symbol 4-4-4 level code has no such design.	Multiplayer-safety doc 09 (child safety).	Flagged in §10.A7; the entry method is open (MRG-03). Builder is Later scope.	§10.A7
REC-25	Coins per qualified round in the fail-forward example	01 §1.9.2 ("Qualified round 2: +10") vs 07 §7.3 retune (5 per round)	The GDD example used the gapfill-C v3 rate (10 Coins per round); 07 proposes 10 + 5/round + 25/win (07-ISS-01).	Progression doc 07 for economy values (the GDD only illustrates the itemised copy).	Example made rate-neutral ("+N Coins, rates per §7.3").	§1.9.2

ID	Topic	Sources in conflict	Conflict	Authority	Resolution in the master	Where
REC-26	Engine stance	01 scope defaults ("engine-neutral", Q-A1) vs 08 §8.1 (recommends Unity 6 LTS)	The GDD keeps the design engine-neutral; 08 recommends Unity 6 LTS with a netcode spike at M0.	Technical architecture doc 08 (systems).	Unity 6 LTS is the systems recommendation for the M0 engine decision record; requirements and game rules stay engine-neutral until Q-A1 is answered (Appendix C).	§1 intro, §8.1

B.2 Duplicate content removed

#	Duplicate	Kept in (authority)	Change in the master
1	Session screen-flow diagram in Section 1 (src 01 §7.1) repeated the navigation map of Section 4	§4.1.2, Figure 4.1	Diagram removed from §1.7.1; a reference and the session rules are kept
2	Quota formula and rounding rule restated in Section 2	§1.4.5 (game rules)	§2.0.1 refers to §1.4.5
3	Per-round phase timeline and fixed overhead restated in Section 2	§1.4.3 (game rules)	§2.0.1 refers to §1.4.3
4	Difficulty-by-slot values restated in the level-building guidelines	§1.10.1 (game rules)	§2.10 rule 4 refers to §1.10.1
5	Readability rule restated in the level-building guidelines	§2.0.3 and §1.10.2	§2.10 rule 7 refers to them
6	Tutorial stations, assists and skip rule restated in the UX flows	§1.8 (game rules)	§4.2.4 refers to §1.8 and keeps only the UI presentation
7	Social rules (parties, friends, quick-chat, reporting) restated in the GDD	Section 9 (child safety)	§1.11 summarises and refers to Section 9
8	Colour values defined separately in the UI tokens and the art palette	§5.2 (art direction)	§4.8.1 tokens now use the §5.2 colours
9	Audio feel briefs in the character feedback table	§6.3 (authoritative event list)	§3.7 marked as a feel brief that points to §6.3
10	Each source's issues list, traceability matrix, open questions and document header	Appendices A, C, D and the Document control page	Moved out of the sections, once each

B.3 Diagram syntax fixes

All Mermaid diagrams were rendered with mermaid-cli. These failed to parse as written and were fixed (meaning unchanged):

Diagram	Problem and fix
d01_04 (Section 1 session flow, removed as duplicate)	Edge label contained a quoted string ("Play now") inside a -- text --> link, which the flowchart parser rejects (got STR). Rewritten as a |label| link. (Diagram then removed as a duplicate.)
Figure 4.3	Node label contained parentheses inside [ ], parsed as a shape token (PS). Label quoted.
Figure 8.2	Chained transition "RoundIntro --> Countdown --> Playing" is not valid in stateDiagram-v2. Split into two transitions.
Figure 9.8	Two node labels contained parentheses (and a semicolon, which ends a statement). Labels quoted.

Appendix C — Open Questions & Sign-offs

These are the decisions the merge could not make. Each row gives the default the design uses until the decision is made. "Needed by" is taken from the sources; where the sources give no date, the cell says so. The last two columns are for the decision and sign-off.

C.1 Open questions and sign-offs

#	Question / sign-off	Source	Current default in this design	Decision by	Needed by	Decision	Signed / date
1	**Q-A1 — What is "MiniMax"?** Engine, AI tool for assets or code, or something else?	Req. v3 Ch. 15; NFR-10-02; DEP-01; §8.1	Requirements stay engine-neutral. Section 8 recommends Unity 6 LTS for the M0 engine decision record (REC-26).	Owner	Before M0 (DEP-01; §8.16)		
2	**Q-A2 — Is the age range 6–12 right?**	Req. v3 Ch. 15; UD-1	6–12, with every player treated as a child (AD-01).	Owner	No date in the sources		
3	**Q-B1 — Money after the MVP.** Stay completely free, or sell cosmetics and a premium Season Path with parent approval?	Req. v3 Ch. 15; AD-12; AD-14; §7.11	Completely free. Any later paid item would be a direct, parent-approved platform purchase, never a currency.	Owner	No date in the sources (post-MVP)		
4	**Q-C1 — PC storefront(s)** at launch (Steam, Epic Games Store, Microsoft Store, own launcher)?	Req. v3 Ch. 15; AD-51; 08-I-10	No store-specific features. The identity service supports platform-agnostic linking; the first login provider follows Q-C1.	Owner	M0 (§8.16); store submission at M4		
5	**Q-C2 — Launch countries and data residency.**	Req. v3 Ch. 15; AD-27; AD-39; 08-I-08; 09 open items	UK, EEA and US; servers in EU-West, EU-Central, NA-East, NA-West; a single EU home region for personal data at MVP, confirmed by the DPIA (US-region store if counsel requires). Languages en-GB and en-US.	Owner, with privacy counsel	M4 gate (req. v3 milestones)		
6	**Q-C3 — Peak-player (CCU) forecast and monthly budget** for servers and anti-cheat.	Req. v3 Ch. 15; D-15; D-16; DEP-09; §8.8	Planning value F = 10,000 peak concurrent players (about $6/h compute and $14/h egress at peak); load test to 10× F. Anti-cheat vendor follows the budget (merged issue 08-I-05 / 09-I-16).	Owner	M0 (§8.16); vendor quotes for M3 (DEP-09)		
7	**Q-D1 — Trademark check** for the name "Wobblebeans" and the bean character.	Req. v3 Ch. 15; DEP-12	Flagged as a launch blocker.	Owner (legal)	M4 — trademark cleared		
8	**Q-D2 — Asset repository: 3D models or 2D images?**	Req. v3 Ch. 15; DEP-11; FR-12-03; 05-I-10; §5.8	All art is placeholder. 2D images can be used directly for UI, banners, patterns and icons; costumes need 3D modelling from 2D references. Section 5 assumes a mixed repository and plans modelling capacity.	Owner	M2 — art direction and asset pipeline (DEP-11)		

#	Question / sign-off	Source	Current default in this design	Decision by	Needed by	Decision	Signed / date
9	**Parental consent (VPC) vendor**	§8.5; §8.16; §9.12; FR-11-02	Verifiable parental consent through a certified vendor (examples given in §8.5: KWS, PRIVO, Veratad); vendor choice open.	Owner, with privacy counsel	M0 (§8.16)		
10	**Counsel sign-off: guest play under the COPPA internal-operations exception**	09-I-01; FR-07-06 AC1; FR-11-02; DEP-10	Guests may play full online shows without consent, relying on the internal-operations exception: no social graph, preset communication only, pseudonymous ID. The "online play" consent purpose covers persistent-profile online features. FR-11-02 wording to be clarified.	Privacy counsel	Before M3 closed beta (DEP-10)		
11	**Counsel sign-off: legal hold on flagged replays**	09-I-05; AD-21; FR-11-17; NFR-10-21	Flagged replay snippets are kept 14 days unless under legal hold, as a "legally required safety record" under NFR-10-21, so evidence for NCMEC/CEOP escalations is preserved.	Privacy counsel	Before M3 closed beta (DEP-10)		
12	**72-hour moderation SLA for Builder level submissions**	10-ISS-09; FR-09-18 AC2; FR-11-17	Design target 72 h for pre-moderation of submissions (reports 24 h), added to the FR-11-17 staffing plan. Builder does not launch until moderator staffing for these SLAs is in place (§10.B6).	Operations	Builder launch gate (Later)		
13	**Tile Drop early-quota rule** in a non-final slot	02-I-5 / 01-ISS-10 (merged issue); FR-01-03 AC4; REC-12	When Tile Drop is the fallback for a non-final slot it runs as a qualifier with a quota (60%, 1:45 limit) and ends at the quota; in the Final slot it is the normal Tile Drop Final. Needs FR-02-01 per-slot overrides.	Owner	No date in the sources		
14	**Parent-gate number pad as an exception to FR-11-03**	04-UX-I2; FR-11-03 AC3; FR-11-07	The parent gate uses a digits-only NumberPad (no text field, no letters). Proposed amendment to FR-11-03 AC3: "no free-text field; the parent-gate numeric pad is excepted".	Requirements analyst / owner	No date in the sources		
15	**XP when a child quits a show**	01-ISS-01; FR-01-09 AC4; FR-06-01 AC3; REC-02	A quit keeps all per-round XP and Coins already earned and forfeits only the +100 participation XP. Proposed amendment to FR-06-01 AC3: "0 participation XP is granted for a quit".	Requirements analyst / owner	No date in the sources		

C.2 Other sign-offs named in the sources

Sign-off	Source	Needed by
Analyst sign-off on new parameter keys and ACs: 03-I-01 (AT wording), 03-I-02, 03-I-05, 03-I-08, 03-I-09	Section 3 issues note	Before M0 movement lock
Change requests to the requirements owner for 10-ISS-01, 10-ISS-02 and 10-ISS-04	Section 10 open items	No date in the sources
Runtime performance thresholds for player-made levels (10-ISS-10)	Section 10 open items	With engineering; no date in the sources
GDD owner to confirm the FR-02-11 level set in Section 1 (MRG-01)	Merge	No date in the sources (M3 needs the 8 MVP levels)
Child-panel playtest of XP per hour and reward copy (gapfill-C §14)	Section 7 open row	No date in the sources
DPIA and children's risk assessment signed (NFR-11-01)	Req. v3 milestones	M3 closed beta
Privacy-counsel review of all legal summaries (not legal advice)	Req. v3 (DEP-10)	M3 closed beta

Appendix D — Traceability

D.1 Traceability tables from the source documents

The traceability matrices of Sections 1, 3, 4, 6, 7, 8 and 10 are kept here unchanged except for section and issue-ID references, which point to this master document. Sections 2, 5 and 9 carry their traceability inline (requirement IDs in headings, tables and rule text). The index in D.2 covers all sections.

D.1.1 Section 1 — Core Game Design

GDD section / decision	Requirement IDs
§1.1 Pillars P1–P5	FR-03-03, FR-04-05, FR-01-05, FR-01-09, FR-08-03, FR-08-19, FR-03-17, FR-08-11, FR-05-01, FR-07-12, FR-11-03, FR-07-14, FR-08-18, FR-11-15, FR-11-19; UD-1, UD-4; AD-07, AD-10, AD-15, AD-24, AD-37
§1.2 Fantasy & audience	FR-12-04, FR-12-10, FR-05-02, FR-05-03, FR-05-17, FR-06-07, AD-01, AD-35, AD-53
§1.3 Core loop	FR-01-01..05, FR-01-09, FR-03-12, FR-03-13, FR-06-01, FR-06-02, FR-06-13, FR-04-01..04, AD-11, AD-14, AD-31, AD-34
§1.4 Show structure	FR-01-02..05, FR-02-01..06, FR-03-14, FR-07-03, FR-07-11, FR-08-04, AD-05, AD-17, AD-28, AD-29, gapfill-B §1–3, D-12
§1.5 Modes	FR-01-01, FR-01-06, FR-01-07, FR-01-08, FR-01-12, FR-02-08, FR-07-03, FR-07-08, FR-09-23, AD-16, AD-18, AD-19, AD-48, AD-51
§1.6 MVP levels	FR-02-11, FR-03-01, FR-02-01, AD-25, AD-38, AD-40
§1.7 Session flow	FR-08-04, FR-08-16, FR-08-17, FR-08-18, FR-11-15, FR-06-13, NFR-10-18, AD-23
§1.8 Onboarding	FR-08-17, FR-08-19, FR-01-12, FR-01-02 AC5, FR-04-01..04, FR-05-06, FR-07-14, AD-44
§1.9 Fail-forward	FR-04-05, FR-03-12, FR-03-13, FR-01-09, FR-06-06, FR-08-03, FR-07-11, FR-01-05, FR-06-13, AD-10, AD-37
§1.10 Difficulty & fairness	FR-03-17, FR-08-11, FR-02-02, FR-02-03, FR-03-04, FR-03-15, FR-05-01, FR-07-12, FR-07-16, NFR-10-13, FR-08-12, FR-08-13, FR-08-20, NFR-10-19, AD-06, AD-09, AD-16, AD-28, AD-31
§1.11 Social	FR-07-03, FR-07-07, FR-07-14, FR-07-15, FR-11-03, FR-11-04, FR-11-06, FR-01-06, AD-03, AD-04, AD-43, AD-52
§1.12 Gates	NFR-10-12, NFR-10-20, plus ACs cited inline

D.1.3 Section 3 — Character Movement, Physics and Controls

Design element	Requirement IDs
State machine, gating	FR-04-01…05, FR-04-07, FR-05-06, AD-30, AD-31, AD-34, AD-07
Parameters	Ch. 13.2 / gapfill-A §8, AD-06, AD-08, DEP-02 (M0)
Net-facing behaviour	FR-07-12, NFR-10-05, gapfill-A §7
Anim slots & rig	FR-12-02, FR-12-04, FR-12-05, FR-12-06, FR-05-02, FR-05-06, UD-3
Input	FR-04-09, FR-04-11 (Won't), FR-04-12, FR-04-15, FR-07-05 (parity, later)
Assists	FR-04-04 AC4, FR-08-12, FR-08-20, FR-01-12, FR-08-16, AD-31
Camera	FR-04-08, FR-08-20
Juice	FR-08-12 AC4, NFR-10-19, FR-08-20, UD-4, FR-12-02
Level constraints fed back	FR-04-03 AC1 gap rules → Ch. 3

Verification hooks: T-FR-04-01…05, 07, 08, 12, 15; T-FR-08-12, T-FR-08-20; T-FR-12-04 seam test.

D.1.4 Section 4 — UX Flows and UI Design

Requirement	Where addressed
FR-08-01 HUD	§4.4
FR-08-02 Warning	§4.4
FR-08-03 Wording	§4.3 SC-06, R7, L7
FR-08-04 Round flow	§4.1.2 N3, §4.2.5

Requirement	Where addressed
FR-07-11 / AD-36 Spectate	§4.2.6, SC-07
FR-08-06 Contrast/tokens	§4.6, §4.8.1
FR-08-08 / FR-08-09 Audio	§4.6, SC-10
FR-08-11 Colour-blind	§4.4, §4.6
FR-08-12 Accessibility	§4.6, SC-10
FR-08-13 Text size / AD-49	R3, §4.8.1
FR-08-14 Localisation / AD-27	§4.7
FR-08-16 Errors	SC-14, SC-20
FR-08-17 / AD-44 Onboarding	§4.2.1, §4.2.4
FR-08-18 Dark patterns	R6, R8, §4.2.6–2.8, 04-UX-I6
FR-08-19 Kid language	R1, R2, R9
FR-08-20 Motion	§4.6
FR-04-12 Rebinding	§4.6
FR-05-17 / FR-05-10 Dressing Room	§4.2.8, SC-09
FR-01-12 / AD-18 Practice	SC-15
FR-07-06 / AD-04 Friends	SC-17, §4.2.3
FR-07-14 Quick-chat	QuickChatWheel
FR-07-15 Block	SC-16
FR-07-16 Reconnect	SC-20
AD-17 Bot disclosure	SC-03
AD-35 Celebration	§4.2.7
AD-01 No age gate	§4.2.2
FR-11-01 / US-11-4 Guest	§4.2.1
FR-11-02 / AD-22 Consent	§4.2.3
FR-11-03 No text/voice	Global constraint, 04-UX-I2
FR-11-04 Names	SC-21
FR-11-06 Reporting	SC-13
FR-11-07 Parent gate	§4.2.2, SC-11
FR-11-08 / FR-11-10 Controls & dashboard	§4.2.10
FR-11-09 / AD-12 No spending	SC-09, Purchases empty state
FR-11-11 Privacy defaults	§4.2.3 P4, R6
FR-11-12 Data deletion	SC-10, §4.2.10
FR-11-13 Privacy notices	SC-22, §4.2.3
FR-11-15 / AD-23 Wellbeing	SC-19
FR-11-19 No marketing notifications	No notification UI to children
AD-26 Original palette	§4.8.1
AD-51 PC-only	R5, 04-UX-I4

D.1.6 Section 6 — Audio Design

Req	What it asks	Where this doc meets it
UD-1	Child-directed, no voice, no countdown pressure	§6.5, §6.6, §6.8
UD-4 / AD-24 / FR-11-18	Bright, non-violent slapstick; PEGI 3 / ESRB E	§6.1, §6.3 (sound-design rules)
AD-53	No spoken presenter	§6.8
AD-02 / FR-11-03 / FR-07-09	No voice for players	§6.8
FR-08-07	Original, toy-like electro-pop, no lyrics; menu/round/Final/victory cues; −16 LUFS ±1	§6.1, §6.2, §6.5

Req	What it asks	Where this doc meets it
FR-08-08	SFX for jump, dive, grab, fall, bounce, qualify, eliminate; event list; visual equivalents	§6.3 (audio event list)
FR-08-09	Master/Music/SFX/UI sliders 0–100, ≤100 ms, persisted	§6.4, §6.6
FR-08-12 AC4 / AD-45	Visual indicator for every gameplay-relevant audio cue	§6.7
FR-08-02 / NFR-10-19	Warning cue; ≤3 flashes/s	§6.3.4, §6.7
FR-08-03 / FR-08-19	Kind outcome tone	§6.3.5 (Out stinger rules)
FR-08-17	Tutorial has no voice-over	§6.8
FR-07-15	Block and mute	§6.3.6
FR-03-01…17, FR-04-01…08	Obstacles, movement, physics	§6.3.1–3.3
FR-01-02…09	Show phases	§6.2
FR-12-01/02/03/07/08	Manifest, naming, formats, provenance, review	§6.9
FR-09 (Builder level settings)	Level music track choice	§6.2.3, Issue 06-A-03

D.1.7 Section 7 — Progression, Rewards and Customisation

Design section	Requirement IDs
§7.1 XP	FR-06-01, FR-01-09, FR-09-21, AD-15, AD-19
§7.2 Season Path	FR-06-02, FR-06-03/04 (Won't), AD-11, AD-14
§7.3 Coins/shop	FR-06-05, FR-06-06, FR-05-09, FR-05-11, FR-05-12 (Won't), FR-06-08 (Won't), AD-12, AD-13
§7.4 Slots	FR-05-01…08, FR-12-01, FR-12-04, FR-12-08, FR-11-05
§7.5 Dressing room	FR-05-10, FR-05-17, AD-22
§7.6 Challenges	FR-06-13, FR-09-23, AD-33
§7.8 Healthy play	FR-11-15, FR-11-08, FR-11-19, FR-08-18, AD-23
§7.9 Data	FR-06-05, FR-06-19, FR-05-10 AC1
§7.11 Later paid	FR-11-09, FR-11-10, FR-05-14/16, FR-06-02 premium, Q-B1

D.1.8 Section 8 — Technical Architecture

Requirement	Section
FR-07-12 Server architecture	§8.2, §8.4
FR-01-02 Matchmaking / FR-07-02 Region / FR-07-03 Party	§8.5, §8.8
FR-07-06 Accounts / FR-11-02,07,08,09,10,12	§8.5, §8.6, §8.7
FR-07-11 Spectating / FR-07-16 Reconnect	§8.3, §8.4
FR-07-14 Quick-chat / FR-07-15 Block / FR-11-03,04,05,06,17	§8.5, §8.9
FR-03-02, FR-03-07, FR-02-02 AC4, FR-01-05	§8.3
FR-01-09 Rewards stored ≤5 s / FR-06-01,02,05,06,07,13,19 / FR-05-10,11,17	§8.5, §8.6, §8.12
FR-01-03 AC5, FR-01-01 AC1, FR-09-19, FR-09-26	§8.10
FR-01-12 Practice Playground offline	§8.2 (client-only; no network calls; CI check)
NFR-10-02 Engine	§8.1, R-01/02/09/10
NFR-10-03 Player count, NFR-10-04 Performance	§8.8, §8.12, R-01
NFR-10-05 Netcode	§8.3
NFR-10-06 Launch scaling	§8.8, §8.10
NFR-10-08/09/12/21, FR-11-11/14/16, NFR-11-01	§8.2, §8.6, §8.9, §8.11
NFR-10-13 Anti-cheat, NFR-10-16 Security	§8.9
NFR-10-17 Reliability	§8.8
NFR-10-18 Load times, NFR-10-19 Flash	§8.11, §8.12
NFR-10-20 Config	§8.10
D-01..D-19	§8.3, §8.4, §8.8, §8.9, §8.12

Requirement	Section
FR-09-01..18 Builder (Later)	Out of scope; reserve an ugc service, object storage and the FR-09-18 moderation queue

D.1.10 Section 10 — Builder (Creative) and Live Ops

FR-09-01…18 → Part A §A2–A8; FR-09-19 → B1; FR-09-21/23 → B3; FR-09-26 → B4; FR-01-03 → B5; FR-01-08 → A7; FR-06-02/13 → B1–B2; FR-11-06/08/11/14/17/18 → A2, A6, B7; FR-12-08 → A4–A6, B4; NFR-10-12 → B7; AD-02/05/11/15/20/33 cited inline.

D.2 Requirement index

Every v3 requirement ID cited in this document, with its v3 title, priority and release, and the sections that cite it (173 IDs).

Requirement	v3 title	Priority	Release	Cited in sections
FR-01-01	Show Picker	Must	MVP	1, 8
FR-01-02	Matchmaking and lobby fill	Must	MVP	1, 6, 8, 9
FR-01-03	Round selection and show sequencing	Must	MVP	1, 2, 6, 8, 10
FR-01-04	Qualification, elimination and disconnects	Must	MVP	1, 2, 6, 9
FR-01-05	Final and winners	Must	MVP	1, 2, 6, 8
FR-01-06	Team scoring (Duos and Squads)	Should	R2	1, 6
FR-01-07	Ranked Show			1, 6, 10
FR-01-08	Level Hop and Builder Picks (continuous play)	Could	Later	1, 2, 6, 10
FR-01-09	End-of-show rewards	Must	MVP	1, 6, 7, 8
FR-01-12	Practice Playground (offline)	Should	MVP	1, 2, 3, 4, 8
FR-02-01	Round registry	Must	MVP	1, 2, 10
FR-02-02	COURSE (race) rounds	Must	MVP	1, 2, 8
FR-02-03	SURVIVAL rounds	Must	MVP	1, 2
FR-02-04	POINTS (hunt) rounds	Must	MVP	1, 2
FR-02-05	LOGIC rounds	Could	R2	1, 2
FR-02-06	TEAM rounds	Should	R2	1
FR-02-08	Peekaboo Teams (asymmetric hide-and-seek)			1
FR-02-11	MVP level set	Must	MVP	1, 2
FR-03-01	Obstacle framework	Must	MVP	1, 2, 6, 8
FR-03-02	Deterministic periodic motion	Must	MVP	2, 6, 8
FR-03-03	Knockback and ragdoll	Must	MVP	1, 2, 6
FR-03-04	Speed variants	Should	MVP	1, 2, 6
FR-03-05	Floor modifiers	Must	MVP (ice, water: R2)	2, 3, 6
FR-03-06	Trick doors	Must	MVP	2, 6
FR-03-07	Drop tiles	Must	MVP	2, 6, 8
FR-03-08	Triggered hazards	Should	R2	2, 6
FR-03-09	Projectile spawners	Should	R2	2, 6
FR-03-10	Launchers	Must	MVP	2, 6
FR-03-11	Volumes	Could	R2	2, 6
FR-03-12	Checkpoints	Must	MVP	1, 2, 6
FR-03-13	Out-of-bounds and respawn	Must	MVP	1, 2, 6
FR-03-14	Rising slime	Must	MVP	1, 2, 6
FR-03-15	Variations	Should	R2	1, 2, 6
FR-03-16	Themes	Could	Later	2, 6

Requirement	v3 title	Priority	Release	Cited in sections
FR-03-17	Readability and telegraphing	Must	MVP	1, 2, 6
FR-04-01	Movement	Must	MVP	1, 3, 6
FR-04-02	Jump	Must	MVP	1, 3, 6
FR-04-03	Dive	Must	MVP	1, 3, 6
FR-04-04	Grab	Must	MVP	1, 3, 6
FR-04-05	Ragdoll, stumble and recovery	Must	MVP	1, 3, 6
FR-04-06	Falling out and respawn (merged)			6
FR-04-07	Player-to-player collision	Must	MVP	3, 6
FR-04-08	Camera	Must	MVP	3, 6
FR-04-09	Show names	Should	MVP	3
FR-04-11	Walk modifier			3
FR-04-12	Input rebinding	Must	MVP (mobile button layout: Later)	1, 3, 4
FR-04-15	Controller support and hot-plug	Must	MVP	3, 6
FR-05-01	Cosmetic-only	Must	MVP	1, 5, 7
FR-05-02	Split upper and lower costumes	Must	MVP	1, 3, 7
FR-05-03	Colours, patterns and faceplates	Must	MVP	1, 7
FR-05-04	Costume traits	Could	R2	7
FR-05-05	Faceplate occlusion	Should	MVP	5, 7
FR-05-06	Emotes	Must	MVP	1, 3, 6, 7, 9
FR-05-07	Celebrations	Should	MVP	7
FR-05-08	Banners and titles	Should	R2	7
FR-05-09	Rarity labels	Could	R2	7
FR-05-10	How items are obtained	Must	MVP	4, 7, 8
FR-05-11	Coin Shop	Should	MVP	7
FR-05-12	Daily personal store	Won't	—	7
FR-05-14	Premium currency	Won't	—	7, 8
FR-05-16	Paid packs			7
FR-05-17	Dressing room	Must	MVP	1, 4, 7, 8
FR-06-01	XP and player level	Must	MVP	1, 7, 8
FR-06-02	Season Path			1, 7, 10
FR-06-03	Buying the pass with currency	Won't	—	7
FR-06-05	Wallet and ledger	Must	MVP	7, 8
FR-06-06	Earning Coins	Must	MVP	1, 7
FR-06-07	Trophies and Trophy Rank	Should	MVP	1
FR-06-08	Second currency (reference: “Crown Shards”)	Won't	—	7
FR-06-13	Challenges	Should	MVP	1, 7, 10
FR-06-19	Economy admin, grant and rollback tools	Must	MVP	7, 8
FR-07-02	Server region	Should	MVP	8, 9
FR-07-03	Party	Should	MVP	1, 8, 9
FR-07-05	Cross-play and input parity	Could	Later (AC1: MVP)	3, 9
FR-07-06	Accounts, child profiles and progression	Must	MVP	4, 8, 9
FR-07-07	Friends	Should	MVP	1, 9
FR-07-08	Private Shows	Should	R2	1, 9
FR-07-09	Voice chat	Won't	—	6, 9
FR-07-10	Text chat	Won't	—	9

Requirement	v3 title	Priority	Release	Cited in sections
FR-07-11	Spectating after elimination	Must	MVP	1, 4, 8, 9
FR-07-12	Server architecture	Must	MVP	1, 3, 8
FR-07-14	Quick-chat	Must	MVP	1, 4, 6, 8, 9
FR-07-15	Block and mute	Must	MVP	1, 4, 6, 8, 9
FR-07-16	Reconnect	Must	MVP	1, 4, 8, 9
FR-08-01	HUD layout	Must	MVP	1, 4
FR-08-02	Warning cue	Should	MVP	1, 2, 4, 6
FR-08-03	Positive wording	Must	MVP	1, 2, 4, 6
FR-08-04	Round flow	Must	MVP	1, 4, 6
FR-08-06	UI palette and contrast	Must	MVP	4
FR-08-07	Music	Should	MVP	6
FR-08-08	Sound effects	Must	MVP	3, 4, 6
FR-08-09	Volume controls	Must	MVP	4, 6
FR-08-11	Colour-blind support	Must	MVP	1, 2, 4, 5, 6
FR-08-12	Accessibility options	Must	MVP	1, 3, 4, 5, 6
FR-08-13	Readable text sizes	Must	MVP	1, 4
FR-08-14	Localisation	Must	MVP (additional languages: R2)	4, 6
FR-08-16	Error, offline and empty states	Must	MVP	1, 3, 4, 9
FR-08-17	First-run onboarding and tutorial	Must	MVP	1, 4, 6
FR-08-18	No dark patterns	Must	MVP	1, 4, 7
FR-08-19	Child-friendly language and icons	Must	MVP	1, 2, 4, 6, 10
FR-08-20	Motion comfort	Should	MVP	1, 2, 3, 4, 6
FR-09-01	Builder access	Could	Later	1, 8, 10
FR-09-02	Object library	Could	Later	1, 8, 10
FR-09-03	Budget	Could	Later	1, 8, 10
FR-09-04	Per-object caps	Could	Later	1, 8, 10
FR-09-05	Transform tools	Could	Later	1, 8, 10
FR-09-06	Overlap rules	Could	Later	1, 8, 10
FR-09-07	Object properties	Could	Later	1, 8, 10
FR-09-08	Logic	Could	Later	1, 8, 10
FR-09-09	Builder game modes	Could	Later	1, 8, 10
FR-09-10	Level settings	Could	Later	1, 8, 10
FR-09-11	Playtest	Could	Later	1, 8, 10
FR-09-12	Verification run	Could	Later	1, 8, 10
FR-09-13	Level codes	Could	Later	1, 8, 10
FR-09-14	Versioning	Could	Later	1, 8, 10
FR-09-15	Level names and tags	Could	Later	1, 8, 10
FR-09-16	Likes	Could	Later	1, 8, 10
FR-09-17	Discovery (Builder Picks and Level Hop)	Could	Later	1, 8, 10
FR-09-18	Level moderation			1, 8, 10
FR-09-19	Season container	Must	MVP	8, 10
FR-09-20	Reference pass (“Fame Pass”) (merged)			10
FR-09-21	XP boost events	Could	R2	7, 10
FR-09-22	Challenges (merged)			10
FR-09-23	Limited-time shows and live events	Should	R2	1, 7, 10

Requirement	v3 title	Priority	Release	Cited in sections
FR-09-24	Round rotation (merged)			10
FR-09-25	Ranked cycles (merged)			10
FR-09-26	News Board	Should	MVP	8, 10
FR-11-01	Child-directed service	Must	MVP	4, 8, 9
FR-11-02	Parent Account and verifiable parental consent	Must	MVP	1, 4, 8, 9
FR-11-03	No open text or voice communication	Must	MVP	1, 4, 6, 8, 9, 10
FR-11-04	Generated display names	Must	MVP	1, 4, 8, 9
FR-11-05	Curated text safety	Must	MVP	4, 7, 8, 9
FR-11-06	Child-friendly safety reporting	Must	MVP	1, 4, 8, 9, 10
FR-11-07	Parent gate	Must	MVP	4, 8, 9
FR-11-08	Parental controls	Must	MVP	1, 4, 7, 8, 9, 10
FR-11-09	Spending controls	Must	MVP	4, 7, 8, 9
FR-11-10	Parent dashboard	Should	MVP	4, 7, 8, 9
FR-11-11	Privacy by default	Must	MVP	4, 8, 9, 10
FR-11-12	Data access, export and deletion	Must	MVP	1, 4, 8, 9, 10
FR-11-13	Privacy notice and transparency	Must	MVP	4, 8, 9
FR-11-14	No advertising, no profiling, no data sale	Must	MVP	8, 9, 10
FR-11-15	Playtime and wellbeing	Should	MVP	1, 4, 7, 8, 9, 10
FR-11-16	No geolocation	Must	MVP	8, 9
FR-11-17	Moderation operations	Must	MVP	1, 8, 9, 10
FR-11-18	Age-appropriate content standard	Must	MVP	1, 6, 8, 9, 10
FR-11-19	No marketing notifications	Must	MVP	1, 4, 7, 8, 9, 10
FR-12-01	Asset manifest	Must	MVP	5, 6, 7
FR-12-02	Naming convention	Must	MVP	3, 5, 6
FR-12-03	Formats	Must	MVP	5, 6
FR-12-04	Base rig and split upper/lower costumes	Must	MVP	1, 3, 5, 7
FR-12-05	Asset budgets	Must	MVP	3, 5
FR-12-06	Placeholder fallback	Must	MVP	3, 5
FR-12-07	Provenance and rights	Must	MVP	5, 6, 8
FR-12-08	Content and originality review	Must	MVP	5, 6, 7, 10
FR-12-09	Import validation tool	Should	MVP	5
FR-12-10	Art direction (pending)	Must	MVP (M2)	1, 5
NFR-10-01	Platforms	Must	MVP	8
NFR-10-02	Engine and tooling	Must	MVP (M0)	5, 8
NFR-10-03	Player count	Must	MVP	1, 8
NFR-10-04	Performance (client and server)			2, 5, 8
NFR-10-05	Netcode quality	Must	MVP	3, 8
NFR-10-06	Launch scaling	Must	MVP	8
NFR-10-08	Children's privacy	Must	MVP	8
NFR-10-09	Communications	Must	MVP	8
NFR-10-12	Analytics and telemetry	Must	MVP	1, 8, 10
NFR-10-13	Anti-cheat	Must	MVP	1, 8, 9
NFR-10-16	Security baseline	Must	MVP	8
NFR-10-17	Reliability and availability	Must	MVP	8
NFR-10-18	Load times	Should	MVP	1, 8
NFR-10-19	Accessibility conformance and photosensitivity	Must	MVP	1, 2, 3, 4, 5, 6, 8

Requirement	v3 title	Priority	Release	Cited in sections
NFR-10-20	Data-driven configuration	Must	MVP	1, 8
NFR-10-21	Data retention	Must	MVP	8, 9, 10
NFR-11-01	Data protection impact assessment and children's risk assessment	Must	MVP	8, 9

D.3 Coverage checks

- **IDs cited but not in v3:** none.
- **Retired (merged) v3 IDs cited:** FR-04-06 (Section 6), FR-09-20 (Section 10), FR-09-22 (Section 10), FR-09-24 (Section 10), FR-09-25 (Section 10). These appear only as "absorbs/retired" notes, not as design anchors, unless stated otherwise in the section.
- **MVP Must requirements not cited by any section:** FR-06-18 (No paid subscription), NFR-10-14 (Age rating).

Appendix E — Merge method and verification

E.1 How the merge was done

1. The ten source documents were read in full together with the core requirements v3 (`Wobblebeans_Requirements_v3.docx`).
2. Each source document became one section, in source order. Document headers moved to the Document control page; issue lists, traceability matrices and open questions moved to Appendices A, D and C.
3. Section and issue references were rewritten to point inside this document (see *How to read this document*). References to the requirements and their gapfill annexes were left unchanged.
4. Conflicts were reconciled by the order of authority (Appendix B.1) and duplicate content was removed (Appendix B.2). No new requirements, numbers or decisions were added; where the sources leave a point open, it is listed in Appendix C or A.2.
5. Mermaid diagrams were rendered with mermaid-cli and embedded as images. Very wide diagrams are placed on landscape pages.

E.2 Checks made during the merge

Check	Result
Progression simulation (`sim.py` in the design folder, Section 7.10) re-run	Output matches the table in §7.10: Casual 30,000 XP/week, tiers by week 2/4/6/8 = 15/27/39/51, Coins 8,000 (v3) / 4,760 (tuned), 4 wins; Regular 58,900 XP/week, 27/50/70/70, 18,000 / 10,920, 8 wins; Hardcore 110,300 XP/week, 47/70/70/70, 43,800 / 25,320, 30 wins
Contrast of Night Ink #1F2140 on Cloud #FFFDF7 (UI token rule "≥ 12:1", §4.8.1)	15.3:1, passes
Mermaid diagrams	24 found in the sources; 4 failed to parse and were fixed (Appendix B.3); 1 removed as a duplicate; 23 embedded
Requirement IDs	Every requirement ID cited was checked against the v3 document (Appendix D.3)
Issue lists	Every source issue maps to a register entry (Appendix A.3)

E.3 Placeholders

All imagery is a placeholder: the cover image, every "(img)" slot in the wireframes, and the colours, icons and fonts named in Sections 4 and 5. Final art comes from the owner's GitHub asset repository (UD-3, Q-D2); the import checklist is in §5.9.